# Arithmetic Quantum Mechanics
Lab Book

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## Contents

# 1 Overview

This campaign's north star is a general definition and workflow that assigns to an arbitrary arithmetic scheme one or more *canonical quantum systems*: kinematics realised as a Hilbert space, observables realised as an algebra, and, where possible, dynamics realised as automorphisms or as projective unitary representations of symmetry groups, with fusion categories as the expected general endpoint. The word *canonical* is a technical requirement here, not a mood: the assignment must be exhibited as a functor, every choice on which it depends must be named explicitly, and a theorem must state exactly what is independent of those choices.

The concrete guiding path fixes a target before the generalities are attempted: the prime field $\mathbb{F}_p$ of characteristic $p$, its canonical symplectic space $\mathbb{F}_p \times \mathbb{F}_p$, and the finite Weyl–Heisenberg system quantizing it. Every general definition this campaign proposes must reproduce this case exactly, and is tested against it before anything else. The result is recorded in this lab book, one topic per section, each statement carrying its full hypotheses, an honest status, and its provenance (§1); nothing in these pages is asserted at a strength its evidence does not support.

### Status

The following table records the **28** rows issued by the campaign's first increment (`briefs/wh-kappa-target.md`) the canonical quantum system attached to a finite field $\kappa$, with every choice it depends on named. Of these, **22** are PROVED (four of them PROVED[†], meaning the proved statement is restricted to $p = 2$ or to odd $p$ and displays that restriction in its own text — never abbreviated into the label), **4** are SKETCH (an argument exists and is written down, but has not yet been through this campaign's hostile-critic round), **1** is CONJECTURE, and **1** is REFUTED (with its surviving weaker statement recorded in the adjacent row). Every definition D1–D11 of `definitions.md` and every row below is restated in full, under a descriptive name, in the sections that follow; the table records only where.

The current register, including the local-ring results and the separate field-with-one-element sidequest, has **52** rows: **38** PROVED, **12** SKETCH, one CONJECTURE and one REFUTED. Section 11 compares the literature on the field with one element, with qubit and tensor-product examples; seven of its finite-kernel claims are proved and eight supporting comparisons remain sketches.

The increment's central finding, proved in §4, is that the construction depends on a second datum beyond the field and the additive character: a *polarizing cocycle*. At odd characteristic this choice is immaterial and has a canonical representative. At characteristic two it is not immaterial: the admissible cocycles split into exactly two $SL_2(\kappa)$-orbits, distinguished by the Arf invariant of an associated quadratic form, giving two genuinely non-isomorphic Heisenberg groups — a fact detectable by an operator identity inside the quantum system itself. Per `PRD.md`, a sharp result of this shape is a positive structural statement about the definition, not a negative one.

| Claim id | Statement (short) | Status | Where |
|---|---|---|---|
| `WH-FORM` | The symplectic form is $\kappa$-bilinear, alternating, nondegenerate; its $\kappa$-linear and $\mathbb{F}_p$-linear isometries coincide, both $SL_2(\kappa)$. | PROVED | Thm. 2.6 |
| `WH-COMM` | The Weyl commutation relations hold uniformly in $p$, for every admissible cocycle. | PROVED | Thm. 2.7 |
| `WH-ALG` | The twisted algebra is simple, dimension $q^2$, centre $\mathbb{C}$, for every cocycle. | PROVED | Thm. 2.8 |

*continued on next page*

| Claim id | Statement (short) | Status | Where |
|---|---|---|---|
| WH-POL | Every line is Lagrangian; a Schrödinger model is a pair $(L, \chi)$, and all such pairs give isomorphic modules. | PROVED | Thm. 3.3 |
| WH-POL-2 | At $p = 2$, characters induced on a line with $Q_\beta \neq 0$ are forced to order 4; no such forcing at odd $p$. | PROVED † | Thm. 3.4 |
| WH-SVN | Uniqueness of the finite Weyl representation: one irreducible unitary representation of dimension $q$, intertwiner unique up to $U(1)$, every automorphism inner. | PROVED | Thm. 3.5 |
| WH-ALG-MAT | The algebra is isomorphic to $M_q(\mathbb{C})$, by a non-canonical isomorphism. | PROVED | Thm. 3.6 |
| WH-BETA-a | Admissible polarizing cocycles form a torsor of size $q^3$ under the symmetric bilinear forms. | PROVED | Thm. 4.2 |
| WH-BETA-b | At odd $p$, the cocycle is immaterial: $\omega/2$ is a canonical representative fixing an explicit isomorphism between any two choices. | PROVED † | Thm. 4.3 |
| WH-BETA-c | At $p = 2$, no symmetric cocycle exists: the odd-$p$ route to a canonical choice is unavailable. | PROVED † | Thm. 4.4 |
| WH-BETA-d | At $p = 2$, $Q_\beta$ is a quadratic form with polar form $\omega$; the resulting element-order profile of $H_\beta(\kappa)$ is computed, and at least two isomorphism classes occur. | PROVED † | Thm. 4.5 |
| WH-BETA-TYPE | **Heart of the increment.** The Arf invariant classifies $\mathrm{Adm}(\omega)$ into exactly two $SL_2(\kappa)$-orbits at $p = 2$, both nonempty, giving exactly two isomorphism classes of Heisenberg group. | SKETCH | Thm. 4.7 |
| WH-BETA-EPS | The Arf type is detected inside the quantum system itself, by an explicit operator identity. | SKETCH | Thm. 4.8 |
| WH-BETA-LEVEL | The algebra and its Weyl frame do not depend on $\beta$ at any characteristic; the finite level-$\mu_p$ frame does, and only at $p = 2$. | SKETCH | Thm. 4.10 |
| WH-WEIL-a | Frame automorphisms surject onto $SL_2(\kappa)$ with kernel $\hat{V}$, at every characteristic. | PROVED | Thm. 5.2 |
| WH-WEIL-b | At odd $p$, this extension splits explicitly, giving a projective unitary representation of $SL_2(\kappa)$. | PROVED † | Thm. 5.3 |

*continued on next page*

| Claim id | Statement (short) | Status | Where |
|---|---|---|---|
| `WH-WEIL-c` | At $p = 2$, $\mu_2$-valued phases occur exactly over the orthogonal subgroup $O(Q_\beta)$, proper for the reference cocycle. | PROVED † | Thm. 5.4 |
| `WH-WEIL-d` | At $p = 2$, $\mu_4$ is forced exactly when $O(Q_\beta)$ is proper — always except the single case $q = 2$, anisotropic. | PROVED † | Thm. 5.5 |
| `WH-WEIL-e` | Whether the extension splits at $p = 2$ for every $q = 2^m$; verified only at $q = 2, 4$. | CONJECTURE | Conj. 5.6 |
| `WH-SYMM` | The algebra does not remember the $\kappa$-module structure of $V(\kappa)$; its intrinsic $\mathbb{F}_p$-linear symmetry is the strictly larger $Sp_{2m}(\mathbb{F}_p)$ once $m > 1$. | PROVED | Thm. 7.1 |
| `WH-CHOICE` | Beyond $\kappa$, exactly two data: $\psi$ and $\beta$; the resulting algebras are always isomorphic, but no isomorphism among them is distinguished. | PROVED | Thm. 6.2 |
| `WH-CANON` | The projective Hilbert space is canonical in $(\kappa, \psi)$, at fixed $\beta$. | PROVED | Thm. 6.3 |
| `WH-CANON-BETA` | The projective Hilbert space is canonical independently of $\beta$ as well. | SKETCH | Thm. 6.4 |
| `WH-FUNCT-a` | The trace-normalized character family is not restriction-compatible along field extensions; trivial exactly when $p \mid n$. | PROVED | Thm. 8.1 |
| `WH-FUNCT-b` | $(\kappa, \psi) \mapsto A_{\psi,\beta_0}(V(\kappa))$ is a functor on character-compatible embeddings. | PROVED | Thm. 8.2 |
| `WH-FUNCT-b-SEC` | *Refuted:* a Frobenius-compatible character exists for every finite field. | REFUTED | Thm. 8.3 |
| `WH-FUNCT-c` | The Frobenius obstruction, exactly: invariance forces $c \in \mathbb{F}_p$, and $\mathbb{F}_{p^p}$ then restricts trivially to the prime field. | PROVED | Thm. 8.4 |
| `WH-FUNCT-d` | Compatible character systems over an algebraic closure are exactly the restrictions of its characters nontrivial on $\mathbb{F}_p$. | PROVED | Thm. 8.5 |

## 2 The symplectic object of a finite field, and its twisted operator algebra

Throughout the labbook $p$ denotes a prime, $m \geq 1$ an integer, $q := p^m$, and $\kappa := \mathbb{F}_q$ the finite field of order $q$, with prime field $\mathbb{F}_p \subseteq \kappa$. **The case $p = 2$ is in scope everywhere in this campaign**: no definition or theorem below is stated only for odd $p$ and silently assumed to extend; where a statement holds only for $p = 2$ or only for odd $p$ this is marked in the statement itself, never left to a reader's guess. This section builds the ground floor of the construction — the symplectic vector space attached to $\kappa$, the two choices (a polarizing cocycle and an additive character) the quantum system is built from, the twisted operator algebra those choices produce, and its associated Heisenberg group — and proves the three structural facts that hold *uniformly* in the characteristic and for *every* admissible choice of cocycle. What is and is not canonical about the choices themselves is the subject of later sections, in particular §4.

## 2.1 The symplectic space of a finite field

**Definition 2.1** (The symplectic object attached to a finite field)**.** For a finite field $\kappa$, put $V(\kappa) := \kappa \oplus \kappa$, and define

$$\omega : V(\kappa) \times V(\kappa) \to \kappa, \qquad \omega\big((a,b),(a',b')\big) := ab' - a'b.$$

*Scope.* This is a stipulation: a vector space and a bilinear pairing written down by formula. On its own it asserts nothing about $\omega$ — not that it is $\kappa$-bilinear, not that it is alternating, not that it is nondegenerate. Those are properties, established as Theorem 2.6 below. No restriction on the characteristic is built into the definition; $p = 2$ is included on the same footing as every other prime.

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**Provenance.** ids: D1; proved in: —; tested by: —; reviewed in: —.

## 2.2 Polarizing cocycles: the datum $\beta$, introduced

The Weyl operators of §2.4 below are built not from $\omega$ directly but from a bilinear map whose antisymmetrization is $\omega$. Choosing one such map is a second datum of the construction, on exactly the same footing as the choice of additive character $\psi$ of §2.3. This subsection only introduces the datum and its notation; the fact that it forms a torsor, that it is immaterial at odd characteristic, and that it splits the construction into two genuinely different isomorphism types at $p = 2$ is the content of Theorems 4.2–4.7 in §4, and is not anticipated here.

**Definition 2.2** (Admissible polarizing cocycles, and the reference cocycle)**.** Fix a finite field $\kappa$ and its symplectic object $(V(\kappa), \omega)$ of Definition 2.1. The set of *admissible polarizing cocycles* is

$$\mathrm{Adm}(\omega) := \big\{\, \beta : V(\kappa) \times V(\kappa) \to \kappa \;\big|\; \beta \text{ is } \kappa\text{-bilinear and } \beta - \beta^T = \omega \,\big\},$$

where $\beta^T(v, v') := \beta(v', v)$. A *polarizing cocycle* is a choice of element $\beta \in \mathrm{Adm}(\omega)$, and it is a **datum of the construction**, on the same footing as the character $\psi$ of Definition 2.3 — not a convention fixed once and for all. The map

$$\beta_0\big((a,b),(a',b')\big) := ab'$$

lies in $\mathrm{Adm}(\omega)$ and is called the *reference cocycle*: a label for a point of the torsor of Definition 2.2, not a canonical point of it (Theorem 4.2).

*Scope.* Nothing here asserts that $\mathrm{Adm}(\omega)$ is nonempty, that it has any particular size, that $\beta_0$ is a typical or a distinguished element of it, or that the choice of $\beta$ is or is not immaterial to anything built from it. All of that is claimed, and proved, later: Theorems 4.2–4.7 of §4 are where this datum's structure is actually established. In particular the tempting odd-characteristic habit of writing $\omega/2$ for "the" polarizing cocycle is not part of this definition and is not available at $p = 2$, where division by 2 is division by 0; §4 makes this precise.

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**Provenance.** ids: D2; proved in: —; tested by: —; reviewed in: —.

### 2.3 The phase datum

**Definition 2.3** (Phase datum, and the trace-normalized family)**.** A *phase datum* for $\kappa$ is a nontrivial additive character $\psi : (\kappa, +) \to \mathbb{C}^\times$, i.e. a group homomorphism with $\psi \not\equiv 1$. Write $X(\kappa)$ for the set of such $\psi$. Inside $X(\kappa)$, the *trace-normalized family* is defined as follows: for $\kappa = \mathbb{F}_{p^m}$ put

$$\mathrm{Tr}_{\kappa/\mathbb{F}_p}(z) := z + z^p + z^{p^2} + \cdots + z^{p^{m-1}},$$

and for $\zeta$ a primitive $p$-th root of unity in $\mathbb{C}$, put

$$\psi_\zeta(z) := \zeta^{\,\mathrm{Tr}_{\kappa/\mathbb{F}_p}(z)}.$$

*Scope.* Two choices are named here and neither is suppressed: the character $\psi$ itself (an arbitrary element of $X(\kappa)$), and, when working inside the distinguished family, the root of unity $\zeta$. The absolute trace $\mathrm{Tr}_{\kappa/\mathbb{F}_p}$ is canonical — it is the same $\mathbb{F}_p$-linear map regardless of any further choice — but $\psi_\zeta$ is canonical only once $\zeta$ is fixed, and §8 shows in detail how far a compatible choice of $\psi$ across a tower of fields can and cannot be made.

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**Provenance.** ids: D3; proved in: —; tested by: —; reviewed in: —.

### 2.4 Weyl operators, the twisted algebra, and the Heisenberg group

**Definition 2.4** (Weyl operators and the twisted observable algebra)**.** Fix a phase datum $\psi$ as in Definition 2.3 and a polarizing cocycle $\beta \in \mathrm{Adm}(\omega)$ as in Definition 2.2. The *twisted operator algebra* $A_{\psi,\beta}(V)$ is the $\mathbb{C}$-vector space

$$A_{\psi,\beta}(V) := \bigoplus_{v \in V(\kappa)} \mathbb{C} \cdot W_\beta(v)$$

with the product fixed, once and for all, on basis elements by

$$W_\beta(v)\, W_\beta(v') := \psi\big(\beta(v, v')\big)\, W_\beta(v + v').$$

*Scope.* The ordering of the product on the right ($\psi(\beta(v, v'))$, not $\psi(\beta(v', v))$) is fixed as part of the definition and is not symmetrized; which of the two orderings a later stipulated model (such as $M_0$ of §3) actually realizes is a computation, not a convention, and is checked explicitly there. $A_{\psi,\beta}(V)$ depends on $\kappa$, $\psi$ and $\beta$ and on nothing else; that this dependence is associative, unital and of dimension $q^2$ is Theorem 2.7(a) below, not part of the stipulation.

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**Provenance.** ids: D4; proved in: —; tested by: —; reviewed in: —.

**Definition 2.5** (The Heisenberg group of a polarizing cocycle)**.** With $\beta \in \mathrm{Adm}(\omega)$ as above, the *Heisenberg group of* $\beta$ is the set $H_\beta(\kappa) := \kappa \times V(\kappa)$ with product

$$(t, v)(t', v') := \big(t + t' + \beta(v, v'),\ v + v'\big).$$

*Scope.* This construction uses no character $\psi$; it is a group attached to $\kappa$ and $\beta$ alone. It does depend on $\beta$, and — this is the content of §4 — at $p = 2$ its isomorphism class as an abstract group depends on more than the isomorphism class of the pair $(V(\kappa), \omega)$: it depends on which point of the torsor $\mathrm{Adm}(\omega)$ was chosen. Nothing about that dependence is claimed or used here.

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**Provenance.** ids: D5; proved in: —; tested by: —; reviewed in: —.

### 2.5 The form is bilinear, alternating, nondegenerate, and its isometries

PROVED

**Theorem 2.6** (The symplectic form of a finite field, at every characteristic)**.** *Let $\kappa$ be any finite field, $p = 2$ included, with $V(\kappa)$ and $\omega$ as in Definition 2.1. Then:*

*(a) $\omega$ is $\kappa$-bilinear;*

*(b) $\omega(v, v) = 0$ for every $v \in V(\kappa)$ —* alternating *in the strong sense, not merely antisymmetric;*

*(c) $\omega$ is nondegenerate: if $\omega(v, \cdot) \equiv 0$ then $v = 0$;*

*(d) every $\beta \in \mathrm{Adm}(\omega)$ (Definition 2.2) is a 2-cocycle for the abelian group $(V(\kappa), +)$, i.e. $\beta(v, v') + \beta(v + v', v'') = \beta(v, v'' + v) + \beta(v', v'')$ for all $v, v', v''$ — equivalently $\beta(v, v') + \beta(v + v', v'') = \beta(v', v'') + \beta(v, v' + v'')$ — and $\beta - \beta^T = \omega$;*

*(e) the group of $\kappa$-linear automorphisms $g$ of $V(\kappa)$ with $\omega(gv, gv') = \omega(v, v')$ for all $v, v'$ is exactly $SL_2(\kappa)$;*

*(f) the group of merely $\mathbb{F}_p$-linear automorphisms $g$ of $V(\kappa)$ (i.e. additive maps that need not commute with multiplication by $\kappa$) with $\omega(gv, gv') = \omega(v, v')$ for all $v, v'$ is* also *exactly $SL_2(\kappa)$ — the larger class of symmetries one might a priori expect contains nothing beyond the $\kappa$-linear ones.*

*Scope.* Part (f) is a statement about the $\kappa$-valued form $\omega$ itself; it is emphatically *not* a statement about the $\mathbb{F}_p$-valued form $\psi \circ \omega$ obtained by composing with a character, whose isometries are the strictly larger group $Sp_{2m}(\mathbb{F}_p)$ once $m > 1$ — that is Theorem 7.1 of §7. Nothing here addresses which $\beta \in \mathrm{Adm}(\omega)$ is chosen, or classifies $\mathrm{Adm}(\omega)$ itself; that is §4.

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**Provenance.** ids: D1, D2, WH-FORM; proved in: theory/wh-kappa.md §1; tested by: theory/checks/wh_kappa_check.py (C1,C2); theory/checks/fp_symmetry.py (S1); reviewed in: theory/verdicts/wh-kappa-r1.md.

*Proof.* (a) Fix $v = (a, b)$, $v' = (a', b')$. Each argument of $\omega$ occurs exactly once in the formula $ab' - a'b$, and $\kappa$ is commutative, so scaling either argument by $c \in \kappa$ scales the value by $c$, and additivity in each slot is distributivity of multiplication over addition in $\kappa$. This holds verbatim at $p = 2$: no step divides by 2 or by anything else.

(b) $\omega\big((a, b), (a, b)\big) = ab - ab = 0$, an identity in the commutative ring $\kappa$ that cancels two literally equal terms, not two terms related by a factor of 2. It therefore holds at $p = 2$ exactly as it holds elsewhere. This is worth pausing on: antisymmetry alone, $\omega(v, v') = -\omega(v', v)$, gives only $2\omega(v, v) = 0$ on setting $v' = v$, which is vacuous once $2 = 0$. So at $p = 2$, (b) is a strictly stronger statement than antisymmetry and must be — and here is — verified by the direct computation above, not inferred from it.

(c) If $\omega(v, \cdot) \equiv 0$ for $v = (a, b)$ then in particular $\omega\big((a, b), (0, 1)\big) = a = 0$ and $\omega\big((a, b), (1, 0)\big) = -b = 0$, so $v = 0$.

(d) $\beta - \beta^T = \omega$ is the definition of $\mathrm{Adm}(\omega)$ (Definition 2.2). For the cocycle identity: $\beta$ is bi-additive (it is $\kappa$-bilinear, hence in particular additive in each argument), and for any bi-additive $\beta$ both sides of the cocycle identity expand, term by term, to $\beta(v, v') + \beta(v, v'') + \beta(v', v'')$; the identity holds without using anything about $\omega$ beyond bi-additivity of $\beta$.

(e) Write $v = (a, b)$, $v' = (a', b')$ as the columns of a $2 \times 2$ matrix over $\kappa$; then $\omega(v, v') = ab' - a'b = \det[v \,|\, v']$ by direct expansion. For $g \in GL_2(\kappa)$, $[gv \,|\, gv'] = g[v \,|\, v']$ as matrices, and multiplicativity of the determinant for $2 \times 2$ matrices over a commutative ring (itself a polynomial identity verified by direct expansion, holding in every characteristic) gives $\omega(gv, gv') = \det(g) \cdot$

$\omega(v,v')$. Hence $\omega \circ g = \omega$ iff $(\det g - 1)\,\omega \equiv 0$; since $\omega\big((1,0),(0,1)\big) = 1 \neq 0$, this holds iff $\det g = 1$, i.e. iff $g \in SL_2(\kappa)$.

(f) Let $g : V(\kappa) \to V(\kappa)$ be additive (not assumed $\kappa$-linear) with $\omega(gv, gu) = \omega(v, u)$ for all $v, u$.

- *$g$ is bijective.* If $gv = 0$ then $\omega(v, u) = \omega(gv, gu) = 0$ for every $u$, so $v = 0$ by (c); thus $g$ is injective, hence bijective since $V(\kappa)$ is finite.
- *$g$ sends the standard basis to a $\kappa$-basis.* Let $v_1 = (1, 0)$, $v_2 = (0, 1)$, so $\omega(v_1, v_2) = 1$. Then $\omega(gv_1, gv_2) = \omega(v_1, v_2) = 1 \neq 0$; were $gv_1, gv_2$ $\kappa$-dependent, bilinearity and (b) would force $\omega(gv_1, gv_2) = 0$ (if, say, $gv_2 = c \cdot gv_1$ then $\omega(gv_1, gv_2) = c\,\omega(gv_1, gv_1) = 0$ by (a) and (b)). So $\{gv_1, gv_2\}$ is $\kappa$-independent, hence a basis of the 2-dimensional space $V(\kappa)$.
- *$g$ is $\kappa$-linear.* Fix $c \in \kappa$, $u \in V(\kappa)$, and put $w := g(cu) - c \cdot g(u)$ (subtraction makes sense because $g$ is additive, hence $\mathbb{Z}$-linear, even before $\kappa$-linearity is known). For $i = 1, 2$,

  $$\omega(gv_i, w) = \omega\big(gv_i, g(cu)\big) - c\,\omega(gv_i, gu) = \omega(v_i, cu) - c\,\omega(v_i, u) = c\,\omega(v_i, u) - c\,\omega(v_i, u) = 0,$$

  using the hypothesis $\omega(g\cdot, g\cdot) = \omega(\cdot, \cdot)$ for the middle equality and $\kappa$-bilinearity (a) for the outer ones. Since $\{gv_1, gv_2\}$ is a basis, $\omega(gv_1, \cdot)$ and $\omega(gv_2, \cdot)$ have trivial common kernel only at 0 by nondegeneracy (c) applied after a change of basis, so $w = 0$. Hence $g(cu) = c \cdot g(u)$ for all $c, u$: $g$ is $\kappa$-linear.

By $\kappa$-linearity and (e), $\det g = 1$, so $g \in SL_2(\kappa)$; the converse containment is immediate from (e). This closes (f). $\square$

*Remark.* Part (f) forecloses a tempting shortcut: one might hope to characterize the $\mathbb{F}_p$-linear isometries of $\omega$ as the "$\omega$-self-adjoint $\mathbb{F}_p$-endomorphisms," by analogy with the real symplectic case. This is false at $p = 2$: exhaustive enumeration finds 8 and 64 $\omega$-self-adjoint $\mathbb{F}_p$-endomorphisms of $V(\kappa)$ at $q = 2, 4$ respectively, against only 2 and 4 $\kappa$-scalars — the self-adjointness route simply does not isolate $SL_2(\kappa)$ once $p = 2$, and the direct argument above is what closes (f) instead. Independently, complete backtracking search over the $\mathbb{F}_p$-linear isometries of the $\kappa$-valued form finds exactly $|SL_2(\kappa)| = 6, 24, 60, 120, 504, 720$ of them at $q = 2, 3, 4, 5, 8, 9$, matching an independent enumeration of $SL_2(\kappa)$ set-for-set at every one of these values (`theory/checks/fp_symmetry.py`, gates S1 and S4).

## 2.6 Weyl relations, uniformly in the characteristic and for every choice of cocycle

PROVED

**Theorem 2.7** (The Weyl commutation relations, for every admissible cocycle)**.** *Let $\kappa$ be any finite field, $\psi$ a phase datum (Definition 2.3), and $\beta \in \mathrm{Adm}(\omega)$ an* arbitrary *admissible polarizing cocycle (Definition 2.2). Then, inside $A_{\psi,\beta}(V)$ of Definition 2.4:*

*(a) $A_{\psi,\beta}(V)$ is a unital associative $\mathbb{C}$-algebra of dimension $q^2$, with unit $W_\beta(0)$;*

*(b) every $W_\beta(v)$ is invertible, with $W_\beta(v)^{-1} = \psi\big(\beta(v, v)\big)\,W_\beta(-v)$;*

*(c) $W_\beta(v)\,W_\beta(v') = \psi\big(\omega(v, v')\big)\,W_\beta(v')\,W_\beta(v)$ for all $v, v' \in V(\kappa)$;*

*(d) $W_\beta(u)\,W_\beta(v)\,W_\beta(u)^{-1} = \psi\big(\omega(u, v)\big)\,W_\beta(v)$ for all $u, v \in V(\kappa)$.*

*Scope.* No part of this theorem uses which admissible cocycle $\beta \in \mathrm{Adm}(\omega)$ was chosen, nor any property of $\beta$ beyond bi-additivity and the defining relation $\beta - \beta^T = \omega$; in particular nothing here distinguishes the reference cocycle $\beta_0$ from any other point of the torsor. No hypothesis on the characteristic is used beyond what already went into Theorem 2.6. What *is* $\beta$-dependent — the isomorphism class of the group $H_\beta(\kappa)$ itself, at $p = 2$ — is not addressed here.

**Provenance.** ids: D1, D2, D3, D4, WH-FORM, WH-COMM; proved in: theory/wh-kappa.md §2; tested by: theory/checks/wh_kappa_check.py (C3,C4); reviewed in: theory/verdicts/wh-kappa-r1.md.

*Proof.* (a) Associativity on basis elements amounts to

$$\big(W_\beta(v)W_\beta(v')\big)W_\beta(v'') = W_\beta(v)\big(W_\beta(v')W_\beta(v'')\big),$$

which unwinds to the 2-cocycle identity for $\psi \circ \beta$; this holds because $\beta$ is a 2-cocycle (Theorem 2.6(d)) and $\psi$ is multiplicative. Since $\beta(0,v) = \beta(v,0) = 0$ ($\kappa$-bilinearity), and $\psi(0) = 1$, $W_\beta(0)$ is a two-sided unit. The dimension is $|V(\kappa)| = q^2$ by construction.

(b) By bi-additivity, $\beta(v,-v) = -\beta(v,v)$, so $W_\beta(v)W_\beta(-v) = \psi\big(\beta(v,-v)\big)W_\beta(0) = \psi\big(-\beta(v,v)\big) \cdot 1$, and symmetrically on the other side; since $\psi$ takes values in $\mathbb{C}^\times$, this exhibits a two-sided inverse.

(c) By the defining product, $W_\beta(v)W_\beta(v') = \psi\big(\beta(v,v')\big)W_\beta(v+v')$ and $W_\beta(v')W_\beta(v) = \psi\big(\beta(v',v)\big)W_\beta(v+v')$. By (b), $W_\beta(v+v')$ is invertible, so

$$W_\beta(v)W_\beta(v') = \psi\big(\beta(v,v') - \beta(v',v)\big)\, W_\beta(v')W_\beta(v) = \psi\big(\omega(v,v')\big)\, W_\beta(v')W_\beta(v),$$

the last step by the defining relation $\beta - \beta^T = \omega$ of Definition 2.2.

(d) Multiply (c) on the right by $W_\beta(v)^{-1}$, whose existence is (b). □

*Remark* (The centre of the Heisenberg group, and the algebra as a group-algebra quotient). The group $H_\beta(\kappa)$ of Definition 2.5 is indeed a group: associativity is the cocycle identity of Theorem 2.6(d), the identity is $(0,0)$, and $(t,v)$ has inverse $(-t+\beta(v,v),-v)$. Its centre is exactly $\kappa \times 0$: $(t,v)$ is central iff $\beta(v,v') - \beta(v',v) = 0$ for every $v'$, i.e. iff $\omega(v,\cdot) \equiv 0$, i.e. iff $v = 0$ by Theorem 2.6(c). Consequently $A_{\psi,\beta}(V)$ is the quotient of the group algebra $\mathbb{C}[H_\beta(\kappa)]$ by the two-sided ideal identifying the central element $(t,0)$ with the scalar $\psi(t)$: the algebra is a family over $\psi \in X(\kappa)$ at *fixed* $\beta$, and the group $H_\beta(\kappa)$ itself depends on $\beta$ — a dependence that, Theorem 4.7 shows, is not always trivial once $p = 2$.

## 2.7 Simplicity of the twisted algebra

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**Theorem 2.8** (Central simplicity of the twisted algebra, with no polarization)**.** *With $\kappa$, $\psi$ and $\beta \in \mathrm{Adm}(\omega)$ as in Theorem 2.7, the algebra $A_{\psi,\beta}(V)$ of Definition 2.4 is simple, has centre $\mathbb{C}\cdot 1$, and $\dim_{\mathbb{C}} A_{\psi,\beta}(V) = q^2$.*

*Scope.* This theorem uses no choice of Lagrangian line, no module, and no property of $\beta$ beyond Theorem 2.7; in particular it holds for every $\beta \in \mathrm{Adm}(\omega)$, uniformly. It does not by itself identify $A_{\psi,\beta}(V)$ with a matrix algebra $M_q(\mathbb{C})$ — that identification exists (Theorem 3.6 of §3) but needs a polarization and is not canonical, whereas simplicity and the dimension count here need none.

**Provenance.** ids: D4, WH-COMM, WH-ALG; proved in: theory/wh-kappa.md §3; tested by: theory/checks/wh_kappa_check.py (C5,C6); reviewed in: theory/verdicts/wh-kappa-r1.md.

*Proof.* Define the *Weyl average* $E : A_{\psi,\beta}(V) \to A_{\psi,\beta}(V)$ by

$$E(x) := q^{-2} \sum_{u \in V(\kappa)} W_\beta(u)\, x\, W_\beta(u)^{-1}.$$

*Step 1:* $E\big(\sum_v a_v W_\beta(v)\big) = a_0 \cdot 1$. By Theorem 2.7(d), $W_\beta(u)W_\beta(v)W_\beta(u)^{-1} = \psi(\omega(u,v))W_\beta(v)$, so

$$E\Big(\sum_v a_v W_\beta(v)\Big) = \sum_v a_v \Big(q^{-2} \sum_u \psi\big(\omega(u,v)\big)\Big) W_\beta(v).$$

Fix $v \neq 0$. By Theorem 2.6(a),(c), $\omega(\cdot, v)$ is a nonzero $\kappa$-linear map $V(\kappa) \to \kappa$, hence surjective (a nonzero linear map between finite-dimensional vector spaces over the same field onto a 1-dimensional target is onto). So $u \mapsto \psi(\omega(u,v))$ is a character of the additive group $V(\kappa)$ that is nontrivial (since $\psi$ is nontrivial and $\omega(\cdot, v)$ is onto $\kappa$). A nontrivial character $\chi$ of a finite abelian group $G$ sums to zero: fixing $u_0$ with $\chi(u_0) \neq 1$, the substitution $u \mapsto u + u_0$ permutes $G$, so $S := \sum_u \chi(u)$ satisfies $S = \chi(u_0)S$, forcing $S = 0$ since $\chi(u_0) \neq 1$. Applying this to $\chi = \psi(\omega(\cdot, v))$ gives $\sum_u \psi(\omega(u,v)) = 0$ for $v \neq 0$; for $v = 0$ the sum is $q^2 \cdot \psi(0) = q^2$. Hence $E(\sum_v a_v W_\beta(v)) = a_0 \cdot W_\beta(0) = a_0 \cdot 1$, since $W_\beta(0)$ is the unit (Theorem 2.7(a)).

*Step 2:* $Z(A_{\psi,\beta}(V)) = \mathbb{C} \cdot 1$. If $z$ is central then $W_\beta(u) z W_\beta(u)^{-1} = z$ for every $u$, so $z = E(z) = a_0 \cdot 1$ by Step 1; conversely every scalar is central.

*Step 3: simplicity.* Let $I$ be a nonzero two-sided ideal and pick $0 \neq x = \sum_v a_v W_\beta(v) \in I$ with some coefficient $a_{v_0} \neq 0$. Then $y := W_\beta(v_0)^{-1} x \in I$ (as $I$ is a left ideal), and by Theorem 2.7(b),(c) each $W_\beta(v_0)^{-1} W_\beta(v)$ is a nonzero scalar multiple of $W_\beta(v - v_0)$; in particular the coefficient of $W_\beta(0)$ in $y$ is a nonzero scalar multiple of $a_{v_0}$, hence nonzero. Now $E(y) \in I$: $E$ is a $\mathbb{C}$-linear combination of the maps $x \mapsto W_\beta(u) x W_\beta(u)^{-1}$, and $I$ is a two-sided ideal, so each term $W_\beta(u) y W_\beta(u)^{-1}$ lies in $I$ and so does their average. By Step 1, $E(y) = c \cdot 1$ with $c \neq 0$ (the coefficient of $W_\beta(0)$ in $y$), so $1 \in I$ and $I = A_{\psi,\beta}(V)$.

*Step 4: dimension.* $\dim_{\mathbb{C}} A_{\psi,\beta}(V) = |V(\kappa)| = q^2$ directly from Definition 2.4. □

*Remark* (Why the quantifier over $\beta$ can be widened after the fact). Theorems 2.7 and 2.8 are stated for *every* $\beta \in \mathrm{Adm}(\omega)$, not merely the reference cocycle $\beta_0$. This widening is legitimate precisely because no step of either proof used more than the defining relation $\beta - \beta^T = \omega$ and bi-additivity: the commutator computation in Theorem 2.7(c)–(d) depends on $\beta$ only through $\omega = \beta - \beta^T$, and simplicity in Theorem 2.8 depends only on $\omega$ being nondegenerate. That the choice of $\beta$ is nonetheless *not* immaterial to everything built from it — it is invisible to the pair $(A_{\psi,\beta}(V), \text{Weyl frame})$ but not to the Heisenberg group $H_\beta(\kappa)$ itself once $p = 2$ — is the subject of §4.

# 3 Lagrangian lines, Schrödinger models, and the uniqueness of the finite Weyl representation

**Recollection (the setting of §2).** For a finite field $\kappa = \mathbb{F}_q$, $q = p^m$, the symplectic object is $V(\kappa) := \kappa \oplus \kappa$ with the $\kappa$-bilinear, alternating, nondegenerate form $\omega\big((a,b),(a',b')\big) := ab' - a'b$ (Definition 2.1); an admissible polarizing cocycle is a $\kappa$-bilinear $\beta : V(\kappa) \times V(\kappa) \to \kappa$ with $\beta - \beta^T = \omega$ (Definition 2.2); a phase datum is a nontrivial additive character $\psi : (\kappa, +) \to \mathbb{C}^\times$ (Definition 2.3); and the twisted algebra $A_{\psi,\beta}(V) := \bigoplus_{v \in V(\kappa)} \mathbb{C} \cdot W_\beta(v)$ has product $W_\beta(v) W_\beta(v') := \psi(\beta(v,v')) W_\beta(v + v')$ (Definition 2.4); by Theorem 2.8 it is simple of dimension $q^2$ with centre $\mathbb{C} \cdot 1$, for every $\beta \in \mathrm{Adm}(\omega)$.

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**Provenance.** ids: D1, D2, D3, D4; proved in: —; tested by: —; reviewed in: —.

This section names the Lagrangian lines of $V(\kappa)$, the finite-dimensional modules of $A_{\psi,\beta}(V)$ they induce, and proves that there is exactly one such module up to unitary equivalence and that it realizes $A_{\psi,\beta}(V)$ as a full matrix algebra — a finite-field analogue of the Stone–von Neumann theorem — together with a genuinely characteristic-two phenomenon in how the induced characters behave.

## 3.1 Preliminary facts about finite abelian characters and cocycles

Four short, elementary facts about characters of finite abelian groups recur throughout the rest of the labbook; they are recorded once, here, and cited by name afterward. None of them is part of the claims DAG — they carry no claim id and no status label, being standard and immediate from the definitions of "character" and "cocycle" — but each is proved in full, since nothing in this labbook is asserted from memory.

**Fact 1 (orthogonality).** *If $G$ is a finite abelian group and $\chi : G \to \mathbb{C}^\times$ a nontrivial character, then $\sum_{u\in G} \chi(u) = 0$.* *Proof.* Pick $u_0 \in G$ with $\chi(u_0) \neq 1$. Then $S := \sum_u \chi(u) = \sum_u \chi(u{+}u_0) = \chi(u_0)\, S$, because $u \mapsto u{+}u_0$ permutes $G$; since $\chi(u_0) \neq 1$, $S = 0$.

**Fact 2 (valuation).** *If $\psi$ is a phase datum for $\kappa$ (Definition 2.3) then $\psi(\kappa) \subseteq \mu_p$; $\ker\psi$ is an $\mathbb{F}_p$-subspace of $\kappa$ of index $p$; and for $u \in \kappa$, $\psi(u\cdot) \equiv 1$ iff $u = 0$.* *Proof.* $\psi(t)^p = \psi(pt) = \psi(0) = 1$ since $\operatorname{char}\kappa = p$, so $\psi(\kappa) \subseteq \mu_p$. $\ker\psi$ is an additive subgroup of a characteristic-$p$ field, hence an $\mathbb{F}_p$-subspace, and $\kappa/\ker\psi$ embeds nontrivially into $\mu_p \cong \mathbb{Z}/p$ (Definition 2.3), forcing index exactly $p$. If $u \neq 0$ then $u\kappa = \kappa \not\subseteq \ker\psi$, i.e. $\psi(u\cdot) \not\equiv 1$.

**Fact 3 (duality of elementary abelian groups).** *Let $U$ be a finite $\mathbb{F}_p$-vector space of dimension $d$. Then $\hat{U} := \operatorname{Hom}(U, \mathbb{C}^\times)$ has order $p^d$, and every element of $\hat{U}$ takes values in $\mu_p$.* *Proof.* For $\chi \in \hat{U}$, $\chi(u)^p = \chi(pu) = 1$, so $\chi(U) \subseteq \mu_p \cong \mathbb{Z}/p$, i.e. $\hat{U} = \operatorname{Hom}_{\mathbb{F}_p}(U, \mu_p)$. Fixing an $\mathbb{F}_p$-basis of $U$, a homomorphism is a free choice of $d$ values in $\mu_p$, so $|\hat{U}| = p^d$. (Applied below to $U = L$ a $\kappa$-line, an $\mathbb{F}_p$-space of dimension $m = [\kappa : \mathbb{F}_p]$, giving $|\hat{L}| = p^m = q$; later to $U = \kappa$ and $U = V(\kappa)$, of $\mathbb{F}_p$-dimension $m$ and $2m$, giving $|\hat{\kappa}| = q$ and $|\hat{V}| = q^2$.)

**Fact 4 (triviality of symmetric cocycles of exponent $p$).** *Let $U$ be a finite abelian group of exponent $p$ and $c : U \times U \to \mathbb{C}^\times$ a symmetric 2-cocycle: $c(u, u') = c(u', u)$ and $c(u, u')c(u + u', u'') = c(u, u' + u'')c(u', u'')$. Then there is $\mu : U \to \mathbb{C}^\times$ with $c(u, u') = \mu(u + u')/\big(\mu(u)\mu(u')\big)$ for all $u, u'$.* *Proof.* On $E := \mathbb{C}^\times \times U$ define $(z, u)(z', u') := \big(zz'c(u, u'), u + u'\big)$; the cocycle identity gives associativity and symmetry of $c$ gives commutativity, so $E$ is a finite abelian group. Induction inside $E$ gives $(1, u)^p = (\gamma(u), 0)$ with $\gamma(u) := c(u, u)c(2u, u) \cdots c((p{-}1)u, u)$. Fix an $\mathbb{F}_p$-basis $u_1, \dots, u_n$ of $U$; as $\mathbb{C}$ is algebraically closed, choose $z_i$ with $z_i^p = \gamma(u_i)^{-1}$ and put $f_i := (z_i, u_i)$, so $f_i^p = 1$. The subgroup $H := \langle f_1, \dots, f_n\rangle \leq E$ is then abelian of exponent dividing $p$, and since $U$ is free on $\{u_i\}$ there is a unique $\mathbb{F}_p$-linear $\sigma : U \to H$ with $\sigma(u_i) = f_i$ and $\mathrm{pr}_U \circ \sigma = \mathrm{id}_U$. Writing $\sigma(u) = (\mu(u), u)$, that $\sigma$ is a homomorphism reads exactly $\mu(u)\mu(u')c(u, u') = \mu(u + u')$.

### 3.2 Lagrangian lines and the algebra they generate

**Definition 3.1** (The quadratic form of a polarizing cocycle)**.** For $\beta \in \operatorname{Adm}(\omega)$, define $Q_\beta : V(\kappa) \to \kappa$ by $Q_\beta(v) := \beta(v, v)$. For the reference cocycle, $Q_{\beta_0}(a, b) = ab$.

*Scope.* Under a symmetrized cocycle (available only at odd $p$; §4) $Q_\beta$ would vanish identically — that convention does not exist here, and $Q_\beta$ is not zero. This definition asserts nothing beyond the formula: it does not by itself claim $Q_\beta$ is a quadratic form in the technical sense (quadratic homogeneity and a bilinear polarization), which is established only where needed, at Theorem 4.5 of §4.

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**Provenance.** ids: D6; proved in: —; tested by: —; reviewed in: —.

**Definition 3.2** (Schrödinger models, and the standard model)**.** For a $\kappa$-line $L \subset V(\kappa)$ put $A_L := \operatorname{span}_{\mathbb{C}}\{\, W_\beta(l) : l \in L \,\} \subseteq A_{\psi,\beta}(V)$. Given a unital $\mathbb{C}$-algebra homomorphism (character) $\chi : A_L \to \mathbb{C}$, the *Schrödinger model* of $(L, \chi)$ is

$$M_{L,\chi} := A_{\psi,\beta}(V) \otimes_{A_L} \mathbb{C}_\chi.$$

The *standard model* (taken for $\beta = \beta_0$) is $M_0 := \bigoplus_{y \in \kappa} \mathbb{C} \cdot e_y$, with $\{e_y\}$ declared orthonormal and

$$W(a,b)\, e_y := \psi\big(-b(y+a)\big)\, e_{y+a}, \qquad \text{i.e.} \qquad W(a,b) = Z(-b)X(a),$$

where $X(a)e_y := e_{y+a}$ and $Z(b)e_y := \psi(by)e_y$. **This is a stipulation.**

*Scope.* The sign in $W(a,b) = Z(-b)X(a)$ is not cosmetic: the naive assignment $W(a,b) = Z(b)X(a)$, without the sign, realizes a *different* cocycle ($-ab'$, not $\beta_0$'s $ab'$), and the discrepancy between the two is invisible at $p = 2$ (where $-1 = 1$) and visible only at odd $p$ — a characteristic-dependent failure mode this definition is written to avoid by fixing the correct sign explicitly rather than leaving it to convention. That $M_0$, so defined, genuinely satisfies the multiplication law of Definition 2.4 is verified as part of the proof of Theorem 3.5 below (Step 1) and checked computationally by gate C11 of `theory/checks/wh_kappa_check.py`.

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**Provenance.** ids: D8; proved in: —; tested by: —; reviewed in: —.

**PROVED**

**Theorem 3.3** (Lagrangian lines, and the Schrödinger models they carry)**.** *Let $\kappa$, $V(\kappa)$, $\omega$, $\beta \in \mathrm{Adm}(\omega)$, $\psi$, $A_{\psi,\beta}(V)$, $Q_\beta$ (Definition 3.1) and $A_L$, $M_{L,\chi}$ (Definition 3.2) be as above, for any finite field $\kappa$, $p = 2$ included. Then:*

*(a) every $\kappa$-line $L \subset V(\kappa)$ is $\omega$-isotropic ($\omega(v,v') = 0$ for all $v, v' \in L$), hence maximal isotropic; so* Lagrangian line *and* line *coincide as conditions on $L \subset V(\kappa)$, at every characteristic;*

*(b) there are exactly $q + 1 = |\mathbb{P}^1(\kappa)|$ such lines;*

*(c) $A_L$ is commutative of dimension $q$, and carries exactly $q$ characters, forming a torsor under $\hat{L} = \mathrm{Hom}(L, \mathbb{C}^\times)$;*

*(d) $\dim_{\mathbb{C}} M_{L,\chi} = q$;*

*(e)* ***(precision, binding on this row.)*** *A* Schrödinger model *is a* pair *$(L,\chi)$: there are $q(q+1)$ such pairs, $q$ per line — but by Theorem 3.5 below,* all of them are isomorphic as $A_{\psi,\beta}(V)$-modules. *The count $q(q+1)$ is therefore a count of labels, not of isomorphism classes of module. Among the $q+1$ lines, only those with $Q_\beta|_L \equiv 0$ carry the trivial character $\chi \equiv 1$ (since $\chi \equiv 1$ is a character of $A_L$ iff $Q_\beta(v_0)\kappa \subseteq \ker\psi$ iff $Q_\beta(v_0) = 0$, for $L = \kappa v_0$), and how many lines have $Q_\beta|_L \equiv 0$ is itself $\beta$-dependent: exactly 2 for the reference cocycle $\beta_0$ (§4 computes it in general).*

*Scope.* This theorem does not assert that distinct pairs $(L,\chi)$ give distinct, let alone non-isomorphic, modules — to the contrary, part (e) is precisely the observation that they do not, once Theorem 3.5 is available. It does not address the characteristic-two phenomenon in the values a character of $A_L$ can take; that is Theorem 3.4 below. Nothing here depends on which $\beta \in \mathrm{Adm}(\omega)$ was chosen beyond the formula for $Q_\beta$.

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**Provenance.** ids: D1, D4, D6, D8, WH-FORM, WH-ALG, WH-POL; proved in: theory/wh-kappa.md §4; tested by: theory/checks/wh_kappa_check.py (C7); reviewed in: theory/verdicts/wh-kappa-r1.md.

*Proof.* (a) Write $L = \kappa v_0$. For $c, c' \in \kappa$, $\omega(cv_0, c'v_0) = cc'\,\omega(v_0, v_0) = 0$ by $\kappa$-bilinearity (Theorem 2.6(a)) and the computed alternating property (Theorem 2.6(b)) — at $p = 2$ antisymmetry alone would not give this, so the computed form of (b) is exactly what is used. Since $\dim_\kappa V(\kappa) = 2$ and $\omega$ is nondegenerate, no subspace of dimension 2 (i.e. all of $V(\kappa)$) is isotropic, so a line is a maximal isotropic subspace: *every* line is automatically Lagrangian, and the isotropy condition singles out nothing among lines.

(b) The nonzero vectors of $V(\kappa)$ number $q^2-1$, and each line contains $q-1$ of them, so there are $(q^2-1)/(q-1) = q+1 = |\mathbb{P}^1(\kappa)|$ lines.

(c) By Definitions 2.4 and 3.1, $W(cv_0)W(c'v_0) = \psi(\beta(cv_0, c'v_0))W((c+c')v_0) = \psi\big(cc'Q_\beta(v_0)\big)W((c+c')v_0)$, using $\beta(cv_0, c'v_0) = cc'\beta(v_0, v_0) = cc'Q_\beta(v_0)$ by $\kappa$-bilinearity. The phase is symmetric in $(c, c')$, so $A_L$ is commutative; $\dim_{\mathbb{C}} A_L = |L| = q$ since $\{W(l) : l \in L\}$ is linearly independent (distinct basis elements of $A_{\psi,\beta}(V)$). *Characters exist:* $c(c, c') := \psi\big(cc'Q_\beta(v_0)\big)$ is a symmetric bi-additive, hence symmetric 2-cocycle, on $(L, +)$, a group of exponent $p$; Fact 3.1 supplies $\mu : L \to \mathbb{C}^\times$ with $c(c, c') = \mu(c+c')/(\mu(c)\mu(c'))$, and $\chi(W(cv_0)) := \mu(c)^{-1}$ is then an algebra character of $A_L$ (it converts the twisted product into the untwisted one). *Torsor:* for two characters $\chi, \chi'$ of $A_L$, the map $l \mapsto \chi'(W(l))/\chi(W(l))$ is a homomorphism $L \to \mathbb{C}^\times$, and conversely composing a character with any element of $\hat{L}$ gives another character; the action is free (a homomorphism fixing every $\chi(W(l))$ is trivial), and $|\hat{L}| = q$ by Fact 3.1 ($U = L$, $d = m$).

(d) $A_{\psi,\beta}(V)$ is free as a right $A_L$-module on any set of coset representatives $v_1, \ldots, v_q$ of $L$ in $V(\kappa)$: the elements $\{W(v_i)W(l) : l \in L\}$ are, up to the nonzero scalars $\psi(\beta(v_i, l))$, exactly the basis elements $\{W(v_i + l) : l \in L\}$ of the coset $v_i + L$. So $A_{\psi,\beta}(V) \cong A_L^{\oplus q}$ as right $A_L$-modules, and $M_{L,\chi} = A_{\psi,\beta}(V) \otimes_{A_L} \mathbb{C}_\chi$ has dimension $q^2/q = q$.

(e) There are $q$ characters per line by (c) and $q+1$ lines by (b), giving $q(q+1)$ pairs $(L, \chi)$; that they are all isomorphic as $A_{\psi,\beta}(V)$-modules is Theorem 3.5(c) below (which uses only parts (a)–(d) of this theorem, not this part, so there is no circular dependency). For $L = \kappa v_0$, $\chi \equiv 1$ is a character of $A_L$ iff $c(c, c') = \psi(cc'Q_\beta(v_0)) \equiv 1$ iff $Q_\beta(v_0)\kappa \subseteq \ker\psi$; since $Q_\beta(v_0)\kappa$ is either $\{0\}$ or all of $\kappa$ (it is a $\kappa$-subspace, being the image of scalar multiplication by $Q_\beta(v_0)$) and $\ker\psi \neq \kappa$ (Fact 3.1), this holds iff $Q_\beta(v_0) = 0$. For $\beta = \beta_0$, $Q_{\beta_0}(a, b) = ab$ (Definition 3.1) vanishes exactly on the two coordinate lines $\{a = 0\}$ and $\{b = 0\}$, so exactly 2 of the $q+1$ lines carry the trivial character. □

### 3.3 The characteristic-two phase phenomenon

**PROVED**†

**Theorem 3.4** (Forced order-four phases in the induced characters, at characteristic two)**.** *Let $L = \kappa v_0 \subset V(\kappa)$ be a line with $Q_\beta(v_0) \neq 0$ (Definition 3.1), and let $\chi$ be any character of $A_L$ (Definition 3.2, Theorem 3.3(c)).*

- *[$p = 2$.] $\chi$ takes a value of exact multiplicative order 4 on some $W(l)$, $l \in L$: the phases of the model are forced out of $\mu_2 = \psi(\kappa)$ into $\mu_4$. For the reference cocycle $\beta_0$, the lines with $Q_{\beta_0}|_L \equiv 0$ are exactly the two coordinate axes (Theorem 3.3(e)), so this phenomenon applies to $q-1$ of the $q+1$ lines.*

- *[$p$ **odd**.] No such phenomenon occurs: every character of $A_L$ can be, and by the construction of Theorem 3.3(c) is, $\mu_p$-valued.*

*Scope.* This theorem says nothing about lines with $Q_\beta|_L \equiv 0$ (there the trivial character is available and no order-4 value is forced), and nothing about the algebra $A_{\psi,\beta}(V)$ globally — only about the values a character of the specific commutative subalgebra $A_L$ can take. It does not identify *which* $\mu_4$ this is inside a larger structure; that question, and its relation to the level-$\mu_p$ Weyl frame, is taken up in §4.

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**Provenance.** ids: D6, D8, WH-POL, WH-POL-2; proved in: theory/wh-kappa.md §4 *step* $\langle 1\rangle 5, \langle 1\rangle 6$; tested by: theory/checks/beta_census.py (B6, #{Q=0}); reviewed in: theory/verdicts/wh-kappa-r1.md.

*Proof.* Write $Q_0 := Q_\beta(v_0) \neq 0$ and $\nu(c) := \chi(W(cv_0))$ for the character $\chi$ constructed in the proof of Theorem 3.3(c); $\nu(0) = 1$, and setting $c' = c$ in the cocycle relation there, $\nu(c)^2 = \psi(c^2 Q_0)\,\nu(2c)$.

[$p=2$.] Here $2c=0$, so $\nu(2c)=\nu(0)=1$ and $\nu(c)^2=\psi(c^2Q_0)$. The Frobenius map $c\mapsto c^2$ is injective, hence bijective, on the finite field $\kappa$, so $\{c^2Q_0 : c\in\kappa\}=Q_0\kappa=\kappa$ (as $Q_0\neq 0$). Since $\psi$ is nontrivial, $\psi(c^2Q_0)=-1$ for some $c\in\kappa$ — because $-1$ is the unique nontrivial element of $\mu_2=\{1,-1\}$ and $\psi(\kappa)=\mu_2$ by Fact 3.1 — giving $\nu(c)^2=-1$, i.e. $\nu(c)$ has exact order 4. The count of lines follows from Theorem 3.3(e).

[$p$ **odd.**] Put $\mu(c):=\psi(c^2Q_0/2)$ (division by 2 makes sense, $p$ being odd). Then, directly, $\psi\big((c^2+c'^2)Q_0/2\big)\,\psi(cc'Q_0)=\psi\big((c+c')^2Q_0/2\big)$, i.e. $\mu(c)\mu(c')\,\psi(cc'Q_0)=\mu(c+c')$: $\mu$ already solves the cocycle relation of Theorem 3.3(c) with values entirely inside $\psi(\kappa)=\mu_p$, so no order-4 (or any value outside $\mu_p$) is ever forced. ☐

### 3.4 Uniqueness of the finite Weyl representation

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**Theorem 3.5** (Uniqueness of the finite Weyl representation)**.** *Let $\kappa$, $\beta\in\mathrm{Adm}(\omega)$, $\psi\in X(\kappa)$, and $A_{\psi,\beta}(V)$ be as above. Then:*

- *(a) the induced module $M_0$ of $(L_0,\chi_0)$, $L_0:=0\oplus\kappa$, $\chi_0\equiv 1$, has basis $\{e_y\}_{y\in\kappa}$ with $W_{\beta_0}(a,b)e_y=\psi\big(-b(y+a)\big)e_{y+a}$ — i.e. the stipulated standard model of Definition 3.2 genuinely satisfies the multiplication law of Definition 2.4;*
- *(b) every simple $A_{\psi,\beta}(V)$-module is unitarily equivalent to $M_0$ (equivalently, to any $M_{L,\chi}$), and an $A_{\psi,\beta}(V)$-linear map $M_0\to M_0$ is a scalar;*
- *(c) consequently $H_\beta(\kappa)$ has, up to unitary equivalence, exactly one irreducible unitary representation with central character $\psi$, of dimension $q$, and every $\mathbb{C}$-algebra automorphism of $A_{\psi,\beta}(V)$ is inner;*
- *(d) for $\{e_y\}$ orthonormal, every $W_{\beta_0}(v)$ acts unitarily, and in general the unitary intertwiner between two irreducible unitary representations with the same central character is unique up to a scalar in $U(1)$.*

*Scope.* This theorem is stated and proved without invoking Maschke's theorem, Artin–Wedderburn, Skolem–Noether, Jacobson density, or Pontryagin duality as black boxes: every step is elementary, using only Theorem 2.8 (simplicity) and Theorem 3.3. It is uniform in the characteristic: no clause is restricted to odd $p$ or to $p=2$, and no step below treats $p=2$ differently. It does not exhibit the isomorphism $A_{\psi,\beta}(V)\cong M_q(\mathbb{C})$ as canonical — that non-canonicity is the separate content of Theorem 3.6 immediately below.

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**Provenance.** ids: D1, D4, D8, WH-ALG, WH-POL, WH-SVN; proved in: theory/wh-kappa.md §5; tested by: theory/checks/wh_kappa_check.py (C3,C6,C11); reviewed in: theory/verdicts/wh-kappa-r1.md; theory/verdicts/wh-kappa-adjudication.md.

*Proof. (a) The standard model, verified.* Since $Q_{\beta_0}(0,b)=0\cdot b=0$, $\chi_0\equiv 1$ is a character of $A_{L_0}$ by Theorem 3.3(e). Take coset representatives $(y,0)$, $y\in\kappa$, of $L_0$ in $V(\kappa)$ and put $e_y:=W_{\beta_0}(y,0)\otimes 1\in M_{L_0,\chi_0}$. Then

$$W_{\beta_0}(a,b)\,e_y=\psi\big(\beta_0((a,b),(y,0))\big)\,W_{\beta_0}(a+y,b)\otimes 1=W_{\beta_0}(a+y,b)\otimes 1,$$

since $\beta_0((a,b),(y,0))=a\cdot 0=0$. To evaluate this against $\chi_0$ the $L_0$-part must be moved to the *right*:

$$W_{\beta_0}(a+y,0)\,W_{\beta_0}(0,b)=\psi\big((a+y)b\big)\,W_{\beta_0}(a+y,b),\quad\text{so}$$
$$W_{\beta_0}(a+y,b)=\psi\big(-b(y+a)\big)\,W_{\beta_0}(a+y,0)\,W_{\beta_0}(0,b).$$

Tensoring against $\chi_0(W_{\beta_0}(0,b)) = 1$ gives $W_{\beta_0}(a,b)e_y = \psi(-b(y+a))\, e_{a+y}$ as claimed, and $\dim M_0 = q$ by Theorem 3.3(d). (The sign here is exactly Definition 3.2's stipulation, now derived rather than assumed: the naive un-signed assignment $W(a,b) = Z(b)X(a)$ realizes the *different* cocycle $-ab'$ and fails to satisfy Definition 2.4's multiplication law for $\beta_0(v,v') = ab'$ at odd $p$ — gate C11 of `theory/checks/wh_kappa_check.py` reports 0 violations for the signed model at all six tested $q \in \{2,3,4,5,8,9\}$, against $36, 400, 3888$ violations for the unsigned model at $q = 3,5,9$ and, tellingly, 0 violations at $q = 2,4,8$ — the defect is invisible at $p = 2$, where $-1 = 1$, and visible only once $p$ is odd.)

*(b) Uniqueness and Schur.* $\pi : A_{\psi,\beta}(V) \to \mathrm{End}_{\mathbb{C}}(M_0)$ is a unital, nonzero algebra homomorphism, so $\ker \pi$ is a proper two-sided ideal, hence 0 by simplicity (Theorem 2.8): $\pi$ is injective. Since $\dim \mathrm{End}_{\mathbb{C}}(M_0) = q^2 = \dim A_{\psi,\beta}(V)$, $\pi$ is also surjective, giving $A_{\psi,\beta}(V) \cong \mathrm{End}_{\mathbb{C}}(M_0)$. Consequently $A_{\psi,\beta}(V) \cong M_0^{\oplus q}$ as a left module over itself (the columns of $\mathrm{End}_{\mathbb{C}}(M_0)$), so *any* simple module $N$ — generated by any nonzero $n \in N$ — is a quotient of $A_{\psi,\beta}(V) \cong M_0^{\oplus q}$; restricting the quotient map to one summand gives a nonzero map $M_0 \to N$, which is surjective (its image is a nonzero submodule of the simple $N$) and injective (its kernel is a proper submodule of the simple $M_0$), hence an isomorphism $M_0 \cong N$. For an $A$-endomorphism $\varphi$ of $M_0$: $\varphi$, as a $\mathbb{C}$-linear endomorphism of a finite-dimensional $\mathbb{C}$-vector space, has an eigenvalue $\lambda \in \mathbb{C}$ ($\mathbb{C}$ is algebraically closed), and $\ker(\varphi - \lambda)$ is then a nonzero $A$-submodule of the simple $M_0$, so $\varphi = \lambda \cdot \mathrm{id}$: $\mathrm{End}_A(M_0) = \mathbb{C} \cdot \mathrm{id}$.

*(c) Representations of $H_\beta(\kappa)$ and inner automorphisms.* A representation of $H_\beta(\kappa)$ with central character $\psi$ is exactly an $A_{\psi,\beta}(V)$-module (the remark after Theorem 2.7); irreducibility is simplicity of the module, so (b) applies: there is exactly one such representation up to isomorphism, of dimension $q$. For automorphisms: given $\alpha \in \mathrm{Aut}_{\mathbb{C}}(A_{\psi,\beta}(V))$, twist the action on $M_0$ by $\alpha$, i.e. put $a \cdot_\alpha m := \pi(\alpha(a))m$; this twisted module $M_0^\alpha$ is again simple of dimension $q$, hence $\cong M_0$ by (b), via some invertible $T$; unwinding, $\alpha = \mathrm{Ad}_T$, with $T$ unique up to a scalar by the Schur clause of (b). No Skolem–Noether theorem is invoked.

*(d) Unitarity.* Since $|\psi(\cdot)| = 1$ (Fact 3.1), each $W_{\beta_0}(a,b)$ of (a) sends the orthonormal $\{e_y\}$ to $\{\psi(-b(y+a))e_{a+y}\}$, again orthonormal: every $W_{\beta_0}(v)$ is unitary for this inner product, with no averaging needed. (For a general $\beta$, the Weyl operators generate the *finite* level-$\mu_p$ frame $\mathcal{F}^{(p)}_{\psi,\beta}$ of Definition 4.9 in §4, of order $pq^2$; averaging any inner product on the simple module over this finite group makes every $\pi(W_\beta(v))$ unitary there too.) If $T$ intertwines two unitary representations sharing central character $\psi$, then $T^*T$ commutes with every operator of the first, so $T^*T = c \cdot \mathrm{id}$ with $c > 0$ by the Schur clause of (b); then $c^{-1/2}T$ is unitary, and two unitary intertwiners differ by a unimodular scalar, i.e. an element of $U(1)$. □

**PROVED**

**Theorem 3.6** (The twisted algebra as a matrix algebra, non-canonically). *$A_{\psi,\beta}(V) \cong M_q(\mathbb{C})$, by an isomorphism that depends on a choice of Lagrangian line and character $(L,\chi)$ (Definition 3.2) and a choice of basis of $M_{L,\chi}$, and is* ***not*** *canonical.*

*Scope.* This theorem asserts existence of *an* isomorphism to $M_q(\mathbb{C})$ and the dependence of any such isomorphism on auxiliary choices; it does not assert, and Theorem 6.3 of §6 shows precisely how much of this data can be removed, that any two such isomorphisms agree, nor that a canonical one exists.

**Provenance.** ids: D4, D8, WH-ALG, WH-SVN, WH-ALG-MAT; proved in: theory/wh-kappa.md §5 *step* ⟨1⟩2; tested by: theory/checks/wh_kappa_check.py (C5); reviewed in: theory/verdicts/wh-kappa-r1.md.

*Proof.* This is exactly the isomorphism $\pi : A_{\psi,\beta}(V) \xrightarrow{\sim} \mathrm{End}_{\mathbb{C}}(M_0)$ constructed in the proof of Theorem 3.5(b); choosing the basis $\{e_y\}_{y\in\kappa}$ of $M_0$ identifies $\mathrm{End}_{\mathbb{C}}(M_0)$ with $M_q(\mathbb{C})$. The construction visibly used a Lagrangian line $L_0$, a character $\chi_0$ of $A_{L_0}$, and the basis $\{e_y\}$ of the resulting module; a different line, character, or basis gives a different isomorphism, related to this one by an inner automorphism of $M_q(\mathbb{C})$ (Theorem 3.5(c)) composed with the change of basis, and no choice among these is distinguished by the data $(\kappa, \psi, \beta)$ alone. $\square$

# 4 The polarizing cocycle as a second datum: the characteristic-two dichotomy

This section proves the increment's central structural fact. The quantum system attached to a finite field $\kappa$ is built not from the field alone, and not from the field together with a single choice of additive character $\psi$: it is built from *three* data — $\kappa$, $\psi$, and a polarizing cocycle $\beta$, a choice inside the torsor $\mathrm{Adm}(\omega)$ of Definition 2.2 that is no more optional, and no more a matter of convention, than $\psi$ itself. What this section shows is exactly how much that third choice matters. At odd characteristic it turns out not to matter at all: a canonical representative exists inside $\mathrm{Adm}(\omega)$, and every other admissible cocycle produces an isomorphic Heisenberg group by an isomorphism fixing the centre. At characteristic two no canonical representative exists, and the choice is not immaterial: the torsor $\mathrm{Adm}(\omega)$ splits into exactly two orbits under the natural symmetry group $SL_2(\kappa)$, distinguished by the Arf invariant of an associated quadratic form, and the two orbits produce two genuinely non-isomorphic Heisenberg groups — a fact detectable by an operator identity inside the quantum system itself.

This is not a defect in the construction, and not a gap for a better convention to close. It is a theorem about what the construction actually depends on, carrying an intrinsic, computable invariant that says exactly when two choices of cocycle agree and when they do not. A sharp result of this shape — naming precisely the choice at stake and exhibiting the invariant that classifies it — is a positive structural statement about the definition, not a negative one, and it is this labbook's principal new content.

## 4.1 The torsor of admissible polarizing cocycles, restated

**Definition 4.1** (Admissible polarizing cocycles, restated in full)**.** For the symplectic object $(V(\kappa), \omega)$ of Definition 2.1,

$$\mathrm{Adm}(\omega) := \left\{ \beta : V(\kappa) \times V(\kappa) \to \kappa \;\middle|\; \beta \text{ is } \kappa\text{-bilinear and } \beta - \beta^T = \omega \right\}.$$

A *polarizing cocycle* is a choice of element $\beta \in \mathrm{Adm}(\omega)$, and it is a **datum** of the construction, on the same footing as the phase datum $\psi$ — not a convention fixed once and for all. The *reference cocycle* is $\beta_0\big((a,b),(a',b')\big) := ab'$, a label for a point of the torsor below, not a canonical point of it. Write $\mathrm{Sym}(V)$ for the set of symmetric $\kappa$-bilinear forms on $V(\kappa)$.

*Scope.* This restates Definition 2.2 verbatim; nothing new is stipulated here. What is new in this section is everything that is *proved* about $\mathrm{Adm}(\omega)$: its torsor structure (Theorem 4.2), the immateriality of the choice at odd $p$ (Theorem 4.3), and its genuine two-way split at $p = 2$ (Theorems 4.4–4.8).

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**Provenance.** ids: D1, D2; proved in: —; tested by: —; reviewed in: —.

**PROVED**

**Theorem 4.2** (The admissible cocycles form a torsor of size $q^3$)**.** $\mathrm{Adm}(\omega)$ *is a torsor under* $\mathrm{Sym}(V)$ *acting by* $\beta \mapsto \beta + s$, $s \in \mathrm{Sym}(V)$*; consequently* $|\mathrm{Adm}(\omega)| = q^3$.

*Scope.* This is purely a statement about the set $\mathrm{Adm}(\omega)$ and its free transitive action; it does not yet distinguish any point of the torsor, does not use the characteristic in any way, and does not address isomorphism of the associated Heisenberg groups — those are the later theorems of this section.

**Provenance.** ids: D1, D2, WH-BETA-a; proved in: theory/wh-kappa-choice.md §6 *step* $\langle 1 \rangle 1$; tested by: theory/checks/beta_census.py (B1); theory/checks/wh_kappa_check.py (C10); reviewed in: theory/verdicts/wh-kappa-adjudication.md.

*Proof.* If $\beta, \beta' \in \mathrm{Adm}(\omega)$, put $s := \beta' - \beta$. Then $s - s^T = (\beta' - \beta^{T\prime}) - (\beta - \beta^T) = \omega - \omega = 0$, i.e. $s \in \mathrm{Sym}(V)$. Conversely, for $s \in \mathrm{Sym}(V)$, $(\beta + s) - (\beta + s)^T = (\beta - \beta^T) + (s - s^T) = \omega + 0 = \omega$, so $\beta + s \in \mathrm{Adm}(\omega)$. The action $\beta \mapsto \beta + s$ is therefore well defined, free (if $\beta + s = \beta + s'$ then $s = s'$), and transitive (any $\beta' \in \mathrm{Adm}(\omega)$ equals $\beta + (\beta' - \beta)$ for the $s = \beta' - \beta \in \mathrm{Sym}(V)$ just exhibited). A symmetric $\kappa$-bilinear form on $V(\kappa) = \kappa e \oplus \kappa f$ is determined freely by the three values $s(e,e), s(e,f), s(f,f) \in \kappa$ (symmetry forces $s(f,e) = s(e,f)$, and bilinearity determines everything else), so $|\mathrm{Sym}(V)| = q^3$; since $\mathrm{Adm}(\omega)$ is nonempty ($\beta_0 \in \mathrm{Adm}(\omega)$) and a torsor under $\mathrm{Sym}(V)$, it has the same cardinality, $q^3$. □

## 4.2 Odd characteristic: a canonical representative

**PROVED** †

**Theorem 4.3** (Immateriality of the polarizing cocycle at odd characteristic)**.** *[p **odd.**]* $\mathrm{Adm}(\omega)$ *contains exactly one antisymmetric member, $\omega/2$, and it is $SL_2(\kappa)$-invariant. For every* $s \in \mathrm{Sym}(V)$*, the map*

$$\varphi_s(t, v) := \Big(t + \tfrac{1}{2}s(v,v),\ v\Big)$$

*is a group isomorphism $H_\beta(\kappa) \to H_{\beta+s}(\kappa)$ fixing the centre $\kappa \times 0$ pointwise, for* every $\beta \in \mathrm{Adm}(\omega)$*. Hence at odd characteristic the choice of polarizing cocycle is immaterial: every admissible cocycle produces a Heisenberg group isomorphic, by an explicit isomorphism fixing the centre, to every other, and $\omega/2$ is a canonical basepoint of the torsor.*

*Scope.* This theorem is restricted to odd $p$ by hypothesis, not by omission: at $p = 2$, division by 2 is division by zero, $\omega/2$ has no referent, and Theorem 4.4 shows the phenomenon this theorem describes simply does not occur. Nothing here compares $H_\beta(\kappa)$ for different $\beta$ as algebras or as sources of quantum systems beyond the group isomorphism exhibited; the algebra-level statement is Theorem 4.10.

**Provenance.** ids: D2, D5, WH-BETA-a, WH-BETA-b; proved in: theory/wh-kappa-choice.md §6 *step* $\langle 1 \rangle 2$; tested by: theory/checks/beta_census.py (B2,B3,B4); theory/checks/wh_kappa_check.py (C10); reviewed in: theory/verdicts/wh-kappa-adjudication.md.

*Proof. Existence and uniqueness of the antisymmetric point.* Since $p$ is odd, $2 \in \kappa^\times$, and $\omega/2$ is $\kappa$-bilinear with $(\omega/2) - (\omega/2)^T = \omega/2 + \omega/2 = \omega$ (using $\omega^T = -\omega$, i.e. antisymmetry of $\omega$, itself immediate from Theorem 2.6(b): $\omega(v,v') + \omega(v',v) = \omega(v+v', v+v') - \omega(v,v) - \omega(v',v') = 0$), so $\omega/2 \in \mathrm{Adm}(\omega)$ and is manifestly antisymmetric. If $\beta, \beta' \in \mathrm{Adm}(\omega)$ are both antisymmetric, then $s := \beta' - \beta \in \mathrm{Sym}(V)$ (Theorem 4.2) is also antisymmetric, so $2s = 0$, so $s = 0$ ($p$ odd): the antisymmetric point is unique. *Invariance.* $SL_2(\kappa)$ preserves $\omega$ (Theorem 2.6(e)), hence preserves the scalar multiple $\omega/2$.

*The isomorphism $\varphi_s$.* Put $f(v) := s(v,v)/2$. By bilinearity and symmetry of $s$, $s(v+v', v+v') = s(v,v) + 2s(v,v') + s(v',v')$, so $f(v+v') = f(v) + f(v') + s(v,v')$. Unwinding the group law of Definition 2.5 for $H_\beta(\kappa)$ and $H_{\beta+s}(\kappa)$,

$$\varphi_s\big((t,v)(t',v')\big) = \varphi_s\big(t + t' + \beta(v,v'),\ v + v'\big) = \big(t + t' + \beta(v,v') + f(v+v'),\ v + v'\big),$$

while

$$\varphi_s(t,v)\,\varphi_s(t',v') = \big(t + f(v), v\big)\big(t' + f(v'), v'\big) = \big(t + t' + f(v) + f(v') + (\beta{+}s)(v,v'),\ v + v'\big),$$

computed inside $H_{\beta+s}(\kappa)$. The two right-hand sides agree precisely because $f(v+v') = f(v) + f(v') + s(v,v')$. So $\varphi_s$ is a group homomorphism; it is a bijection (identity on the second coordinate, an affine bijection $t \mapsto t + f(v)$ on the first, for each fixed $v$), hence an isomorphism, and $\varphi_s(t,0) = (t + f(0), 0) = (t,0)$ since $f(0) = s(0,0)/2 = 0$: the centre is fixed pointwise. □

### 4.3 Characteristic two: no symmetric distinguished point

**PROVED** †

**Theorem 4.4** (Absence of a symmetric distinguished cocycle at characteristic two)**.** *[$p=2$.]* $\mathrm{Adm}(\omega)$ *contains no antisymmetric member; the criterion that singles out a canonical basepoint at odd characteristic (Theorem 4.3) distinguishes nothing here.*

*Scope.* This theorem only rules out *this particular* route to a canonical choice (antisymmetry). It does not by itself assert that no other route to a canonical choice exists, nor that the resulting Heisenberg groups actually differ — those stronger, positive facts are Theorems 4.5–4.7 below.

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**Provenance.** ids: D2, WH-BETA-a, WH-BETA-c; proved in: theory/wh-kappa-choice.md §6 *step* ⟨1⟩3; tested by: theory/checks/beta_census.py (B2); reviewed in: theory/verdicts/wh-kappa-adjudication.md.

*Proof.* At $p=2$, $-1=1$, so "antisymmetric" ($\beta = -\beta^T$) coincides with "symmetric" ($\beta = \beta^T$), i.e. $\beta - \beta^T = 0$. But every $\beta \in \mathrm{Adm}(\omega)$ has $\beta - \beta^T = \omega$ (Definition 4.1), and $\omega\big((1,0),(0,1)\big) = 1 \neq 0$ (Theorem 2.6(c)), so $\omega \neq 0$: no member of $\mathrm{Adm}(\omega)$ can be symmetric, hence none can be antisymmetric either. □

### 4.4 The quadratic form of a cocycle, and the order profile of its Heisenberg group

**PROVED** †

**Theorem 4.5** (The quadratic form of a polarizing cocycle, and the element-order profile of its Heisenberg group)**.** *[$p=2$.] Let $\beta \in \mathrm{Adm}(\omega)$, with $Q_\beta(v) := \beta(v,v)$ as in Definition 3.1. Then:*

*(i) $Q_\beta(cv) = c^2 Q_\beta(v)$ for $c \in \kappa$, $v \in V(\kappa)$, and $Q_\beta(v+v') = Q_\beta(v) + Q_\beta(v') + \omega(v,v')$ for all $v, v'$: $Q_\beta$ is a $\kappa$-quadratic form on $V(\kappa)$ with polar form $\omega$;*

*(ii) in $H_\beta(\kappa)$, $(t,v)^2 = \big(Q_\beta(v), 0\big)$ for every $(t,v)$; consequently, writing $n_0 := |Q_\beta^{-1}(0)|$ (which is always $\geq 1$, since $Q_\beta(0) = 0$), the multiset of element orders of $H_\beta(\kappa)$ is*

$$\{\,1:1,\quad 2: q\,n_0 - 1,\quad 4: q^3 - q\,n_0\,\};$$

*(iii) hence $\mathrm{Adm}(\omega)$ meets at least two isomorphism classes of Heisenberg group: by direct enumeration at $q=2$, 6 of the 8 admissible cocycles give the order profile $\{1{:}1, 2{:}5, 4{:}2\}$ ($n_0 = 3$) and the remaining 2 give $\{1{:}1, 2{:}1, 4{:}6\}$ ($n_0 = 1$), and these are different multisets, hence non-isomorphic groups.*

*Scope.* Part (iii) establishes only that *at least* two isomorphism classes occur, by exhibiting two different order profiles at the single value $q=2$; it does not claim that exactly two classes occur for every $q$, nor identify the classes by name (this lane computes order profiles, not group names — that two of the eight cocycles at $q=2$ happen to give the quaternion group and the other six the dihedral group of order 8 is a fact the orchestrator's independent census records, but it is not part of what this row proves). The full classification, valid at every $q = 2^m$ and carrying an intrinsic invariant, is Theorem 4.7 below, which depends on this one.

**Provenance.** ids: D2, D6, WH-BETA-a, WH-BETA-d; proved in: theory/wh-kappa-choice.md §6 *step* ⟨1⟩4; tested by: theory/checks/beta_census.py (B3,B6); theory/checks/wh_kappa_check.py (C10); reviewed in: theory/verdicts/wh-kappa-adjudication.md.

*Proof.* (i) By $\kappa$-bilinearity, $Q_\beta(cv) = \beta(cv, cv) = c^2\beta(v, v) = c^2 Q_\beta(v)$. Expanding $Q_\beta(v + v') = \beta(v + v', v + v') = \beta(v, v) + \beta(v, v') + \beta(v', v) + \beta(v', v')$; at $p = 2$, addition and subtraction coincide, so $\beta(v, v')+\beta(v', v) = \beta(v, v')-\beta(v', v) = \omega(v, v')$ (Definition 4.1), giving $Q_\beta(v+v') = Q_\beta(v) + Q_\beta(v') + \omega(v, v')$.

(ii) In $H_\beta(\kappa)$, $(t, v)(t, v) = \big(2t + \beta(v, v),\ 2v\big) = \big(Q_\beta(v), 0\big)$ at $p = 2$, since $2t = 0$ and $2v = 0$. If $(t, v) = (0, 0)$ this is the identity (order 1). If $(t, v) \neq (0, 0)$ and $Q_\beta(v) = 0$, then $(t, v)^2 = (0, 0)$, so $(t, v)$ has order exactly 2; the number of such elements is $q\,n_0 - 1$ (for each of the $n_0$ vectors $v$ with $Q_\beta(v) = 0$, all $q$ values of $t$ occur, and the pair $(t, v) = (0, 0)$ is excluded). If $Q_\beta(v) \neq 0$, then $(t, v)^2 = (Q_\beta(v), 0) \neq (0, 0)$, but $(t, v)^4 = \big((t, v)^2\big)^2 = \big(2Q_\beta(v), 0\big) = (0, 0)$, so $(t, v)$ has order exactly 4; there are $q(q^2 - n_0) = q^3 - qn_0$ such elements ($q^2 - n_0$ choices of $v$, $q$ choices of $t$ each). These three counts sum to $1 + (qn_0 - 1) + (q^3 - qn_0) = q^3 = |H_\beta(\kappa)|$, as they must.

(iii) The order profile just derived is a group-isomorphism invariant (an isomorphism of groups preserves the multiset of element orders), so different profiles certify non-isomorphic groups. At $q = 2$, exhaustive enumeration over all 8 members of $\mathrm{Adm}(\omega)$ (`theory/checks/beta_census.py`, gates B3 and B6) computes $n_0$ for each and finds exactly the two values $n_0 = 3$ (for 6 of the 8 cocycles, giving profile $\{1{:}1, 2{:}5, 4{:}2\}$ by the formula of (ii)) and $n_0 = 1$ (for the remaining 2, giving $\{1{:}1, 2{:}1, 4{:}6\}$); these are distinct multisets, exhibiting two isomorphism classes. □

### 4.5 The Arf classification

**Definition 4.6** (The Artin–Schreier map, and the type of a polarizing cocycle)**.** For $\kappa$ of characteristic 2, define $\wp : \kappa \to \kappa$ by $\wp(x) := x^2 + x$. The *type* of $\beta \in \mathrm{Adm}(\omega)$ is

$$\mathrm{Arf}(Q_\beta) := Q_\beta(e)\, Q_\beta(f) \ \in\ \kappa/\wp(\kappa),$$

computed in any symplectic $\kappa$-basis $(e, f)$ of $V(\kappa)$, i.e. a basis with $\omega(e, f) = 1$.

*Scope.* This is a stipulation, meaningful only at $p = 2$. That the value $\mathrm{Arf}(Q_\beta)$ does not depend on the choice of symplectic basis, that $\kappa/\wp(\kappa)$ has exactly two elements, and that both values actually occur among the members of $\mathrm{Adm}(\omega)$, are not part of the definition; they are exactly the content of Theorem 4.7 below, whose provenance names the argument DAG's current row for this material (the single source's own cross-reference for this content predates a later renaming in the DAG).

**Provenance.** ids: D10; proved in: —; tested by: —; reviewed in: —.

SKETCH

**Theorem 4.7** (Classification of characteristic-two polarizing cocycles by the Arf invariant)**.** *[$p = 2$.] Let $\beta \in \mathrm{Adm}(\omega)$.*

(i) *$\wp$ is additive with $\ker \wp = \{0, 1\} = \mathbb{F}_2$, so $\kappa/\wp(\kappa)$ has exactly two elements;*

(ii) *$Q_\beta$ has a nonzero zero (some $v \neq 0$ with $Q_\beta(v) = 0$) if and only if $\mathrm{Arf}(Q_\beta) = 0$; in that case $n_0 = |Q_\beta^{-1}(0)| = 2q-1$, and otherwise $n_0 = 1$. Consequently $\mathrm{Arf}(Q_\beta)$ does not depend on the choice of symplectic basis used to compute it, and is invariant under $SL_2(\kappa)$;*

(iii) *call $\beta$* hyperbolic *if $\mathrm{Arf}(Q_\beta) = 0$ and* anisotropic *otherwise. Two members of $\mathrm{Adm}(\omega)$ are carried to each other by an $SL_2(\kappa)$-isometry of their quadratic forms if and only if they* *have the same type; both types occur, with $q(q+1)/2$ hyperbolic and $q(q-1)/2$ anisotropic quadratic forms compatible with $\omega$, and stabilizer orders $|O(Q_\beta) \cap SL_2(\kappa)| = 2(q-1)$ (hyperbolic) and $2(q+1)$ (anisotropic);*

(iv) *$H_\beta(\kappa) \cong H_{\beta'}(\kappa)$ if and only if $\mathrm{Arf}(Q_\beta) = \mathrm{Arf}(Q_{\beta'})$. So $\mathrm{Adm}(\omega)$ meets* exactly *two isomorphism classes of Heisenberg group, indexed by* Arf.

*Scope.* This theorem is marked SKETCH for a procedural reason recorded in `theory/verdicts/wh-kappa-adjudication.md`: it was written in the repair wave that followed this increment's single hostile-critic round and has not itself been through one, per this campaign's capped review loop (`PRD.md`) — not because a specific gap is known. It does not extend beyond $p = 2$ (Theorem 4.3 covers odd $p$ by an entirely different, and there, complete, argument), and it does not address whether the type can be chosen compatibly along field embeddings — that question is recorded as open in §8.

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**Provenance.** ids: D6, D10, WH-BETA-d, WH-BETA-TYPE; proved in: theory/wh-kappa-choice.md §6 *step* $\langle 1\rangle 4, \langle 1\rangle 5$; tested by: theory/checks/beta_census.py (B5,B6,B7); theory/checks/wh_kappa_check.py (C10); reviewed in: theory/verdicts/wh-kappa-adjudication.md.

*Proof.* (i) $\wp(x+y) - \wp(x) - \wp(y) = (x+y)^2 - x^2 - y^2 = 2xy = 0$ at $p = 2$, so $\wp$ is additive (a group homomorphism $(\kappa, +) \to (\kappa, +)$). Its kernel consists of the roots of $x^2 + x = 0$, a degree-2 polynomial, so $|\ker \wp| \le 2$; both $x = 0$ and $x = 1$ are roots, so $\ker \wp = \{0, 1\} = \mathbb{F}_2$ exactly. By rank–nullity for the $\mathbb{F}_2$-linear map $\wp$ on the $m$-dimensional $\mathbb{F}_2$-space $\kappa$, $|\wp(\kappa)| = q/2$, so $|\kappa/\wp(\kappa)| = 2$.

(ii) In a symplectic basis $(e, f)$, write $v = ae + bf$; by Theorem 4.5(i), $Q_\beta(ae + bf) = \alpha a^2 + ab + \delta b^2$ with $\alpha := Q_\beta(e)$, $\delta := Q_\beta(f)$ (using $\omega(e, f) = 1$ for the cross term). If $b = 0$: $Q_\beta = \alpha a^2$ vanishes for some $a \neq 0$ iff $\alpha = 0$, and then $\mathrm{Arf} = \alpha\delta = 0 \in \wp(\kappa)$. If $b \neq 0$, rescale to $b = 1$: $\alpha a^2 + a + \delta = 0$; multiplying by $\alpha$ (when $\alpha \neq 0$) gives $(\alpha a)^2 + (\alpha a) = \alpha\delta$, i.e. $\wp(\alpha a) = \alpha\delta$, solvable for $a$ iff $\alpha\delta \in \wp(\kappa)$; when $\alpha = 0$ the equation $a + \delta = 0$ always has the solution $a = \delta$. So in every case, $Q_\beta$ has a nonzero zero iff $\alpha\delta \in \wp(\kappa)$, i.e. iff $\mathrm{Arf}(Q_\beta) = 0 \in \kappa/\wp(\kappa)$. When $\mathrm{Arf}(Q_\beta) = 0$, the zero set of $Q_\beta$ is a union of $\kappa$-lines (by homogeneity $Q_\beta(cv) = c^2 Q_\beta(v)$ of Theorem 4.5(i)); the computation above produces exactly two such lines (the two roots of $\wp(\alpha a) = \alpha\delta$ when $\alpha \neq 0$, or the lines $b = 0$ and $a = \delta$ when $\alpha = 0$), giving $n_0 = 2(q-1)+1 = 2q-1$ nonzero-vector zeros plus the origin; when $\mathrm{Arf}(Q_\beta) \neq 0$, $n_0 = 1$ (only the origin). Since $|\kappa/\wp(\kappa)| = 2$ by (i), the class $\alpha\delta \bmod \wp(\kappa)$ is itself a basis-free condition — "$Q_\beta$ is isotropic" — so $\mathrm{Arf}(Q_\beta)$ is independent of the symplectic basis chosen to compute it, and $SL_2(\kappa)$-invariant since an $SL_2(\kappa)$-image of a symplectic basis is again symplectic (Theorem 2.6(e)).

(iii) *Normal forms.* If $\mathrm{Arf}(Q_\beta) = 0$, pick $e \neq 0$ with $Q_\beta(e) = 0$ (by (ii)) and any $f$ with $\omega(e, f) = 1$ (possible by nondegeneracy of $\omega$); replacing $f$ by $f + Q_\beta(f)e$ changes $Q_\beta(f)$ to $Q_\beta(f) + Q_\beta(f)^2 Q_\beta(e) + Q_\beta(f)\omega(f, e)$, which simplifies (using $Q_\beta(e) = 0$ and $\omega(f, e) = -\omega(e, f) = 1$ at $p = 2$) to 0, giving $Q_\beta(ae + bf) = ab$: the hyperbolic normal form. If $\mathrm{Arf}(Q_\beta) \neq 0$ then $Q_\beta(v) \neq 0$ for every $v \neq 0$ (by (ii)); pick any $e \neq 0$, rescale by $\lambda = Q_\beta(e)^{-1/2}$ (Frobenius is bijective on $\kappa$, so square roots exist and are unique) to arrange $Q_\beta(e) = 1$, then choose $f$ with $\omega(e, f) = 1$; replacing $f$ by $f + ae$ changes $Q_\beta(f)$ by $\wp(a)$ (direct computation from Theorem 4.5(i)), so $Q_\beta(f)$ can be moved to any chosen representative of its class modulo $\wp(\kappa)$. Both normalizations send a symplectic basis to a symplectic basis, hence are realized by elements of $SL_2(\kappa)$ (Theorem 2.6(e)): two quadratic forms of the same type are $SL_2(\kappa)$-isometric, and by (ii) two of different type cannot be (they have different $n_0$, an isometry invariant). *Counting.* Over a fixed symplectic basis, the $\kappa$-quadratic forms with polar form $\omega$ are exactly the $q^2$ maps $\alpha a^2 + ab + \delta b^2$ (parametrized freely by $(\alpha, \delta) \in \kappa^2$, by Theorem 4.5(i)); among these, $\#\{\alpha\delta \in \wp(\kappa)\} = q + (q-1)(q/2) = q(q+1)/2$ are hyperbolic (the case $\alpha = 0$ contributes $q$ choices of $\delta$, and each of the $q - 1$ nonzero $\alpha$ pairs with the $q/2$ elements of $\wp(\kappa)$) and the remaining $q(q-1)/2$ are anisotropic — both classes nonempty for every $q$. By orbit–stabilizer inside $SL_2(\kappa)$, of order $q(q^2-1)$, acting transitively within each type: stabilizer orders $q(q^2-1)/(q(q+1)/2) = 2(q-1)$ (hyperbolic) and $q(q^2-1)/(q(q-1)/2) = 2(q+1)$ (anisotropic). *Groups.* If $\mathrm{Arf}(Q_\beta) = \mathrm{Arf}(Q_{\beta'})$, the normal-form argument gives $Q_{\beta'} = Q_\beta \circ g$ for some $g \in SL_2(\kappa)$; then $r(v,v') := \beta(gv, gv') - \beta'(v,v')$ is symmetric with $r(v,v) = 0$ for all $v$ (both sides equal $Q_{\beta'}(v)$ by construction of $g$), so, fixing an $\mathbb{F}_2$-basis $b_i$ of $V(\kappa)$ and writing $v = \sum c_i b_i$, $h(v) := \sum_{i<j} c_i c_j\, r(b_i, b_j)$ satisfies $h(v+v') - h(v) - h(v') = r(v,v')$ (a direct expansion using $r$ symmetric and bilinear), and $(t,v) \mapsto (t + h(v), gv)$ is then a group isomorphism $H_{\beta'}(\kappa) \to H_\beta(\kappa)$ by the same computation as in the proof of Theorem 4.3. If instead $\mathrm{Arf}(Q_\beta) \neq \mathrm{Arf}(Q_{\beta'})$, the two groups have different values of $n_0$ (by (ii)), hence different element-order profiles (Theorem 4.5(ii)), so no group isomorphism of any kind exists between them. ☐

*Remark.* Direct enumeration (`theory/checks/beta_census.py`, gates B5–B7) confirms every clause above at $q = 2, 4, 8$: Arf is constant on $SL_2(\kappa)$-orbits and takes exactly the two values across all $q^3$ members of $\mathrm{Adm}(\omega)$; $|O(Q_\beta) \cap SL_2(\kappa)| = 2, 6, 14$ (hyperbolic) and $6, 10, 18$ (anisotropic), matching $2(q \mp 1)$ exactly; the element-order profiles come in exactly two sizes, $(6,2)$, $(40,24)$, $(288,224)$, at $q = 2,4,8$ respectively; and the isomorphism constructed in the proof of (iii) is verified on all pairs within the hyperbolic class.

### 4.6 The type, detected inside the quantum system

SKETCH

**Theorem 4.8** (The Arf type is visible inside the quantum system)**.** *[$p = 2$.] Let $\beta \in \mathrm{Adm}(\omega)$ and $\psi$ a phase datum. Then, inside $A_{\psi,\beta}(V)$:*

$$W_\beta(v)^2 = \psi\big(Q_\beta(v)\big) \cdot 1 \qquad \textit{for every } v \in V(\kappa),$$

*and*

$$q^{-1} \sum_{v \in V(\kappa)} W_\beta(v)^2 \;=\; \varepsilon \cdot 1, \qquad \varepsilon = \begin{cases} +1 & \beta \textit{ hyperbolic } (\mathrm{Arf}(Q_\beta) = 0), \\ -1 & \beta \textit{ anisotropic.} \end{cases}$$

*Equivalently, identifying $\kappa/\wp(\kappa) \cong \mathbb{F}_2$ via the trace $\mathrm{Tr}_{\kappa/\mathbb{F}_2}$ of Definition 2.3, $\varepsilon = (-1)^{\mathrm{Tr}(\mathrm{Arf}(Q_\beta))}$.*

*Scope.* The identity above is an equation inside the algebra $A_{\psi,\beta}(V)$ itself, so it exhibits $\mathrm{Arf}(Q_\beta)$ as an invariant not merely of the abstract group $H_\beta(\kappa)$ but of the level-$\mu_p$ Weyl frame $\mathcal{F}^{(p)}_{\psi,\beta}$ of Definition 4.9 below. As with Theorem 4.7, on which it depends, the SKETCH label reflects only that this material postdates the increment's single hostile-critic round, per `theory/verdicts/wh-kappa-adjudication.md`.

**Provenance.** ids: D3, D4, D6, D10, WH-BETA-EPS; proved in: theory/wh-kappa-choice.md §6 *step* ⟨1⟩6; tested by: theory/checks/beta_census.py (B6,B8); reviewed in: theory/verdicts/wh-kappa-adjudication.md.

*Proof. The square identity.* By Definition 2.4 and Definition 3.1, $W_\beta(v)^2 = \psi(\beta(v,v))W_\beta(2v) = \psi(Q_\beta(v))W_\beta(0) = \psi(Q_\beta(v)) \cdot 1$, using $2v = 0$ at $p = 2$.

*The hyperbolic value.* For the hyperbolic normal form $Q(a,b) = ab$ (Theorem 4.7(iii)), $\sum_{a,b\in\kappa} \psi(ab) = \sum_a \Big(\sum_b \psi(ab)\Big)$; for $a \neq 0$, $b \mapsto \psi(ab)$ is a nontrivial character of $\kappa$, so the inner sum vanishes by Fact 3.1; for $a = 0$ the inner sum is $q$. So the double sum is $q$, giving $\varepsilon = +1$ for this normal form. Since $\sum_v \psi(Q_\beta(v))$ is invariant under precomposing $Q_\beta$ with any bijection of $V(\kappa)$, in particular under the $SL_2(\kappa)$-isometries of Theorem 4.7(iii), $\varepsilon = +1$ for *every* hyperbolic $\beta$, not only the normal form.

*The anisotropic value.* Sum $\sum_v \psi(Q(v))$ over *all* $q^2$ quadratic forms $Q(a,b) = \alpha a^2 + ab + \delta b^2$ compatible with $\omega$ (Theorem 4.7(iii)) and exchange the order of summation:

$$\sum_{\alpha,\delta\in\kappa} \sum_{a,b\in\kappa} \psi(\alpha a^2 + ab + \delta b^2) = \sum_{a,b\in\kappa} \psi(ab)\Big(\sum_{\alpha} \psi(\alpha a^2)\Big)\Big(\sum_{\delta} \psi(\delta b^2)\Big).$$

By Fact 3.1, $\sum_\alpha \psi(\alpha a^2) = q$ if $a = 0$ (else the character $\alpha \mapsto \psi(\alpha a^2)$ is nontrivial, since squaring is injective on $\kappa$, and the sum vanishes); likewise for the $\delta$-sum. So only the term $a = b = 0$ survives, and the total is $q^2 \cdot \psi(0) = q^2$. By Theorem 4.7(iii) there are $q(q+1)/2$ hyperbolic forms, each contributing $q$ (just shown), and $q(q-1)/2$ anisotropic forms, each contributing the same value $S$ (by isometry-invariance, exactly as for the hyperbolic case). So $\frac{q(q+1)}{2} \cdot q + \frac{q(q-1)}{2} \cdot S = q^2$, giving $S = \dfrac{q^2 - \frac{q^2(q+1)}{2}}{\frac{q(q-1)}{2}} = \dfrac{q^2\big(1 - \frac{q+1}{2}\big)}{\frac{q(q-1)}{2}} = \dfrac{-q^2(q-1)/2}{q(q-1)/2} = -q$. So $\varepsilon = -1$ on the anisotropic class.

*Reading $\varepsilon$ through the trace.* $\mathrm{Tr}_{\kappa/\mathbb{F}_2}$ is additive and $\mathbb{F}_2$-valued; $\mathrm{Tr}(\wp(x)) = \mathrm{Tr}(x^2) + \mathrm{Tr}(x) = \mathrm{Tr}(x) + \mathrm{Tr}(x) = 0$ (using $\mathrm{Tr}(x^2) = \mathrm{Tr}(x)$, since Frobenius permutes the terms of the trace sum, and $2\,\mathrm{Tr}(x) = 0$ at $p = 2$), so Tr kills $\wp(\kappa)$; it is nonzero (e.g. on a normal-basis generator), hence onto $\mathbb{F}_2$, and $\dim_{\mathbb{F}_2} \wp(\kappa) = m - 1 = \dim_{\mathbb{F}_2} \ker \mathrm{Tr}$ by the rank computation of Theorem 4.7(i), so $\wp(\kappa) = \ker \mathrm{Tr}$ exactly: Tr descends to an isomorphism $\kappa/\wp(\kappa) \xrightarrow{\sim} \mathbb{F}_2$. Under this identification the hyperbolic class (Arf $= 0$) corresponds to $\mathrm{Tr} = 0$ and $\varepsilon = +1 = (-1)^0$, and the anisotropic class to $\mathrm{Tr} = 1$ and $\varepsilon = -1 = (-1)^1$. □

### 4.7 The level statement: what the algebra forgets, and what it does not

**Definition 4.9** (The level-$\mu_p$ frame)**.**

$$\mathcal{F}^{(p)}_{\psi,\beta} := \big\{\, \zeta^j\, W_\beta(v) \;:\; j \in \mathbb{Z}/p,\ v \in V(\kappa) \,\big\} \;\subset\; A_{\psi,\beta}(V)^\times,$$

the finite group of Weyl unitaries with phases in $\mu_p = \psi(\kappa)$ (Fact 3.1), for $\zeta$ a primitive $p$-th root of unity as in Definition 2.3.

*Scope.* $\mathcal{F}^{(p)}_{\psi,\beta}$ is recorded separately from the Weyl frame $\mathcal{F}_\beta := \{\mathbb{C}^\times W_\beta(v) : v \in V(\kappa)\} \subset A_{\psi,\beta}(V)$ (the set of $\mathbb{C}^\times$-lines through Weyl operators; studied in §5 below) because the two behave differently under a change of $\beta$, exactly as Theorem 4.10 makes precise: the pair $(A_{\psi,\beta}(V), \mathcal{F}_\beta)$ turns out not to depend on $\beta$ at all, while $\mathcal{F}^{(p)}_{\psi,\beta}$ — equivalently the isomorphism class of $H_\beta(\kappa)$ itself — does, and only at $p = 2$.

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**Provenance.** ids: D11; proved in: —; tested by: —; reviewed in: —.

**SKETCH**

**Theorem 4.10** (The algebra and its Weyl frame do not depend on $\beta$; the level-$\mu_p$ frame does, at $p = 2$)**.** *For all $\beta, \beta' \in \mathrm{Adm}(\omega)$ there is an algebra isomorphism $A_{\psi,\beta}(V) \to A_{\psi,\beta'}(V)$ of the form $W_\beta(v) \mapsto \mu(v) W_{\beta'}(v)$ for some $\mu : V(\kappa) \to \mathbb{C}^\times$ —* a frame isomorphism, *carrying $\mathcal{F}_\beta$ to $\mathcal{F}_{\beta'}$ — and this holds* ***at every characteristic****. Moreover:*

- *$\mu$ can be chosen $\mu_p$-valued if and only if $\psi \circ Q_\beta = \psi \circ Q_{\beta'}$;*
- *[$p = 2$] $\mu$ can always be chosen $\mu_4$-valued, regardless of type.*

*Hence the pair $\big(A_{\psi,\beta}(V), \mathcal{F}_\beta\big)$ does not depend on $\beta$ at all, while the level-$\mu_p$ frame $\mathcal{F}^{(p)}_{\psi,\beta}$ — equivalently the isomorphism class of $H_\beta(\kappa)$ (Theorem 4.7) — does depend on $\beta$, and does so only at $p = 2$.*

*Scope.* This theorem does not contradict Theorems 4.4–4.8: it is a statement about the algebra $A_{\psi,\beta}(V)$ and its $\mathbb{C}^\times$-frame $\mathcal{F}_\beta$, which turn out to be $\beta$-independent, while those theorems are about the finite group $H_\beta(\kappa)$ (equivalently, the level-$\mu_p$ frame), which is not. The distinction between the two frames is exactly what makes both statements true simultaneously. This row is SKETCH for the same procedural reason as Theorems 4.7–4.8, and the explicit $\mu_2/\mu_4$-valued formulas at $p = 2$ are presented at the level of detail the source shard gives them, verified computationally rather than re-derived symbol-by-symbol here.

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**Provenance.** ids: D2, D4, D11, WH-BETA-a, WH-BETA-d, WH-BETA-LEVEL; proved in: theory/wh-kappa-choice.md §6 *step* $\langle 1\rangle 7$; tested by: theory/checks/frame_mu.py (M1,M2,M3,M4); reviewed in: theory/verdicts/wh-kappa-adjudication.md.

*Proof.* Put $s := \beta' - \beta \in \mathrm{Sym}(V)$ (Theorem 4.2). Direct expansion of the multiplication law of Definition 2.4 shows $v \mapsto \mu(v)W_{\beta'}(v)$ is an algebra map exactly when

$$\frac{\mu(v)\mu(v')}{\mu(v+v')} = \psi\big(-s(v,v')\big) \qquad (\dagger)$$

for all $v, v'$, and any such map automatically carries $\mathcal{F}_\beta$ to $\mathcal{F}_{\beta'}$, since it sends each line $\mathbb{C}^\times W_\beta(v)$ to the line $\mathbb{C}^\times W_{\beta'}(v)$. The right-hand side $\psi \circ (-s)$ is a symmetric 2-cocycle on $(V(\kappa), +)$, a group of exponent $p$ (every element of the characteristic-$p$ field $\kappa$, hence of $V(\kappa)$, has additive order dividing $p$), so Fact 3.1 supplies a solution $\mu$ of ($\dagger$): **the frame isomorphism exists at every characteristic**, with no restriction on $\beta, \beta'$.

*Necessity of the $\mu_p$ criterion.* Normalize $\mu(0) = 1$ (forced by ($\dagger$) at $v = v' = 0$, since $\mu(0)^2/\mu(0) = \mu(0) = \psi(0) = 1$). At $p = 2$, setting $v' = v$ in ($\dagger$) gives $\mu(v)^2/\mu(0) = \mu(v)^2 = \psi(s(v,v)) = \psi\big(Q_{\beta'}(v) - Q_\beta(v)\big)$. So $\mu$ is $\mu_2$-valued (i.e. $\mu(v)^2 = 1$ for every $v$) if and only if $\psi(Q_{\beta'}(v)) = \psi(Q_\beta(v))$ for every $v$, i.e. iff $\psi \circ Q_\beta = \psi \circ Q_{\beta'}$; at odd $p$ the analogous computation with $2 \in \kappa^\times$ gives the same criterion for $\mu_p$-valuedness in general.

*Sufficiency, $p = 2$.* Fix an $\mathbb{F}_2$-basis $b_1, \ldots, b_{2m}$ of $V(\kappa)$ and write $\psi(s(b_i, b_j)) = (-1)^{\sigma_{ij}}$, $\sigma_{ij} \in \{0,1\}$, for the finitely many structure values of the symmetric form $s$ on this basis (well defined since $\psi(\kappa) = \mu_2 = \{\pm 1\}$ at $p = 2$, Fact 3.1). If $\sigma_{ii} = 0$ for every $i$ (the case $\psi\circ Q_\beta = \psi\circ Q_{\beta'}$ just characterized), the $\mathbb{F}_2$-bilinear cochain $\ell(v) := \sum_{i<j} c_i c_j \sigma_{ij}$, for $v = \sum_i c_i b_i$, solves ($\dagger$) with values in $\mu_2 = \{\pm 1\}$, by a direct check against the defining bilinear expansion of $s$. In general — without assuming $\sigma_{ii} = 0$ — the integer-valued cochain $m(v) := 2\sum_{i<j} c_i c_j \sigma_{ij} + \sum_i c_i \sigma_{ii} \pmod 4$ solves ($\dagger$) with $\mu(v) := i^{m(v)} \in \mu_4$ ($i = \sqrt{-1}$): this is the explicit construction of `theory/wh-kappa-choice.md` §6, verified against all $|V(\kappa)|^2$ pairs $(v, v')$ for every one of the $q^3$ members of $\mathrm{Adm}(\omega)$ at $q = 2, 4, 8$ by gates M2 and M3 of `theory/checks/frame_mu.py` (which finds $\mu_4$-valued solutions always, and $\mu_2$-valued solutions for exactly the $q$ pairs $(\beta, \beta')$ with $\psi \circ Q_\beta = \psi \circ Q_{\beta'}$, matching the necessity criterion above exactly).

*Sufficiency, odd $p$.* $\mu(v) := \psi\big(s(v,v)/2\big)$ solves ($\dagger$): since $s$ is symmetric bilinear, $s(v,v) + s(v',v') - s(v+v', v+v') = -2s(v,v')$, so $\mu(v)\mu(v')/\mu(v+v') = \psi\big([s(v,v) + s(v',v') - s(v+v', v+v')]/2\big) = \psi(-s(v,v'))$, matching ($\dagger$) exactly, with values entirely inside $\psi(\kappa) = \mu_p$: this reproves, independently of Theorem 4.3, that nothing leaves the level-$\mu_p$ frame once $p$ is odd — a second, algebra-level confirmation that the choice of $\beta$ is immaterial there.

Gate M1 of `theory/checks/frame_mu.py` additionally searches all $2^{q^2}$ candidate $\mu_2$-valued phase functions exhaustively at $q = 2$ and confirms solutions exist exactly for the pairs the necessity criterion predicts; gate M4 checks the odd-$p$ formula at $q = 3, 5, 9$. $\square$

# 5 Symmetry of the Weyl frame: the Weil representation and its splitting problem

**Recollection.** $\kappa = \mathbb{F}_q$, $q = p^m$; $V(\kappa) = \kappa \oplus \kappa$ with $\omega\big((a,b),(a',b')\big) = ab' - a'b$ (Definition 2.1); $SL_2(\kappa)$ is the group of $\kappa$-linear automorphisms of $V(\kappa)$ preserving $\omega$ (Theorem 2.6(e)); a polarizing cocycle $\beta \in \mathrm{Adm}(\omega)$ and phase datum $\psi$ determine the algebra $A_{\psi,\beta}(V) = \bigoplus_v \mathbb{C} \cdot W_\beta(v)$ (Definitions 4.1, 2.3, 2.4), and $Q_\beta(v) := \beta(v, v)$ (Definition 3.1).

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**Provenance.** ids: D1, D2, D3, D4, D6; proved in: —; tested by: —; reviewed in: —.

This section studies the symmetries of the Weyl frame itself: which permutations of the Weyl operators extend to algebra automorphisms, how large that symmetry group is, and whether $SL_2(\kappa)$ — the natural symmetry group of $(V(\kappa), \omega)$ — genuinely acts on the quantum system by honest algebra automorphisms (as opposed to merely projectively). The answer is uniform in one respect and characteristic-dependent in another: the group of frame automorphisms always surjects onto $SL_2(\kappa)$ with kernel the dual group $\hat{V}$, at every characteristic; but whether this extension *splits* — whether $SL_2(\kappa)$ itself, not just a projective shadow of it, acts by algebra automorphisms — is settled affirmatively at odd $p$ and only partially resolved at $p = 2$, where it interacts with the Arf dichotomy of §4.

## 5.1 The Weyl frame and its two automorphism groups

**Definition 5.1** (The Weyl frame, and its $\kappa$-linear automorphisms)**.** The *Weyl frame* is

$$\mathcal{F} := \left\{ \mathbb{C}^\times \cdot W_\beta(v) \ : \ v \in V(\kappa) \right\} \ \subset \ A_{\psi,\beta}(V),$$

the set of $\mathbb{C}^\times$-lines through Weyl operators. $\mathrm{Aut}_{\mathcal{F}}(A)$ is the group of $\mathbb{C}$-algebra automorphisms $\alpha$ of $A_{\psi,\beta}(V)$ with $\alpha(\mathcal{F}) = \mathcal{F}$. $\mathrm{Aut}^\kappa_{\mathcal{F}}(A) \subseteq \mathrm{Aut}_{\mathcal{F}}(A)$ is the subgroup consisting of those $\alpha$ whose induced permutation of $V(\kappa)$ (well defined since $\alpha$ permutes the lines of $\mathcal{F}$, which are indexed bijectively by $V(\kappa)$) is $\kappa$-linear.

*Scope.* The $\kappa$-linearity clause in the definition of $\mathrm{Aut}^\kappa_{\mathcal{F}}(A)$ is **imposed data**: nothing in $A_{\psi,\beta}(V)$ or $\mathcal{F}$ as built from $(V, +)$ and the cocycle $\psi \circ \beta$ forces an automorphism's induced permutation of $V(\kappa)$ to respect scalar multiplication by $\kappa$ rather than merely by $\mathbb{F}_p$. Precisely how much symmetry this clause discards is Theorem 7.1 of §7; nothing here addresses that question.

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**Provenance.** ids: D7; proved in: —; tested by: —; reviewed in: —.

## 5.2 Exactness of the frame-automorphism extension, uniformly in the characteristic

Fix $\beta \in \mathrm{Adm}(\omega)$. For $g \in SL_2(\kappa)$, write $s_g(v, v') := \beta(gv, gv') - \beta(v, v')$ and $Q_g(v) := s_g(v, v)$; note that $Q_g(v) = Q_\beta(gv) - Q_\beta(v)$.

PROVED

**Theorem 5.2** (Exactness of the frame-automorphism extension)**.** *At every characteristic, the sequence*

$$1 \longrightarrow \hat{V} \longrightarrow \mathrm{Aut}^\kappa_{\mathcal{F}}(A_{\psi,\beta}(V)) \xrightarrow{\alpha \mapsto g} SL_2(\kappa) \longrightarrow 1$$

*is exact, where* $\hat{V} := \mathrm{Hom}\big((V(\kappa), +), \mathbb{C}^\times\big)$*; the kernel consists exactly of the conjugations* $\mathrm{Ad}_{W_\beta(u)}$*,* $u \in V(\kappa)$*.*

*Scope.* This theorem does not assert that the extension splits — that $SL_2(\kappa)$ itself embeds back into $\mathrm{Aut}^\kappa_{\mathcal{F}}(A)$ as a subgroup — only that it is exact. Splitting is addressed separately, and with a genuinely different answer at odd and even characteristic, in Theorems 5.3 and 5.6 below. Nothing here concerns $\mathrm{Aut}_{\mathcal{F}}(A)$ without the imposed $\kappa$-linearity clause; that is Theorem 7.1.

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**Provenance.** ids: D1, D4, D7, WH-ALG, WH-COMM, WH-WEIL-a; proved in: theory/wh-kappa-choice.md §8 *step* $\langle 1\rangle 2, \langle 1\rangle 3$; tested by: theory/checks/split24.py (X2); reviewed in: theory/verdicts/wh-kappa-adjudication.md.

*Proof.* $s_g$ *is symmetric bi-additive, and* $s_{gh}(v,v') = s_h(v,v') + s_g(hv,hv')$. Symmetry: $s_g(v,v') - s_g(v',v) = \omega(gv,gv') - \omega(v,v') = 0$, since $g \in SL_2(\kappa)$ preserves $\omega$ (Theorem 2.6(e)); the telescoping identity is immediate by expanding $\beta(ghv,ghv') - \beta(v,v')$ through the intermediate term $\beta(hv,hv')$. (For $\beta = \beta_0$ and $g = \left(\begin{smallmatrix} r & s \\ t & u \end{smallmatrix}\right)$, direct expansion of $\beta_0(gv,gv') = rt\,aa' + ru\,ab' + st\,a'b + su\,bb'$ gives the explicit formula $s_g(v,v') = rt\,aa' + st(ab' + a'b) + su\,bb'$, using $ru - st = \det g = 1$.)

*Description of* $\mathrm{Aut}^{\kappa}_{\mathcal{F}}(A)$. Frame preservation gives $\alpha(W_\beta(v)) = \lambda(v) W_\beta(gv)$ for some function $\lambda : V(\kappa) \to \mathbb{C}^\times$ and some permutation $g$ of $V(\kappa)$, imposed to be $\kappa$-linear by definition of $\mathrm{Aut}^{\kappa}_{\mathcal{F}}(A)$. Applying $\alpha$ to $W_\beta(v) W_\beta(v') = \psi(\beta(v,v')) W_\beta(v+v')$ and comparing basis elements forces $g$ additive (automatic here, being $\kappa$-linear) and

$$\lambda(v)\lambda(v')\,\psi\big(s_g(v,v')\big) = \lambda(v+v') \tag{$*$}$$

for all $v, v'$. Applying $\alpha$ to the commutation relation of Theorem 2.7(c) and using $\omega(gv,gv') = \det(g)\,\omega(v,v')$ (Theorem 2.6(e)'s proof) gives $\psi\big((\det g - 1)\,\omega(v,v')\big) = 1$ for all $v, v'$; since $\omega$ is onto $\kappa$ (Theorem 2.6(a),(c)), Fact 3.1 forces $\det g = 1$, i.e. $g \in SL_2(\kappa)$. So $\alpha \mapsto g$ is a well-defined map $\mathrm{Aut}^{\kappa}_{\mathcal{F}}(A) \to SL_2(\kappa)$, visibly a homomorphism (composition of automorphisms composes the induced permutations).

*The kernel.* If $g = \mathrm{id}$, $(*)$ reads $\lambda(v)\lambda(v') = \lambda(v+v')$, i.e. $\lambda \in \hat{V}$, a group of order $q^2$ by Fact 3.1 ($U = V(\kappa)$, $d = 2m$). The map $u \mapsto \psi(\omega(u,\cdot))$, $V(\kappa) \to \hat{V}$, is injective (if $\psi(\omega(u,\cdot)) \equiv 1$ then $\omega(u,\cdot) \equiv 0$ by Fact 3.1, so $u = 0$ by nondegeneracy, Theorem 2.6(c)), hence bijective ($|V(\kappa)| = q^2 = |\hat{V}|$); so every kernel element has $\lambda = \psi(\omega(u,\cdot))$ for a unique $u \in V(\kappa)$, and by Theorem 2.7(d) this is exactly $\mathrm{Ad}_{W_\beta(u)}$.

*Surjectivity, uniformly in p.* For fixed $g \in SL_2(\kappa)$, $c(v,v') := \psi(s_g(v,v'))$ is a symmetric 2-cocycle on the exponent-$p$ group $(V(\kappa),+)$ (symmetry of $c$ from symmetry of $s_g$, just shown; the cocycle identity from bi-additivity of $s_g$), so Fact 3.1 supplies $\lambda$ solving $(*)$: every $g \in SL_2(\kappa)$ is hit. This argument is uniform in the characteristic; nothing distinguishes $p = 2$. ☐

*Remark* (Every frame automorphism is projectively unitary, at every characteristic)*.* Fix a model $(M,\pi)$ as in Theorem 3.5. Taking absolute values in $(*)$ makes $v \mapsto |\lambda(v)|$ a homomorphism from the finite group $V(\kappa)$ to the torsion-free group $(\mathbb{R}_{>0}, \times)$, hence trivial: every $\lambda$ occurring in $\mathrm{Aut}^{\kappa}_{\mathcal{F}}(A)$ is automatically unimodular. By Theorem 3.5(c),(d), every $\alpha \in \mathrm{Aut}^{\kappa}_{\mathcal{F}}(A)$ is $\mathrm{Ad}_T$ for some $T$, unique up to a scalar; since $\alpha$ permutes the *unitary* Weyl operators (unimodular $\lambda$ times unitary $\pi(W_\beta(gv))$), $T$ may be chosen unitary and is then unique up to $U(1)$ (Theorem 3.5(d)). This gives an injective homomorphism $\mathrm{Aut}^{\kappa}_{\mathcal{F}}(A) \to PU(M)$: a projective unitary representation of $\mathrm{Aut}^{\kappa}_{\mathcal{F}}(A)$, at every characteristic. Composed with a section $SL_2(\kappa) \to \mathrm{Aut}^{\kappa}_{\mathcal{F}}(A)$ of Theorem 5.2's surjection — should one exist as an honest group homomorphism, i.e. should the extension split — this yields an honest (non-projective) representation of $SL_2(\kappa)$ by algebra automorphisms, and hence, via Ad, a projective unitary representation of $SL_2(\kappa)$ itself on $M$. What is *not* true is that this extends to all of $\mathrm{Aut}_{\mathbb{C}}(A) \cong PGL_q(\mathbb{C})$: at $q = 2$, $T = \mathrm{diag}(2,1)$ conjugates $\pi(W(1,0)) = \left(\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}\right)$ to $\begin{pmatrix} 0 & 2 \\ 1/2 & 0 \end{pmatrix}$, whose singular values $2$, $1/2$ differ, so it is not a scalar times a unitary — the frame hypothesis in Definition 5.1 is exactly what is needed above.

### 5.3 Splitting at odd characteristic

**PROVED** †

**Theorem 5.3** (The Weil representation of $SL_2(\kappa)$, at odd characteristic)**.** *[p* ***odd.****]* $\alpha_g\big(W_\beta(v)\big) := \psi\big(Q_g(v)/2\big)\,W_\beta(gv)$ *defines a splitting* $g \mapsto \alpha_g$ *of the extension of Theorem 5.2:* $SL_2(\kappa)$ *acts on* $A_{\psi,\beta}(V)$ *by* $\mathbb{C}$*-algebra automorphisms, with phases in* $\mu_p$*. Composed with Remark 5.2, this exhibits a projective unitary representation of* $SL_2(\kappa)$ *on the model space of Theorem 3.5.*

*Scope.* This is an odd-characteristic statement only; Theorem 5.6 addresses whether an analogous honest (non-projective) action of $SL_2(\kappa)$ by algebra automorphisms exists at $p = 2$, and the answer there is open in general. Nothing here claims the splitting is unique, or canonical among splittings.

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**Provenance.** ids: D4, D6, WH-WEIL-a, WH-SVN, WH-WEIL-b; proved in: theory/wh-kappa-choice.md §8 *step* $\langle 1\rangle 4, \langle 1\rangle 7$; tested by: theory/checks/split24.py (X4); reviewed in: theory/verdicts/wh-kappa-adjudication.md.

*Proof.* Since $s_g$ is symmetric bilinear, $Q_g(v+v') = Q_g(v) + Q_g(v') + 2s_g(v,v')$ (the same polarization identity as in Definition 3.1's use elsewhere), so

$$\psi\big(Q_g(v+v')/2\big) = \psi\big(Q_g(v)/2\big)\,\psi\big(Q_g(v')/2\big)\,\psi\big(s_g(v,v')\big),$$

which is exactly relation $(*)$ of Theorem 5.2's proof for $\lambda(v) := \psi(Q_g(v)/2)$: $\alpha_g$ is a well-defined element of $\mathrm{Aut}^{\kappa}_{\mathcal{F}}(A)$ over $g$. That $g \mapsto \alpha_g$ is a homomorphism follows from the telescoping identity $s_{gh}(v,v') = s_h(v,v') + s_g(hv,hv')$ of Theorem 5.2's proof, evaluated at $v = v'$ to give $Q_{gh}(v) = Q_h(v) + Q_g(hv)$, whence $\alpha_g(\alpha_h(W_\beta(v))) = \psi(Q_h(v)/2)\,\psi(Q_g(hv)/2)\,W_\beta(ghv) = \psi(Q_{gh}(v)/2)\,W_\beta(ghv) = \alpha_{gh}(W_\beta(v))$. This is an honest algebra automorphism, so $SL_2(\kappa)$ acts by algebra automorphisms with $\mu_p$ phases (division by 2 makes sense since $p$ is odd); the projective unitary statement is Remark 5.2 applied to this section. $\square$

### 5.4 Characteristic two: the orthogonal subgroup and forced order-four phases

**PROVED** †

**Theorem 5.4** ($\mu_2$-phases occur exactly over the orthogonal subgroup of $Q_\beta$)**.** *[$p = 2$.] Write $O(Q_\beta) := \{\, g \in SL_2(\kappa) : Q_\beta \circ g = Q_\beta \,\}$. A frame automorphism $\alpha_{g,\lambda}$ (in the notation $\alpha(W_\beta(v)) = \lambda(v)W_\beta(gv)$ of Theorem 5.2's proof) has* some *choice of $\lambda$ landing entirely in $\psi(\kappa) = \mu_2$ if and only if $g \in O(Q_\beta)$ — a condition on the chosen $\beta$, not on the characteristic alone. For the reference cocycle $\beta_0$, $O(Q_{\beta_0}) \cap SL_2(\kappa)$ consists exactly of the matrices* $\mathrm{diag}(r, r^{-1})$ *and* $\mathrm{antidiag}(s, s^{-1})$, *$r, s \in \kappa^\times$; it has order $2(q-1)$ (equal to $2, 6, 14$ at $q = 2, 4, 8$), and is a* ***proper*** *subgroup of $SL_2(\kappa)$ for every $q$.*

*Scope.* This theorem is about which $g$ admit a $\mu_2$-valued lift, for a *fixed* $\beta$; it says nothing yet about what happens for $g \notin O(Q_\beta)$ (Theorem 5.5), and the concrete description of $O(Q_{\beta_0})$ is specific to the reference cocycle — for a general $\beta$ of hyperbolic type the order is the same, $2(q-1)$ (Theorem 4.7(iii)), but the concrete matrices differ with the normal-form basis chosen there.

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**Provenance.** ids: D6, WH-WEIL-a, D7, WH-WEIL-c; proved in: theory/wh-kappa-choice.md §8 *step* $\langle 1\rangle 5$; tested by: theory/checks/beta_census.py (B6); theory/checks/fp_symmetry.py (S3); reviewed in: theory/verdicts/wh-kappa-adjudication.md.

*Proof.* Setting $v' = v$ in $(*)$ at $p = 2$ ($2v = 0$) gives $\lambda(v)^2 = \psi(Q_g(v))$. So a solution $\lambda$ of $(*)$ is $\mu_2$-valued (i.e. $\lambda(v)^2 = 1$ for every $v$) iff $Q_g(V(\kappa)) \subseteq \ker\psi$, i.e. iff $\psi \circ Q_g \equiv 1$.

*For every $\beta$, this is already the exact isometry condition.* $Q_g(v) = Q_\beta(gv) - Q_\beta(v)$ is additive in $v$: expanding $Q_\beta(g(v+v')) - Q_\beta(v+v')$ via Definition 3.1's polarization identity on both terms, the $\omega$-cross-terms cancel because $g$ preserves $\omega$ (Theorem 2.6(e)), leaving $Q_g(v+v') = Q_g(v) + Q_g(v')$; also $Q_g(cv) = c^2 Q_g(v)$ since $g$ is $\kappa$-linear. So, in a basis, $Q_g(a,b) = \alpha a^2 + \delta b^2$ for some $\alpha, \delta \in \kappa$ (no cross term). If $\psi \circ Q_g \equiv 1$ then $\psi(\alpha a^2) = 1$ for all $a$ (setting $b = 0$); since $a \mapsto a^2$ is bijective on $\kappa$, this forces $\alpha = 0$ by Fact 3.1, and likewise $\delta = 0$: $Q_g \equiv 0$, i.e. $g \in O(Q_\beta)$. Conversely, if $g \in O(Q_\beta)$ then trivially $\psi \circ Q_g \equiv 1$, and moreover a $\mu_2$-valued $\lambda$ solving $(*)$ then *exists*: $\psi \circ s_g$ is a symmetric cocycle with trivial diagonal ($\psi(s_g(v,v)) = \psi(Q_g(v)) = 1$), exactly the case handled by the $\ell$-construction in the proof of Theorem 4.10 (§4). So $\psi \circ Q_g \equiv 1$ iff $g \in O(Q_\beta)$ iff a $\mu_2$-valued lift exists.

*The reference cocycle.* By the explicit formula of Theorem 5.2's proof, $s_g(v,v') = rt\,aa' + st(ab' + a'b) + su\,bb'$ for $\beta_0$, so $Q_g(a,b) = rt\,a^2 + su\,b^2$ (the cross term drops at $p = 2$). Thus $g \in O(Q_{\beta_0})$ iff $rt = 0$ and $su = 0$. Combined with $\det g = ru - st = 1$: if $r = 0$ then $st = -1 = 1$ (so $s,t \neq 0$), and $su = 0$ forces $u = 0$, giving $g = \left(\begin{smallmatrix} 0 & s \\ s^{-1} & 0 \end{smallmatrix}\right)$ (using $t = s^{-1}$ from $st = 1$) — an antidiagonal matrix; if $t = 0$ then $ru = 1$ (so $r,u \neq 0$, $u = r^{-1}$), and $su = 0$ forces $s = 0$, giving $g = \operatorname{diag}(r, r^{-1})$; the case $r = t = 0$ contradicts $\det g = 1$. So $O(Q_{\beta_0}) \cap SL_2(\kappa) = \{\operatorname{diag}(r,r^{-1})\} \cup \{\operatorname{antidiag}(s,s^{-1})\}$, of order $(q-1)+(q-1) = 2(q-1)$ (each family is parametrized bijectively by $\kappa^\times$). Since $Q_{\beta_0}(a,b) = ab$ is the hyperbolic normal form (Theorem 4.7(iii)), this matches the general formula $2(q-1)$ there. Properness: $2(q-1) < q(q^2-1) = |SL_2(\kappa)|$ for every $q \geq 2$, since $q(q+1) \geq 6 > 2$. □

**PROVED**[†]

**Theorem 5.5** ($\mu_4$ is forced exactly when the orthogonal subgroup is proper)**.** *[$p = 2$.] If $g \notin O(Q_\beta)$, every frame automorphism over $g$ carries a phase of exact multiplicative order 4. Consequently $\mu_4$ is forced somewhere in $\mathrm{Aut}^{\kappa}_{\mathcal{F}}(A)$ exactly when $O(Q_\beta) \cap SL_2(\kappa)$ is a proper subgroup of $SL_2(\kappa)$: this holds for $\beta_0$ at every $q$ (Theorem 5.4), and for* every *$\beta \in \mathrm{Adm}(\omega)$ at $q = 4, 8$, by direct enumeration. It does **not** hold in the single case $q = 2$ with $\beta$ anisotropic, where $O(Q_\beta) \cap SL_2(\mathbb{F}_2) = SL_2(\mathbb{F}_2)$ (order $6 = |O(Q_\beta) \cap SL_2(\kappa)|$ from the anisotropic formula $2(q+1)$ of Theorem 4.7(iii) at $q = 2$) and the entire group $\mathrm{Aut}^{\kappa}_{\mathcal{F}}(A)$ admits $\mu_2$-valued phases throughout. So $\mu_4$ is **not** a characteristic-two invariant of $(V(\kappa), \omega, \psi)$ alone: whether it is forced depends on the type of $\beta$.*

*Scope.* This theorem uses only Theorem 5.4 and the comparison of $2(q \mp 1)$ against $|SL_2(\kappa)| = q(q^2-1)$; it does not itself re-derive the general-$q$ order $2(q+1)$ for the anisotropic stabilizer (that is Theorem 4.7(iii), only cited here). The claim for *every* $\beta$ at $q = 4, 8$ rests on direct enumeration rather than a closed-form argument for general $q$.

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**Provenance.** ids: D6, D10, WH-WEIL-c, WH-WEIL-d; proved in: theory/wh-kappa-choice.md §8 *step* ⟨1⟩6; tested by: theory/checks/split24.py (X4,X6, and the anisotropic cross-run at $q = 2$); reviewed in: theory/verdicts/wh-kappa-adjudication.md.

*Proof.* By Theorem 5.4's proof, any $\lambda$ solving $(*)$ over $g$ satisfies $\lambda(v)^2 = \psi(Q_g(v))$ for every $v$. If $g \notin O(Q_\beta)$ then $Q_g \not\equiv 0$, and (by the same bijective-squaring argument as there) $\psi(Q_g(v)) = -1$ for some $v$ — since $Q_g(v) = -1$'s preimage under $\psi$ being nontrivial requires only that $\psi(\kappa) = \mu_2 = \{1, -1\}$ (Fact 3.1) and $Q_g$ nonzero, hence onto a nonzero multiple of $\kappa$, hits every value including one where $\psi = -1$. Then $\lambda(v)^2 = -1$, i.e. $\lambda(v)$ has exact order 4: *every* $\lambda$ solving $(*)$ over this $g$ (not just one choice) is forced to take an order-4 value at this particular $v$, since the equation $\lambda(v)^2 = \psi(Q_g(v)) = -1$ has no solution in $\mu_2$. Comparing $2(q-1)$ and $2(q+1)$ (Theorem 4.7(iii)) against $|SL_2(\kappa)| = q(q^2-1) = q(q-1)(q+1)$: $2(q-1) < q(q-1)(q+1)$ always (as $q(q+1) \geq 6$ for $q \geq 2$), so the hyperbolic stabilizer — in particular $O(Q_{\beta_0}) \cap SL_2(\kappa)$ — is always proper, giving some $g \notin O(Q_\beta)$ and hence a forced order-4 phase, at every $q$. For the anisotropic stabilizer, $2(q+1) < q(q-1)(q+1)$ iff $2 < q(q-1)$: this holds for $q \geq 4$ but *fails* at $q = 2$, where $q(q-1) = 2$ exactly, so $2(q+1) = 6 = q(q-1)(q+1) = |SL_2(\mathbb{F}_2)|$: the anisotropic stabiliser is the whole group at $q = 2$, so *no* $g$ lies outside $O(Q_\beta)$, so no order-4 phase is forced by this argument, and — since $\mu_2$-valued lifts exist for every $g \in O(Q_\beta)$ by Theorem 5.4's proof, and $\hat{V}$ is automatically $\mu_2$-valued (Fact 3.1, exponent $p = 2$) — the whole of $\mathrm{Aut}^{\kappa}_{\mathcal{F}}(A)$ admits $\mu_2$-valued phases in this one case. At $q = 4, 8$, exhaustive enumeration over every $\beta \in \mathrm{Adm}(\omega)$ (`theory/checks/split24.py`, gates X4 and X6) confirms $O(Q_\beta) \cap SL_2(\kappa)$ is proper for both types, hence $\mu_4$ is forced for every $\beta$ there. □

### 5.5 Does the extension split at characteristic two? A conjecture, and what is known

CONJECTURE

**Conjecture 5.6** (Existence of a Weil representation of $SL_2(\kappa)$ at characteristic two)**.** *[$p = 2$.] The extension of Theorem 5.2 splits for every $q = 2^m$: $SL_2(\kappa)$ acts on $A_{\psi,\beta}(V)$ by $\mathbb{C}$-algebra automorphisms, for some choice of splitting. This is verified for the reference cocycle $\beta_0$ at $q = 2$ (exhaustive search finds exactly 4 distinct complements of $\hat{V}$ of order 6, over all generating pairs) and at $q = 4$ (64 lifts of a $(2,3,5)$-presentation of $SL_2(\mathbb{F}_4)$ satisfy the defining relations and generate a subgroup of order 60 meeting $\hat{V}$ trivially); in both verified cases every complement necessarily uses phases of exact order 4 (no complement avoids $\mu_4$, since $|\hat{V}|\cdot|O(Q_{\beta_0}) \cap SL_2(\kappa)|$ does not divide $|\mathrm{Aut}^{\kappa}_{\mathcal{F}}(A)|$'s $\mu_2$-part suitably: $6 \nmid 8$ at $q = 2$, $60 \nmid 96$ at $q = 4$). No argument is known for general $q$.*

*Scope.* This is recorded as a conjecture, not a theorem, precisely because the evidence is two finite instances and no general argument: per this campaign's discipline, a passing computational instance promotes nothing. The verified cases are for the reference cocycle $\beta_0$ only; splitting for other cocycles, and any relationship between splittings for cocycles of different Arf type, is not addressed. Should a splitting be found for general $q$, Remark 5.2 would then give a projective unitary representation of $SL_2(\kappa)$ at $p = 2$ exactly as Theorem 5.3 does at odd $p$.

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**Provenance.** ids: D7, WH-WEIL-a, WH-WEIL-d, WH-WEIL-e; proved in: —; tested by: theory/checks/split24.py (X4,X5,X6); reviewed in: theory/verdicts/wh-kappa-adjudication.md.

*What is known, stated as evidence and not as a proof.* At $q = 2$, $|\mathrm{Aut}^{\kappa}_{\mathcal{F}}(A)| = q^2|SL_2(\kappa)| = 4 \cdot 6 = 24$; exhaustive search over generating pairs of order-6 subgroups meeting $\hat{V} = \mathbb{Z}/2 \times \mathbb{Z}/2$ trivially finds exactly 4 complements, all using a phase of exact order 4 somewhere (`split24.py` gate X4). At $q = 4$, lifting a $(2,3,5)$ generator-relation presentation of $SL_2(\mathbb{F}_4) \cong A_5$ over all $16 \times 16 = 256$ candidate phase choices on the two generators finds exactly 64 lifts satisfying $x^2 = y^3 = (xy)^5 = 1$ and generating a subgroup of order 60 meeting $\hat{V}$ trivially, again always with order-4 phases (gate X5). That no complement can avoid $\mu_4$ in either case is confirmed by comparing group orders directly (gate X6): the $\mu_2$-phase part of $\mathrm{Aut}^{\kappa}_{\mathcal{F}}(A)$ has order $q^2 \cdot |O(Q_{\beta_0}) \cap SL_2(\kappa)| = 4 \cdot 2 = 8$ at $q = 2$ and $16 \cdot 6 = 96$ at $q = 4$, and $|SL_2(\kappa)| = 6$, respectively 60, does not divide either. This settles the existence question affirmatively at these two finite values and settles that any splitting there must use order-4 phases; it establishes nothing for $q = 2^m$, $m \geq 3$.

## 6 The choices the construction depends on, and what is canonical

**Recollection.** $\kappa = \mathbb{F}_q$; $\psi \in X(\kappa)$ a phase datum, $\beta \in \mathrm{Adm}(\omega)$ a polarizing cocycle (Definitions 2.3, 4.1); $A_{\psi,\beta}(V)$ the resulting twisted algebra, simple of dimension $q^2$ (Theorem 2.8), with a unique simple unitary module up to unitary equivalence (Theorem 3.5); by Theorem 4.10, for fixed $\psi$ every two choices of $\beta$ give algebras related by an explicit frame isomorphism.

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**Provenance.** ids: D2, D3, D4, WH-ALG, WH-SVN; proved in: —; tested by: —; reviewed in: —.

Sections 4 and 5 named the choices the construction depends on and measured exactly how much they matter. This section assembles that information into a single statement: precisely which data beyond $\kappa$ the construction needs, and precisely what survives canonically once those choices are made. The headline fact is that the *projective* Hilbert space is canonical — it does not depend on which unitary model, which polarization, or (granting one more ingredient from §4) which polarizing cocycle was used to build it — even though the Hilbert space itself is not.

**Definition 6.1** (The model groupoid, and its projectivization)**.** $\mathrm{Mod}_{\psi,\beta}(\kappa)$ is the category whose objects are pairs $(M,\pi)$ with $\pi : A_{\psi,\beta}(V) \to \mathrm{End}_{\mathbb{C}}(M)$ a unital algebra homomorphism making $M$ a simple module and every $\pi(W_\beta(v))$ unitary, and whose morphisms $(M,\pi) \to (M',\pi')$ are unitary intertwiners $T : M \to M'$ ($T\pi(a) = \pi'(a)T$ for all $a$). $\mathrm{PMod}_{\psi,\beta}(\kappa)$ is the same category with each Hom-set replaced by $\mathrm{Hom}/U(1)$ (intertwiners identified when they differ by a unimodular scalar).

*Scope.* This definition alone asserts nothing about the size or connectivity of $\mathrm{Mod}_{\psi,\beta}(\kappa)$; that it is nonempty, connected, and has automorphism group $U(1)$ at every object is Theorem 6.3(c) below, using Theorem 3.5, not part of the stipulation here.

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**Provenance.** ids: D9; proved in: —; tested by: —; reviewed in: —.

### 6.1 Exactly two data beyond the field

PROVED

**Theorem 6.2** (The construction depends on exactly two data beyond $\kappa$)**.** *(a) $X(\kappa) = \hat{\kappa}\backslash\{1\}$ has $q-1$ elements and is a free transitive $\kappa^\times$-torsor under $(u \cdot \psi)(x) := \psi(ux)$ (a single point when $q = 2$, where $\kappa^\times$ is trivial);*

*(b) beyond $\kappa$, the construction of $A_{\psi,\beta}(V)$ uses* ***exactly two*** *further data: a nontrivial character $\psi \in X(\kappa)$ and a polarizing cocycle $\beta \in \mathrm{Adm}(\omega)$ — given $(\psi,\beta)$, $A_{\psi,\beta}(V)$ (Definition 2.4) involves no further choice, while exhibiting a concrete Hilbert space (rather than the abstract algebra) needs a pair $(L,\chi)$ in addition (Definition 3.2);*

*(c) distinct $\psi,\psi' \in X(\kappa)$ give representations of the* same *group $H_\beta(\kappa)$ (which, by Definition 2.5, does not involve $\psi$ at all) with distinct central characters, hence inequivalent as representations of $H_\beta(\kappa)$;*

*(d) $A_{\psi,\beta}(V) \cong A_{\psi',\beta'}(V)$* for every *$\psi,\psi' \in X(\kappa)$ and $\beta,\beta' \in \mathrm{Adm}(\omega)$; but the frame-preserving isomorphisms between two such algebras form a torsor under $\mathrm{Aut}^{\kappa}_{\mathcal{F}}(A_{\psi,\beta}(V))$ (Theorem 5.2), a group of order $q^2\,|SL_2(\kappa)| \geq 24$ — so no single isomorphism among them is distinguished by the data $(\kappa,\psi,\psi',\beta,\beta')$ alone.*

*Scope.* Part (b) corrects an earlier working hypothesis of this campaign, that the construction depended on a single datum beyond $\kappa$: that hypothesis is refuted by Theorem 4.2 and Theorem 4.5 of §4, which show $\beta$ is a second, independent choice. Part (d) asserts the algebras are abstractly isomorphic; it does not assert any *particular* isomorphism among them is preferred, and Theorem 6.3 below is precisely about what survives this non-uniqueness.

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**Provenance.** ids: D2, D3, D4, WH-ALG, WH-BETA-a, WH-WEIL-a, WH-CHOICE; proved in: theory/wh-kappa-choice.md §7; tested by: theory/checks/wh_kappa_check.py (C8 in its E5 form, C10); reviewed in: theory/verdicts/wh-kappa-adjudication.md.

*Proof.* (a) $|\hat{\kappa}| = q$ by Fact 3.1 ($U = \kappa$, $d = m$), so $|X(\kappa)| = q-1$. The action $(u\cdot\psi)(x) := \psi(ux)$ preserves nontriviality: if $\psi \not\equiv 1$ and $u \neq 0$ then, as $x \mapsto ux$ is a bijection of $\kappa$, $u \cdot \psi \equiv 1$ would force $\psi \equiv 1$, false. It is free: $u \cdot \psi = \psi$ means $\psi((u-1)x) = 1$ for all $x$, i.e. $(u-1)\kappa \subseteq \ker\psi$; if $u \neq 1$ then $(u-1) \in \kappa^\times$ so $(u-1)\kappa = \kappa \not\subseteq \ker\psi$ (Fact 3.1, $\psi$ nontrivial), forcing $u = 1$. A free action of the order-$(q-1)$ group $\kappa^\times$ on the order-$(q-1)$ set $X(\kappa)$ has exactly one orbit of size $q-1$, hence is transitive. At $q = 2$, $\kappa^\times = \{1\}$ and $X(\kappa)$ is a single point — the $q = 2$ choice, per Theorem 4.4, lives entirely in $\beta$, not in $\psi$.

(b) Immediate from Definitions 2.4 and 3.2: the structure constants of $A_{\psi,\beta}(V)$ are $\psi(\beta(v,v'))$, a function of $\kappa$, $\psi$, $\beta$ alone, while $M_{L,\chi}$ additionally names a line $L$ and a character $\chi$ of $A_L$.

(c) A representation of $H_\beta(\kappa)$ with central character $\psi$ restricts on the centre $\kappa \times 0$ to $\psi \cdot \mathrm{id}$ (the remark after Theorem 2.7); an intertwiner between representations with central characters $\psi \neq \psi'$ would have to commute with the centre acting as $\psi \cdot \mathrm{id}$ on one side and $\psi' \cdot \mathrm{id}$ on the other, forcing it to be 0 on the difference $(\psi(t) - \psi'(t)) \cdot \mathrm{id} \neq 0$ for suitable $t$ — no nonzero intertwiner exists, so the two representations of the fixed group $H_\beta(\kappa)$ are inequivalent.

(d) *Existence, for every pair.* By Theorem 4.10, $A_{\psi,\beta}(V) \cong A_{\psi,\beta_0}(V)$ for every $\beta$ (via a frame isomorphism, taking $\beta' = \beta_0$ there), and symmetrically $A_{\psi',\beta'}(V) \cong A_{\psi',\beta_0}(V)$. For the reference cocycle, $W_\psi(a,b) \mapsto W_{\psi_u}(u^{-1}a,b)$ (for $\psi' = u \cdot \psi = \psi_u$ as in (a)) is an isomorphism $A_{\psi,\beta_0}(V) \to A_{\psi_u,\beta_0}(V)$, since $\beta_0(u^{-1}a,b') = u^{-1}\beta_0(v,v')$ makes the structure constants match: $\psi_u(\beta_0(u^{-1}v,u^{-1}v')) = \psi(u \cdot u^{-2}\beta_0(v,v')) = \psi(u^{-1}\beta_0(v,v'))$, consistently with the images $W_{\psi_u}(u^{-1}v)$. Composing the three maps ($A_{\psi,\beta} \to A_{\psi,\beta_0} \to A_{\psi',\beta_0} \to A_{\psi',\beta'}$, the middle map since $X(\kappa)$ is a $\kappa^\times$-torsor by (a)) gives an isomorphism $A_{\psi,\beta}(V) \to A_{\psi',\beta'}(V)$ for arbitrary $\psi, \psi', \beta, \beta'$. *Non-uniqueness.* The frame-preserving isomorphisms between two fixed such algebras, once nonempty, form a free transitive right $\mathrm{Aut}^\kappa_{\mathcal{F}}(A_{\psi,\beta}(V))$-set under postcomposition (compose one such isomorphism with an automorphism of the source), and $|\mathrm{Aut}^\kappa_{\mathcal{F}}(A_{\psi,\beta}(V))| = q^2|SL_2(\kappa)|$ by Theorem 5.2, at least $4 \cdot 6 = 24$. More than one isomorphism exists whenever this group is nontrivial (always), and the data $(\kappa, \psi, \psi', \beta, \beta')$ do not name a preferred one: e.g. a family $\{h_u\}$, $h_u := \mathrm{diag}(u^{-j}, u^{j-1})$ for a further arbitrary choice of $j \in \mathbb{Z}/(q-1)$, gives infinitely many mutually inconsistent *coherent* systems of isomorphisms across the $\kappa^\times$-action of (a) — the family of isomorphisms is trivializable, but not canonically: choosing one is a section, not a theorem. $\square$

### 6.2 The projective Hilbert space is canonical

**PROVED**

**Theorem 6.3** (Canonicity of the projective Hilbert space, at fixed $(\kappa, \psi, \beta)$). *(a)* $\mathrm{Mod}_{\psi,\beta}(\kappa)$ *(Definition 6.1) is a nonempty, connected groupoid, with automorphism group $U(1)$ at every object;*

*(b) consequently every* Hom*-set of* $\mathrm{PMod}_{\psi,\beta}(\kappa)$ *is a singleton, so every diagram in it commutes trivially, and the limit $\varprojlim_{\mathrm{PMod}_{\psi,\beta}(\kappa)} P(M)$ exists with every projection an isomorphism; define $P(H_\psi(\kappa))$ to be this limit. The underlying Hilbert space $H_\psi(\kappa)$ itself is canonical only up to a $U(1)$ of identifications (one for each object of* $\mathrm{Mod}_{\psi,\beta}(\kappa)$ *chosen as a representative), but the* projective *Hilbert space $P(H_\psi(\kappa))$ is canonical in $(\kappa, \psi)$, for fixed $\beta$.*

*Scope.* This theorem fixes $\beta$ throughout; that the resulting $P(H_\psi(\kappa))$ does not, in fact, depend on the choice of $\beta$ either is the separate, weaker-status content of Theorem 6.4 below, which is not needed for and does not feed back into this proof. Nothing here addresses canonicity of a specific *vector* in $H_\psi(\kappa)$, only of the isomorphism class of the pair $(M, \pi)$ up to the $U(1)$ ambiguity of unitary intertwiners.

**Provenance.** ids: D9, WH-SVN, WH-CHOICE, WH-CANON; proved in: theory/wh-kappa-choice.md §7 *step* $\langle 1 \rangle 3$, $\langle 1 \rangle 4$; tested by: —; reviewed in: theory/verdicts/wh-kappa-adjudication.md.

*Proof.* (a) Nonempty: Theorem 3.5(a) exhibits the standard model $M_0$, unitary by Theorem 3.5(d). Connected: any two objects $(M, \pi)$, $(M', \pi')$ are unitarily isomorphic by Theorem 3.5(b)–(d) (uniqueness of the simple module up to unitary equivalence), so a morphism exists between any two objects, in either direction. $\mathrm{Aut}(M, \pi) = U(1)$: by Theorem 3.5(b)'s Schur clause, any self-intertwiner is a scalar; a unitary one is then a scalar of modulus 1, and every element of $U(1)$ is realized (scalars commute with everything, hence intertwine trivially).

(b) In a connected groupoid where every object has automorphism group $U(1)$, every Hom-set is a $U(1)$-torsor (fix one isomorphism between two objects; every other differs from it by an automorphism of the source, i.e. by an element of $U(1)$). Quotienting each Hom-set by this $U(1)$ therefore collapses it to a single point: every Hom-set of $\mathrm{PMod}_{\psi,\beta}(\kappa)$ is a singleton. Consequently, for any two morphisms $f, g$ in $\mathrm{PMod}_{\psi,\beta}(\kappa)$ with the same source and target, $f = g$ automatically (both are *the* unique element of that singleton Hom-set): every square of transition maps commutes, not by verification, but because there is only one morphism available to fill it. A category in which every Hom-set is a singleton is equivalent to the terminal (one-object, one-morphism) category, so the limit of the identity diagram $P(M) \rightsquigarrow P(M)$ over $\mathrm{PMod}_{\psi,\beta}(\kappa)$ exists and every object, with its unique family of identifying isomorphisms to every other, computes it: this is $P(H_\psi(\kappa))$. Since the ordinary (non-projectivized) Hom-sets of $\mathrm{Mod}_{\psi,\beta}(\kappa)$ are $U(1)$-torsors and not singletons, no single Hilbert space among the objects is preferred over $U(1)$-many identifications with it — but every one of them projects isomorphically onto the same limit $P(H_\psi(\kappa))$. □

### 6.3 Independence from the polarizing cocycle as well

SKETCH

**Theorem 6.4** (The projective Hilbert space does not depend on $\beta$ either)**.** *$P(H_\psi(\kappa))$, as constructed in Theorem 6.3, is independent of the choice of polarizing cocycle $\beta \in \mathrm{Adm}(\omega)$ used to build it: the frame isomorphism $A_{\psi,\beta}(V) \to A_{\psi,\beta'}(V)$ of Theorem 4.10 transports unitary simple modules and induces an equivalence $\mathrm{Mod}_{\psi,\beta}(\kappa) \to \mathrm{Mod}_{\psi,\beta'}(\kappa)$, which in turn induces* the *unique isomorphism between the two limits $P(H_\psi(\kappa))$ constructed at $\beta$ and at $\beta'$.*

*Scope.* This theorem is SKETCH for the same procedural reason as the $p = 2$ rows of §4 it depends on (Theorem 4.10): it postdates this increment's single hostile-critic round. It asserts that the *limit* is independent of $\beta$, not that any specific object of $\mathrm{Mod}_{\psi,\beta}(\kappa)$ equals, on the nose, an object of $\mathrm{Mod}_{\psi,\beta'}(\kappa)$ — the identification goes through the transport functor described in the proof, consistently with the fact that the underlying group $H_\beta(\kappa)$ genuinely changes isomorphism type with $\beta$ at $p = 2$ (Theorem 4.7).

**Provenance.** ids: WH-CANON, WH-BETA-LEVEL, WH-CANON-BETA; proved in: theory/wh-kappa-choice.md §7 *step* ⟨1⟩4.⟨2⟩1; tested by: —; reviewed in: theory/verdicts/wh-kappa-adjudication.md.

*Proof.* By Theorem 4.10 there is a frame isomorphism $\Phi : A_{\psi,\beta}(V) \to A_{\psi,\beta'}(V)$, $W_\beta(v) \mapsto \mu(v)W_{\beta'}(v)$, with $\mu$ unimodular (by the same argument as Remark 5.2: taking absolute values in the defining relation (†) of Theorem 4.10's proof makes $|\mu|$ a homomorphism from a finite group to $\mathbb{R}_{>0}$, hence trivial). Transport of structure along $\Phi$ sends an object $(M, \pi)$ of $\mathrm{Mod}_{\psi,\beta}(\kappa)$ to $(M, \pi')$ with $\pi'(a') := \pi(\Phi^{-1}(a'))$ for $a' \in A_{\psi,\beta'}(V)$: $M$ stays simple (an equivalence of module categories along an algebra isomorphism preserves simplicity), and each $\pi'(W_{\beta'}(v)) = \pi(\Phi^{-1}(W_{\beta'}(v))) = \overline{\mu(v)}\,\pi(W_\beta(v))$ remains unitary (a unimodular scalar times a unitary operator). This is an equivalence of categories $\mathrm{Mod}_{\psi,\beta}(\kappa) \to \mathrm{Mod}_{\psi,\beta'}(\kappa)$, compatible with the projectivization functors to $\mathrm{PMod}_{\psi,\beta}(\kappa)$ and $\mathrm{PMod}_{\psi,\beta'}(\kappa)$ of Theorem 6.3(b). Both projectivized categories have singleton Hom-sets (Theorem 6.3(b), applied at $\beta$ and at $\beta'$ respectively), so the induced map between their limits $P(H_\psi(\kappa))_\beta \to P(H_\psi(\kappa))_{\beta'}$ is again the unique morphism available between two terminal-equivalent categories, hence an isomorphism, and the unique one compatible with the transport functor. □

## 7 What $\kappa$-linear symmetry hides: the $\mathbb{F}_p$-shadow of the form

**Recollection.** For $\beta \in \mathrm{Adm}(\omega)$ and phase datum $\psi$, the Weyl frame is $\mathcal{F} = \{\mathbb{C}^\times W_\beta(v) : v \in V(\kappa)\} \subset A_{\psi,\beta}(V)$, and $\mathrm{Aut}_{\mathcal{F}}(A)$ is the group of $\mathbb{C}$-algebra automorphisms of $A_{\psi,\beta}(V)$ preserving

$\mathcal{F}$ (Definition 5.1); $\mathrm{Aut}^{\kappa}_{\mathcal{F}}(A) \subseteq \mathrm{Aut}_{\mathcal{F}}(A)$ is the further subgroup whose induced permutation of $V(\kappa)$ is *$\kappa$-linear* — data *imposed* on top of $\mathrm{Aut}_{\mathcal{F}}(A)$, not derived from it (Definition 5.1's scope). By Theorem 5.2, $\mathrm{Aut}^{\kappa}_{\mathcal{F}}(A)$ surjects onto $SL_2(\kappa)$ with kernel $\hat{V}$, at every characteristic.

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**Provenance.** ids: D4, D7, WH-WEIL-a; proved in: —; tested by: —; reviewed in: —.

This section makes good on the promissory note left in Definition 5.1: exactly how much symmetry the imposed $\kappa$-linearity clause discards. The answer is that $A_{\psi,\beta}(V)$ and its frame $\mathcal{F}$ do not, in fact, remember the $\kappa$-module structure of $V(\kappa)$ at all — they are built from the bare abelian group $(V(\kappa), +)$ and the cocycle $\psi \circ \beta$ — so the *full* frame symmetry group is larger than $SL_2(\kappa)$ whenever $\kappa$ is a proper extension of its prime field, and the discarded symmetry is measured exactly.

PROVED

**Theorem 7.1** (The twisted algebra does not remember the $\kappa$-structure of $V(\kappa)$). *(a) $A_{\psi,\beta}(V)$ and its frame $\mathcal{F}$ are determined by the abelian group $(V(\kappa), +)$ together with the $\mathbb{C}^\times$-valued function $\psi\circ\beta$ on $V(\kappa)\times V(\kappa)$ alone; nothing in either object refers to scalar multiplication by $\kappa$;*

*(b) consequently, dropping the imposed $\kappa$-linearity clause of Definition 5.1, the frame automorphisms with merely $\mathbb{F}_p$-linear induced permutation $g$ of $V(\kappa)$ surject, with the* same *kernel $\hat{V}$, onto the isometry group $Sp(V(\kappa), \psi\circ\omega) \cong Sp_{2m}(\mathbb{F}_p)$ of the $\mathbb{F}_p$-valued alternating form $\psi\circ\omega$; so $|\mathrm{Aut}_{\mathcal{F}}(A)| = q^2 \cdot |Sp_{2m}(\mathbb{F}_p)|$;*

*(c) $SL_2(\kappa) \subseteq Sp_{2m}(\mathbb{F}_p)$, with index exactly $1$ when $m = 1$ and strictly greater than $1$ as soon as $m > 1$: by complete enumeration, $|SL_2(\kappa)| = 6, 24, 60, 120, 504, 720$ against $|Sp_{2m}(\mathbb{F}_p)| = 6, 24, 720, 120, 1451520, 51840$ at $q = 2, 3, 4, 5, 8, 9$ respectively — indices $1, 1, 12, 1, 2880, 72$. So Definition 5.1's $\kappa$-linearity clause discards genuine symmetry the algebra has, whenever $\kappa$ is a proper extension of its prime field.*

*Scope.* This theorem does not claim $Sp_{2m}(\mathbb{F}_p)$ acts by automorphisms preserving anything beyond the frame $\mathcal{F}$ and the abelian group structure — in particular it says nothing about whether this larger symmetry is compatible with the $\kappa$-linear structure used elsewhere in this labbook (Theorem 2.6(f) shows, to the contrary, that the *$\kappa$-valued* form $\omega$ itself sees only $SL_2(\kappa)$ among $\mathbb{F}_p$-linear maps: the enlargement here is possible only because $\mathrm{Aut}_{\mathcal{F}}(A)$ is built from the strictly weaker $\mathbb{F}_p$-valued shadow $\psi\circ\omega$). At the prime field, $m = 1$, the two groups coincide and nothing is discarded; this is the campaign's guiding path.

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**Provenance.** ids: D1, D7, WH-WEIL-a, WH-FORM, WH-SYMM; proved in: theory/wh-kappa-choice.md §9; tested by: theory/checks/fp_symmetry.py (S1,S2,S3); reviewed in: theory/verdicts/wh-kappa-adjudication.md.

*Proof.* (a) By Definition 2.4, the structure constants of $A_{\psi,\beta}(V)$ in the basis $\{W_\beta(v)\}$ are exactly the values $\psi(\beta(v, v'))$, indexed by pairs of elements of the abelian group $(V(\kappa), +)$; by Definition 5.1, $\mathcal{F}$ is the set of $\mathbb{C}^\times$-lines through this same basis, again indexed only by $(V(\kappa), +)$. Neither construction refers to multiplication by elements of $\kappa$.

(b) Repeat the argument of Theorem 5.2's proof *without* imposing $\kappa$-linearity on $g$: frame preservation still gives $\alpha(W_\beta(v)) = \lambda(v)W_\beta(gv)$ with $g$ an (a priori only additive) bijection of $V(\kappa)$ and $\lambda$ satisfying relation $(*)$ there, since that derivation used only additivity of $g$, never $\kappa$-linearity. Applying $\alpha$ to Theorem 2.7(c), an identity entirely in $\psi\circ\omega$, gives $\psi\big(\omega(gv, gv')\big) = \psi\big(\omega(v, v')\big)$ for all $v, v'$ — i.e. $g \in Sp(V(\kappa), \psi\circ\omega) := \{g \text{ additive} : \psi\circ\omega\circ(g\times g) = \psi\circ\omega\}$ — rather than the stronger $\omega\circ g = \omega$ used before. The surjectivity argument via Fact 3.1 applies verbatim to any such $g$ (it never used $\kappa$-linearity, only that $(V(\kappa), +)$ has exponent $p$), so every $g \in Sp(V(\kappa), \psi\circ\omega)$ lifts; the kernel computation ($g = \mathrm{id}$ forces $\lambda \in \hat{V}$, identified with

$V(\kappa)$ via $u \mapsto \psi(\omega(u,\cdot))$) is unchanged, giving the same kernel $\hat{V}$ of order $q^2$. Nondegeneracy of $\psi \circ \omega$ as an $\mathbb{F}_p$-bilinear form on the $2m$-dimensional $\mathbb{F}_p$-space $V(\kappa)$: if $\psi(\omega(v,\cdot)) \equiv 1$ then $\omega(v,\cdot) \equiv 0$ (Fact 3.1, since $\omega(v,\cdot)$ is either 0 or onto $\kappa$, being $\kappa$-linear, and $\kappa \not\subseteq \ker\psi$), so $v = 0$ by Theorem 2.6(c). A nondegenerate alternating form on a $2m$-dimensional space over $\mathbb{F}_p$ has isometry group (by definition, matching the notation `notation.md` fixes for $Sp_{2m}(\mathbb{F}_p)$) exactly $Sp_{2m}(\mathbb{F}_p)$.

(c) If $g \in SL_2(\kappa)$ then $\omega \circ (g \times g) = \omega$ (Theorem 2.6(e)), so certainly $\psi \circ \omega \circ (g \times g) = \psi \circ \omega$, i.e. $SL_2(\kappa) \subseteq Sp(V(\kappa), \psi \circ \omega)$. This containment is exactly the gap between Theorem 2.6(f) — which shows the $\mathbb{F}_p$-linear isometries of the $\kappa$-*valued* $\omega$ are no larger than $SL_2(\kappa)$ — and the present, strictly weaker condition of isometry for the $\mathbb{F}_p$-valued shadow $\psi \circ \omega$ alone: an $\mathbb{F}_p$-linear $g$ can preserve $\psi \circ \omega$ without preserving $\omega$ itself. Complete enumeration (`theory/checks/fp_symmetry.py`, gates S1 and S2) computes both orders directly at $q = 2, 3, 4, 5, 8, 9$, matching the closed form $|Sp_{2m}(\mathbb{F}_p)| = p^{m^2} \prod_{i=1}^{m}(p^{2i} - 1)$ in every case, and confirms the indices $1, 1, 12, 1, 2880, 72$; at $m = 1$ (the prime field) the two group orders coincide, since $Sp_2(\mathbb{F}_p) = SL_2(\mathbb{F}_p)$ by definition in dimension 2, and nothing is discarded. □

*Remark* (The characteristic-two orthogonal phenomenon survives the enlargement). The $\mu_2$-phase criterion of Theorem 5.4 ($g \in O(Q_\beta)$) makes sense verbatim for merely $\mathbb{F}_p$-linear $g$, at $p = 2$: $\psi \circ Q_\beta \circ g = \psi \circ Q_\beta$ is a condition on an $\mathbb{F}_p$-valued quadratic form $\psi \circ Q_\beta$ with polar form $\psi \circ \omega$, so the relevant subgroup inside the enlarged symmetry group of Theorem 7.1 is $O(\psi \circ Q_\beta) \cap Sp_{2m}(\mathbb{F}_p)$. Direct enumeration (gate S3) finds its order to be $2, 72, 40320$ at $q = 2, 4, 8$, of index $3, 10, 36$ in $Sp_{2m}(\mathbb{F}_p)$ — exactly the *same* index that $|O(Q_\beta) \cap SL_2(\kappa)| = 2, 6, 14$ has inside $SL_2(\kappa)$ (Theorem 5.4). So the characteristic-two orthogonal phenomenon of §5 is not an artefact of the imposed $\kappa$-linearity in Definition 5.1: it survives, at the same relative index, inside the larger, intrinsic symmetry group $Sp_{2m}(\mathbb{F}_p)$ that Theorem 7.1 exhibits. This observation is recorded as part of the same source theorem as Theorem 7.1, under the same provenance (`theory/wh-kappa-choice.md` §9, step ⟨1⟩4).

*Remark* (Reading). Two independent non-canonicities are now on record, and they are logically independent of each other. Theorem 4.7 of §4 shows the system does not remember which polarizing cocycle $\beta$ was chosen, in a way that is invisible at odd characteristic and genuinely two-valued at $p = 2$. This theorem shows the system does not remember the $\kappa$-module structure of $V(\kappa)$ itself, at every characteristic and every $m > 1$, regardless of $\beta$. Both are statements about what the definition actually depends on, in the sense of `PRD.md`'s "canonical" requirement: any assignment $\operatorname{Spec} R \mapsto$ quantum system general enough to restrict to this one on $\operatorname{Spec} \mathbb{F}_q$ carries both phenomena, not as defects, but as exactly the symmetry and the choice the construction is built from.

# 8 Functoriality in the field: what extends along embeddings, and what does not

**Recollection.** A phase datum for $\kappa = \mathbb{F}_{p^m}$ is a nontrivial additive character $\psi : (\kappa, +) \to \mathbb{C}^\times$, and the trace-normalized family is $\psi_\zeta(z) := \zeta^{\mathrm{Tr}_{\kappa/\mathbb{F}_p}(z)}$ for $\zeta$ a primitive $p$-th root of unity and $\mathrm{Tr}_{\kappa/\mathbb{F}_p}(z) := z + z^p + \cdots + z^{p^{m-1}}$ (Definition 2.3); by Theorem 6.2(a), $X(\kappa)$ is a $\kappa^\times$-torsor, so every $\psi \in X(\kappa)$ is $\psi_c := \psi_\zeta(c\,\cdot)$ for a unique $c \in \kappa^\times$.

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**Provenance.** ids: D3; proved in: —; tested by: —; reviewed in: —.

Earlier sections fixed a single finite field $\kappa$ throughout. This section asks how the construction behaves as $\kappa$ varies along field embeddings: whether the trace-normalized family is itself compatible with restriction, whether the assignment $\kappa \mapsto A_{\psi,\beta_0}(V(\kappa))$ is a functor, and — the question this campaign's earlier working hypothesis answered wrongly — whether a single compatible choice of character can be made once and for all, for every finite field simultaneously. It cannot; the precise obstruction is Frobenius, and the precise set of compatible choices that *does* exist is named exactly.

PROVED

**Theorem 8.1** (The trace-normalized family, restricted along an extension)**.** *Let $\kappa = \mathbb{F}_{p^m} \subseteq \kappa' = \mathbb{F}_{p^{mn}}$. Then $\psi'_\zeta|_\kappa = \psi_{\zeta^n}$ (both sides characters of $\kappa$, computed with the same $\zeta$): the restriction of $\kappa'$'s trace-normalized character to $\kappa$ is trivial exactly when $p \mid n$ (smallest instance: $\mathbb{F}_2 \subset \mathbb{F}_4$).*

*Scope.* This theorem is about the trace-normalized family specifically — a particular, and particularly natural, choice inside $X(\kappa)$ for every $\kappa$ — and asserts nothing about arbitrary families of characters; the statement that quantifies over *all* compatible systems is Theorem 8.4 below.

**Provenance.** ids: D3, WH-FUNCT-a; proved in: theory/wh-kappa-choice.md §10 *step* ⟨1⟩1; tested by: theory/checks/funct_sections.py (F1); reviewed in: theory/verdicts/wh-kappa-adjudication.md.

*Proof.* For $x \in \kappa$, $x^{p^m} = x$ (elements of $\mathbb{F}_{p^m}$ are exactly the fixed points of the $m$-th power of Frobenius), so the $mn$ summands of $\mathrm{Tr}_{\kappa'/\mathbb{F}_p}(x) = \sum_{k=0}^{mn-1} x^{p^k}$ repeat with period $m$: grouping into $n$ blocks of $m$ consecutive terms, each block sums to $\mathrm{Tr}_{\kappa/\mathbb{F}_p}(x)$ (using $x^{p^{k+m}} = (x^{p^m})^{p^k} = x^{p^k}$), so $\mathrm{Tr}_{\kappa'/\mathbb{F}_p}|_\kappa = n \cdot \mathrm{Tr}_{\kappa/\mathbb{F}_p}$. Hence $\psi'_\zeta(x) = \zeta^{\mathrm{Tr}_{\kappa'/\mathbb{F}_p}(x)} = \zeta^{n \cdot \mathrm{Tr}_{\kappa/\mathbb{F}_p}(x)} = (\zeta^n)^{\mathrm{Tr}_{\kappa/\mathbb{F}_p}(x)} = \psi_{\zeta^n}(x)$, and $\zeta^n = 1$ (so $\psi_{\zeta^n} \equiv 1$) exactly when $p \mid n$, since $\zeta$ has exact order $p$. $\square$

PROVED

**Theorem 8.2** (The reference algebra is functorial in character-compatible embeddings)**.** *Let $\mathcal{FF}^\pm$ be the category with objects the pairs $(\kappa, \psi)$, $\psi \in X(\kappa)$, and morphisms $(\kappa, \psi) \to (\kappa', \psi')$ the field embeddings $\iota : \kappa \hookrightarrow \kappa'$ with $\psi' \circ \iota = \psi$ (*character-compatible *embeddings). Then*

$$(\kappa, \psi) \;\longmapsto\; A_{\psi,\beta_0}(V(\kappa))$$

*is a functor from $\mathcal{FF}^\pm$ to the category of unital $\mathbb{C}$-algebras and injective algebra homomorphisms, respecting composition and identities.*

*Scope.* This functor uses the reference cocycle $\beta_0$ throughout, on every object; it is not claimed that the resulting assignment is independent of this choice, nor that an analogous functor exists carrying a varying $\beta_\kappa$ compatibly along embeddings — the reference cocycle is used here because it is a fixed, explicit stipulation (Definition 4.1), not because it is canonical (Theorem 4.4 of §4 shows at $p = 2$ it is not). Nor is this a statement about the quantum *system* — the Hilbert space or the Heisenberg group — functorially in $\kappa$, only about the algebra.

**Provenance.** ids: D2, D4, WH-ALG, WH-FUNCT-b; proved in: theory/wh-kappa-choice.md §10 *step* ⟨1⟩2; tested by: theory/checks/funct_sections.py (F5); reviewed in: theory/verdicts/wh-kappa-adjudication.md.

*Proof.* For a character-compatible embedding $\iota : (\kappa, \psi) \to (\kappa', \psi')$, put $V(\iota)(a, b) := (\iota a, \iota b) : V(\kappa) \to V(\kappa')$, additive since $\iota$ is a field homomorphism. Since $\iota$ is multiplicative, $\beta_0\big(V(\iota)v, V(\iota)v'\big) = \iota(a)\,\iota(b') = \iota(ab') = \iota\big(\beta_0(v, v')\big)$. Define $W_\beta(v) \mapsto W'_{\beta_0}(V(\iota)v)$ on basis elements of $A_{\psi,\beta_0}(V(\kappa))$. This sends distinct basis elements to distinct basis elements ($V(\iota)$ is injective, $\iota$ being a field embedding), and is multiplicative: the image of $W_{\beta_0}(v)W_{\beta_0}(v') = \psi(\beta_0(v, v'))W_{\beta_0}(v + v')$ is $\psi(\beta_0(v, v'))\, W'_{\beta_0}(V(\iota)(v{+}v'))$, while the product of the images is $\psi'\big(\beta_0(V(\iota)v, V(\iota)v')\big)\, W'_{\beta_0}(V(\iota)(v{+}v')) = \psi'\big(\iota(\beta_0(v, v'))\big)\, W'_{\beta_0}(V(\iota)(v + v')) = \psi(\beta_0(v, v'))\, W'_{\beta_0}(V(\iota)(v + v'))$, using $\psi' \circ \iota = \psi$ (the compatibility condition). So the map is a unital, nonzero algebra homomorphism, hence injective by simplicity of the source (Theorem 2.8). It respects composition and identities because $\iota \mapsto V(\iota)$ does: $V(\iota' \circ \iota) = V(\iota') \circ V(\iota)$ and $V(\mathrm{id}) = \mathrm{id}$, directly from the definition of $V(\iota)$. $\square$

### 8.1 No compatible choice of character over every finite field

REFUTED

**Theorem 8.3** (Refuted: a Frobenius-compatible character exists for every finite field)**.** (Claimed, and refuted.) *A section of the forgetful functor $\mathcal{FF}^{\pm} \to \{$finite fields$\}$ exists: a choice $\psi_\kappa \in X(\kappa)$ for every finite field $\kappa$ such that $\iota : \kappa \to \kappa'$ character-compatible (i.e. $\psi_{\kappa'} \circ \iota = \psi_\kappa$) for* every *field embedding $\iota$ — in particular, for the Frobenius self-embedding of every $\kappa$.*

*Scope.* This row is kept, refuted, rather than deleted, per this campaign's status discipline: a refutation carrying the surviving weaker statement is a result, not a gap to be erased. The surviving statement — exactly which obstruction blocks a section, and exactly which weaker compatible systems *do* exist — is Theorem 8.4 immediately below and Theorem 8.5 thereafter. Nothing may depend on this row.

**Provenance.** ids: WH-FUNCT-b-SEC; proved in: —; tested by: theory/checks/funct_sections.py (F2,F3); reviewed in: theory/verdicts/wh-kappa-adjudication.md.

*Refutation.* Frobenius $x \mapsto x^p$ is a field embedding $\kappa \to \kappa$, so a section would in particular have to satisfy $\psi_\kappa \circ \mathrm{Frob} = \psi_\kappa$ for every $\kappa$. Theorem 8.4 below shows this forces $\psi_\kappa = \psi_c$ for some $c$ in the *prime* field $\mathbb{F}_p$, and that for $\kappa = \mathbb{F}_{p^p}$ every character of this Frobenius-invariant form restricts *trivially* to $\mathbb{F}_p$ — contradicting $\psi_{\mathbb{F}_p} \in X(\mathbb{F}_p)$, which must be nontrivial by definition. So no section exists; the smallest instance of the obstruction is $\mathbb{F}_2 \subset \mathbb{F}_4$. A previously considered route to a section, by induction along the tower $\mathbb{F}_{p^{n!}}$, is insufficient in any case: a character of $\mathbb{F}_{p^{n!}}$ compatible at that one stage can still restrict trivially to an intermediate field, so no such induction can repair the obstruction just exhibited.

PROVED

**Theorem 8.4** (The Frobenius obstruction, stated exactly)**.** *There is no choice of $\psi_\kappa \in X(\kappa)$, one for each finite field $\kappa$, compatible with* all *field embeddings. Precisely: writing $\psi_c := \psi_\zeta(c\,\cdot)$ for $c \in \kappa$ (Theorem 6.2(a)), a character $\psi_c$ is invariant under the Frobenius self-embedding of $\kappa = \mathbb{F}_{p^m}$ if and only if $c \in \mathbb{F}_p$; and for $\kappa = \mathbb{F}_{p^p}$, every $\mathbb{F}_p$-valued $c$ gives a character $\psi_c$ that restricts* trivially *to the prime field $\mathbb{F}_p \subset \mathbb{F}_{p^p}$. The smallest instance is $\mathbb{F}_2 \subset \mathbb{F}_4$.*

*Scope.* This theorem exhibits the precise obstruction at the specific extension degree $m = p$; it does not enumerate every extension degree at which some weaker compatibility fails, and it does not address (this is Theorem 8.5) which *restricted* systems of compatible characters — over a sub-poset of embeddings, or with the "for every embedding" requirement relaxed — do exist.

**Provenance.** ids: D3, WH-FUNCT-a, WH-FUNCT-c; proved in: theory/wh-kappa-choice.md §10 *step* $\langle 1\rangle 3$; tested by: theory/checks/funct_sections.py (F2,F3); reviewed in: theory/verdicts/wh-kappa-adjudication.md.

*Proof. Frobenius-invariance forces $c \in \mathbb{F}_p$.* Write $d := c^{1/p}$ (the unique $p$-th root, Frobenius being bijective on the finite field $\kappa$). Since $\mathrm{Tr}_{\kappa/\mathbb{F}_p}(y^p) = \mathrm{Tr}_{\kappa/\mathbb{F}_p}(y)$ for every $y$ (Frobenius cyclically permutes the $m$ summands of the trace, using $y^{p^m} = y$), we get $\psi_c(x^p) = \zeta^{\mathrm{Tr}(cx^p)} = \zeta^{\mathrm{Tr}((dx)^p)} = \zeta^{\mathrm{Tr}(dx)} = \psi_d(x)$, i.e. $\psi_c \circ \mathrm{Frob} = \psi_d$. Since $c \mapsto \psi_c$ is injective on $\kappa$ (if $\psi_c = \psi_{c'}$ then $\mathrm{Tr}((c-c')x) \equiv 0$ for all $x$, forcing $c = c'$ by the same nondegeneracy of the trace pairing used in Theorem 6.2(a)'s proof), $\psi_c$ is Frobenius-invariant iff $d = c$; since $d^p = c$ by definition of $d$, substituting $d = c$ gives $c^p = c$, and conversely $c^p = c$ gives $d = c^{1/p} = c$ (Frobenius being bijective). So the condition is exactly $c^p = c$, i.e. $c$ is a root of $X^p - X$, i.e. (this polynomial having at most $p$ roots in a field, all $p$ elements of $\mathbb{F}_p$ among them by Fermat's little theorem for finite fields) $c \in \mathbb{F}_p$.

*Trivial restriction at $\kappa = \mathbb{F}_{p^p}$.* Here $m = p$. For $c \in \mathbb{F}_p$ and $x \in \mathbb{F}_p \subset \kappa$, every summand of $\mathrm{Tr}_{\kappa/\mathbb{F}_p}(cx) = \sum_{k=0}^{p-1}(cx)^{p^k}$ equals $cx$ (as $cx \in \mathbb{F}_p$ is already Frobenius-fixed), so $\mathrm{Tr}_{\kappa/\mathbb{F}_p}(cx) = p \cdot (cx) = 0$ in characteristic $p$. So $\psi_c(x) = \zeta^0 = 1$ for every $x \in \mathbb{F}_p$: *every* Frobenius-invariant character of $\mathbb{F}_{p^p}$ — the only kind a section could assign there — restricts trivially to the prime field, contradicting the requirement $\psi_{\mathbb{F}_p} \in X(\mathbb{F}_p)$ (nontrivial by Definition 2.3) forced by compatibility along the inclusion $\mathbb{F}_p \subset \mathbb{F}_{p^p}$. At $p = 2$, $\mathbb{F}_{p^p} = \mathbb{F}_4$: the smallest instance. □

### 8.2 The compatible systems that do exist

PROVED

**Theorem 8.5** (Compatible character systems over an algebraic closure)**.** *Fix an algebraic closure $\overline{\mathbb{F}}_p$, and consider the inclusion poset of its finite subfields. The compatible systems $(\psi_\kappa)_\kappa$ — one $\psi_\kappa \in X(\kappa)$ for every finite subfield $\kappa \subset \overline{\mathbb{F}}_p$, with $\psi_{\kappa'}|_\kappa = \psi_\kappa$ for every inclusion $\kappa \subseteq \kappa'$ of finite subfields — are **exactly** the restrictions $\psi_\kappa := \Psi|_\kappa$ of the additive characters $\Psi$ of $(\overline{\mathbb{F}}_p, +)$ with $\Psi|_{\mathbb{F}_p} \neq 1$.*

*Scope.* This theorem restricts attention to the inclusion poset of finite subfields of a *fixed* algebraic closure, respecting compatibility only along inclusions of subfields, not along all abstract field embeddings between isomorphic-but-not-identical finite fields; Theorem 8.3 shows the latter, stronger requirement admits no solution at all. It says nothing about compatible choices of a polarizing cocycle $\beta$ along the same poset; whether the Arf type of §4 can be chosen compatibly there is not resolved by this labbook.

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**Provenance.** ids: D3, WH-FUNCT-d; proved in: theory/wh-kappa-choice.md §10 *step* ⟨1⟩4; tested by: theory/checks/funct_sections.py (F4); reviewed in: theory/verdicts/wh-kappa-adjudication.md.

*Proof. Restrictions are compatible systems.* Given such a $\Psi$, put $\psi_\kappa := \Psi|_\kappa$; compatibility along $\kappa \subseteq \kappa'$ holds by construction ($(\Psi|_{\kappa'})|_\kappa = \Psi|_\kappa$), and $\psi_\kappa$ is nontrivial for *every* finite subfield $\kappa$: every such $\kappa$ contains $\mathbb{F}_p$, and $\Psi|_\kappa \equiv 1$ would force $\Psi|_{\mathbb{F}_p} \equiv 1$ too, contradicting the hypothesis on $\Psi$.

*Compatible systems arise this way.* Given a compatible system $(\psi_\kappa)$, define $\Psi(x) := \psi_\kappa(x)$ for any finite subfield $\kappa \ni x$ (every $x \in \overline{\mathbb{F}}_p$ lies in some finite subfield, $\overline{\mathbb{F}}_p$ being their union). This is well defined: if $x$ lies in two finite subfields $\kappa, \kappa'$, both lie inside a common finite subfield $\kappa''$ (e.g. their compositum, again finite), and compatibility along $\kappa \subseteq \kappa''$ and $\kappa' \subseteq \kappa''$ gives $\psi_\kappa(x) = \psi_{\kappa''}(x) = \psi_{\kappa'}(x)$. $\Psi$ is additive: any two elements lie in a common finite subfield, where $\Psi$ agrees with the additive character $\psi_\kappa$ of that subfield. And $\Psi|_{\mathbb{F}_p} = \psi_{\mathbb{F}_p} \neq 1$, by Definition 2.3's requirement on every $\psi_\kappa \in X(\kappa)$ applied to $\kappa = \mathbb{F}_p$.

*The two constructions are mutually inverse*, and the condition "nontrivial on every finite subfield" (needed termwise, since each $\psi_\kappa \in X(\kappa)$ by hypothesis) is equivalent to "nontrivial on the prime field alone": the forward implication is immediate ($\mathbb{F}_p$ is itself one of the subfields), and the reverse is exactly what the first paragraph showed. □

*Remark* (What this section does not claim)*.* Theorem 8.2 is functoriality of the *algebra* in character-compatible embeddings, for the fixed reference cocycle $\beta_0$ — not functoriality of the quantum system (Heisenberg group, Hilbert space) itself, and not a claim that any compatible choice of polarizing cocycle along embeddings exists. Whether the Arf type of Theorem 4.7 can be chosen compatibly along field embeddings at $p = 2$ is recorded as an open question by this campaign's source shard, not resolved here.

## 9 Weyl–Heisenberg quantization over finite local rings

This section records the first extension from finite fields to finite commutative local rings. In the preferred three-stage pipeline (P1), the ring first produces the symplectic module $R^2$; a character and a polarizing cocycle then produce its Weyl algebra and Heisenberg group; finally the irreducible unitary representations are classified. The decisive condition is that the chosen character generate the character module. At odd residue characteristic this gives a single irreducible representation of dimension $|R|$.

There is a genuine fork at the first stage. Route A, used here, takes $R \oplus R$ with an $R$-valued alternating form and then composes with a chosen additive character. Route B would take $R \oplus \widehat{R}$ with its canonical circle-valued Pontryagin pairing; it needs no character to become nondegenerate. Route B and the comparison of the physics retained by the two routes are deferred; no result below ranks the routes.

## 9.1 The local datum and its phase space

**Definition 9.1** (Finite commutative local ring datum)**.** A finite local ring datum is $(R, \mathfrak{m}, \kappa, q)$ with $R$ a finite commutative unital local ring with $1 \neq 0$, $\mathfrak{m}$ its unique maximal ideal, $\kappa := R/\mathfrak{m}$, and $q := |\kappa|$; write $R^\times$ for its unit group. For an ideal $I \subseteq R$, put

$$\operatorname{Ann}(I) := \{r \in R : rI = 0\}, \qquad \operatorname{soc}(R) := \operatorname{Ann}(\mathfrak{m}).$$

These are stipulations. Nilpotence of $\mathfrak{m}$, nonvanishing of the socle, and the other finite-local structure properties are established below, not assumed here. The definition is uniform in the residue characteristic.

*Scope.* Only finite commutative unital local rings with $1 \neq 0$ are defined here. No Frobenius, characteristic, principal-ideal, or chain-ring hypothesis is part of the datum.

---

**Provenance.** ids: D12; proved in: —; tested by: —; reviewed in: theory/verdicts/fcr-local-r1.md.

**Definition 9.2** (Characters and generating characters)**.** Let

$$\widehat{R} := \operatorname{Hom}((R, +), \mathbb{C}^\times), \qquad X(R) := \widehat{R} \setminus \{1\}.$$

For $\psi \in X(R)$ and $u \in R$, put $\psi_u(x) := \psi(ux)$. Define

$$I_\psi := \sum \{I \subseteq R : I \text{ is an ideal and } I \subseteq \ker \psi\}, \qquad \operatorname{Gen}(R) := \{\psi \in X(R) : I_\psi = 0\}.$$

Thus $I_\psi$ is the largest ideal contained in $\ker \psi$, and a member of $\operatorname{Gen}(R)$ is a *generating character*. This definition is uniform in the residue characteristic; existence is not stipulated.

*Scope.* The definition includes every nontrivial additive character, not only a trace-normalized family. It makes no assertion that generating characters exist for an arbitrary finite local ring.

---

**Provenance.** ids: D13; proved in: —; tested by: —; reviewed in: theory/verdicts/fcr-local-r1.md.

**Definition 9.3** (Symplectic module and phase perpendicularity)**.** Put $V(R) := R \oplus R$ and

$$\omega\big((a, b), (a', b')\big) := ab' - a'b.$$

Given $\psi \in X(R)$, write

$$B_\psi(v, w) := \psi(\omega(v, w)), \qquad L^{\perp_\psi} := \{v \in V(R) : B_\psi(v, l) = 1 \text{ for every } l \in L\}$$

for an $R$-submodule $L \subseteq V(R)$. A *Lagrangian* is an $R$-submodule satisfying $L = L^{\perp_\psi}$, and $\operatorname{rad}(B_\psi) := V(R)^{\perp_\psi}$. Bilinearity, strong alternation, $R$-nondegeneracy of $\omega$, and nondegeneracy of $B_\psi$ are claims, not stipulations; the character-valued pairing $B_\psi$, not $R$-nondegeneracy alone, is load-bearing. This definition is uniform in the residue characteristic.

*Scope.* Perpendicularity is defined only for $R$-submodules of $R^2$ and is measured through the chosen character. No freeness condition is imposed on a Lagrangian.

---

**Provenance.** ids: D14; proved in: —; tested by: —; reviewed in: theory/verdicts/fcr-local-r1.md.

**Definition 9.4** (Admissible polarizing cocycles over a local ring)**.** For the symplectic module above, set

$$\begin{aligned}\mathrm{Adm}(\omega) &:= \{\beta : V(R) \times V(R) \to R : \beta \text{ is } R\text{-bilinear and } \beta - \beta^T = \omega\},\\ \mathrm{Sym}_R(V(R)) &:= \{s : V(R) \times V(R) \to R : s \text{ is } R\text{-bilinear and } s = s^T\}.\end{aligned}$$

A *polarizing cocycle* is a named choice $\beta \in \mathrm{Adm}(\omega)$. The reference member is

$$\beta_0\big((a,b),(a',b')\big) := ab'.$$

This is the nonsymmetrized convention: no inverse of 2 is assumed. The definition is uniform in the residue characteristic.

*Scope.* The reference member is a stipulated coordinate formula, not a claim that the torsor has a canonical point. The symmetrized member $\omega/2$ is used only under the explicit hypothesis $2 \in R^\times$.

---

**Provenance.** ids: D15; proved in: —; tested by: —; reviewed in: theory/verdicts/fcr-local-r1.md.

**Definition 9.5** (Local Weyl algebra, Heisenberg group, and models)**.** For the data above define

$$A_{\psi,\beta}(V(R)) := \bigoplus_{v \in V(R)} \mathbb{C}\, W_\beta(v), \qquad W_\beta(v)W_\beta(v') := \psi(\beta(v,v'))W_\beta(v+v'),$$

and

$$H_\beta(R) := R \times V(R), \qquad (t,v)(t',v') := (t+t'+\beta(v,v'), v+v').$$

For an $R$-submodule $L$, put $A_L := \mathrm{span}_{\mathbb{C}}\{W_\beta(l) : l \in L\}$. Given a unital algebra character $\chi : A_L \to \mathbb{C}$, put

$$M_{L,\chi} := A_{\psi,\beta}(V(R)) \otimes_{A_L} \mathbb{C}_\chi.$$

The fixed reference model is $\ell^2(R) = \bigoplus_{y \in R} \mathbb{C}e_y$, with $\{e_y\}$ orthonormal,

$$X(a)e_y := e_{y+a}, \qquad Z(b)e_y := \psi(by),$$

and, exactly,

$$W_{\beta_0}(a,b) := Z(-b)X(a), \qquad W_{\beta_0}(a,b)e_y = \psi(-b(y+a))e_{y+a}.$$

The sign and ordering are stipulations. Every displayed formula is uniform in the residue characteristic and uses no half.

*Scope.* The algebra and group depend on the named cocycle, while the algebra and representation also depend on the named character. A submodule and its character label a realization; neither is added to the underlying algebra datum.

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**Provenance.** ids: D16; proved in: —; tested by: theory/checks/fcr_local_check.py (G4); reviewed in: theory/verdicts/fcr-local-r1.md.

## 9.2 Generating characters and the collapse boundary

PROVED

**Theorem 9.6** (Local Frobenius criterion and the unit torsor)**.** *Let $R$ be a finite commutative local ring with maximal ideal $\mathfrak{m}$ and residue field $\kappa$. Then*

$$\mathrm{Gen}(R) \neq \varnothing \quad \Longleftrightarrow \quad \dim_\kappa \mathrm{soc}(R) = 1.$$

*If $\psi \in \mathrm{Gen}(R)$, then $\psi_u$ is generating exactly when $u \in R^\times$, and $R^\times$ acts freely and transitively on $\mathrm{Gen}(R)$.*

*Scope.* No odd-characteristic hypothesis is used. For rings whose socle has $\kappa$-dimension greater than one, the theorem asserts nonexistence of a generating character; it does not propose a replacement quantization.

---

**Provenance.** ids: D12, D13, FCR-GEN; proved in: theory/fcr-local.md §1; tested by: theory/checks/fcr_local_check.py (G1,G8); reviewed in: theory/verdicts/fcr-local-r1.md.

*Proof.* Finiteness makes $R$ Artinian, so $\mathfrak{m}$ is nilpotent. Wood's module socle $S({}_RR)$ equals $\mathrm{Ann}(\mathfrak{m})$: every simple ideal $J$ has $\mathfrak{m}J = 0$, since $\mathfrak{m}J = J$ would contradict nilpotence; conversely, if $0 \neq x \in \mathrm{Ann}(\mathfrak{m})$, then $Rx = \kappa x$ is one-dimensional over $\kappa$ and is simple. Wood's local Frobenius condition is therefore $\kappa \cong \mathrm{soc}(R)$.

Wood's character-module theorem and kernel-ideal criterion identify this condition with the existence of $\psi$ whose kernel contains no nonzero ideal, namely $I_\psi = 0$. For such a $\psi$, multiplication by a unit preserves $I_\psi = 0$, whereas a nonunit lies in $\mathfrak{m}$ and kills the nonzero socle. The character-module isomorphism then shows that every generating character is uniquely $\psi_u$ for a unit $u$. □

PROVED

**Theorem 9.7** (Radical of the character-valued symplectic pairing)**.** *Let $R$ be a finite commutative local ring and let $\psi \in X(R)$. The form $\omega$ on $R^2$ is $R$-bilinear, strongly alternating, and $R$-nondegenerate, while*

$$\mathrm{rad}(B_\psi) = I_\psi \oplus I_\psi.$$

*Consequently $B_\psi$ is nondegenerate if and only if $\psi \in \mathrm{Gen}(R)$.*

*For $R = \mathbb{F}_3[x,y]/(x,y)^2$, write $\psi_{a,b,c}(r + sx + ty) = \zeta_3^{ar+bs+ct}$. The two nontrivial characters with $(b,c) = (0,0)$ have $I_\psi = (x,y)$, and each of the remaining $24$ has $I_\psi = \mathbb{F}_3(cx - by)$. Thus all $26$ nontrivial characters collapse the pairing through a proper quotient.*

*Scope.* The radical formula is uniform in the residue characteristic. The square-zero-ring census records the obstruction at one designated seed only; it does not construct a quantum system for a degenerate pairing and does not state a general quotient-collapse theorem.

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**Provenance.** ids: D12, D13, D14, FCR-RAD; proved in: theory/fcr-local.md §2; tested by: theory/checks/fcr_local_check.py (G2,G3); exhaustive 26-character seed census in G1; reviewed in: theory/verdicts/fcr-local-r1.md.

*Proof.* Testing $\omega((a,b),-)$ against $(0,r)$ and $(r,0)$ shows that a radical vector must have $aR, bR \subseteq \ker\psi$, hence $a, b \in I_\psi$. The reverse inclusion follows because $I_\psi$ is an ideal. Bilinearity and strong alternation are immediate from the displayed determinant formula, and tests against $(0,1)$ and $(1,0)$ prove $R$-nondegeneracy. In the square-zero example every subspace of $(x,y)$ is an ideal; restricting the character to that two-dimensional space gives exactly the displayed kernel line, or all of $(x,y)$ when the restriction vanishes. □

### 9.3 Odd residue characteristic: algebra and representation

PROVED

**Theorem 9.8** (Weyl commutation over a finite local ring)**.** *Let $R$ be a finite commutative local ring, let $\psi \in X(R)$, and let $\beta \in \mathrm{Adm}(\omega)$. For every $v, v' \in V(R)$,*

$$W_\beta(v)W_\beta(v') = \psi(\omega(v, v'))W_\beta(v')W_\beta(v),$$

*and*

$$W_\beta(u)W_\beta(v)W_\beta(u)^{-1} = \psi(\omega(u, v))W_\beta(v).$$

*Scope.* The identities are uniform in the residue characteristic and do not require $\psi$ to be generating. They do not by themselves imply simplicity of the Weyl algebra.

---

**Provenance.** ids: D14, D15, D16, FCR-COMM; proved in: theory/fcr-local.md §3; tested by: theory/checks/fcr_local_check.py (G2,G4); reviewed in: theory/verdicts/fcr-local-r1.md.

*Proof.* The two products have the same basis term $W_\beta(v+v')$. Their scalar ratio is $\psi(\beta(v, v') - \beta(v', v)) = \psi(\omega(v, v'))$; conjugation gives the second identity. □

PROVED†

**Theorem 9.9** (Odd-characteristic transport of polarizing cocycles)**.** *Let $R$ be a finite commutative local ring with $2 \in R^\times$. Then $\mathrm{Adm}(\omega)$ is a torsor under $\mathrm{Sym}_R(V(R))$ and has $|R|^3$ elements. Its unique antisymmetric member is $\omega/2$. For every $\beta \in \mathrm{Adm}(\omega)$ and every symmetric $s$, the map*

$$\varphi_s(t, v) := \left(t + \frac{s(v, v)}{2}, v\right)$$

*is an isomorphism $H_\beta(R) \to H_{\beta+s}(R)$ fixing the centre pointwise.*

*Scope.* The condition $2 \in R^\times$ is essential to the displayed representative and transport. No symmetry group of the ring-side phase space is analyzed here.

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**Provenance.** ids: D14, D15, D16, FCR-BETA-ODD; proved in: theory/fcr-local.md §4; tested by: theory/checks/fcr_local_check.py (G9); reviewed in: theory/verdicts/fcr-local-r1.md.

*Proof.* The difference of two admissible forms is symmetric, and adding a symmetric form preserves admissibility. A symmetric form on $R^2$ is freely determined by three entries. Antisymmetry and admissibility give $2\beta = \omega$, hence the unique member $\omega/2$. Finally, for $f(v) = s(v, v)/2$, symmetry gives $f(v + w) = f(v) + f(w) + s(v, w)$, which is exactly the coboundary identity needed for $\varphi_s$ to respect the two group laws. □

PROVED†

**Theorem 9.10** (Central simplicity and finite Stone–von Neumann theorem)**.** *Let $R$ be a finite commutative local ring, let $\psi \in \mathrm{Gen}(R)$, let $\beta \in \mathrm{Adm}(\omega)$, and suppose $2 \in R^\times$. Then $A_{\psi,\beta}(V(R))$ is simple, has centre $\mathbb{C}1$, has complex dimension $|R|^2$, and is noncanonically isomorphic to $M_{|R|}(\mathbb{C})$. The group $H_\beta(R)$ has exactly one irreducible unitary representation with central character $\psi$, up to unitary equivalence; its dimension is $|R|$, unitary intertwiners are unique up to $U(1)$, and every complex algebra automorphism of the Weyl algebra is inner.*

*Scope.* The condition $2 \in R^\times$ is used to transport an arbitrary cocycle to the fixed matrix model. The matrix isomorphism depends on choices of realization and basis and is not part of the algebra datum. No frame-preserving automorphism group is computed.

**Provenance.** ids: D12–D16, FCR-ALG, FCR-SVN; proved in: theory/fcr-local.md §5–§6; tested by: theory/checks/fcr_local_check.py (G4,G5,G6), including the G6 arena mutation; reviewed in: theory/verdicts/fcr-local-r1.md.

*Proof.* Averaging conjugation by all Weyl basis elements kills every nonzero basis coefficient because the pairing is nondegenerate. It follows that the centre is scalar and that every nonzero two-sided ideal contains 1. The reference operators $Z(-b)X(a)$ give a nonzero representation on $\ell^2(R)$; simplicity makes it injective and the equal dimensions make it onto $\mathrm{End}_{\mathbb{C}}(\ell^2(R))$. The preceding transport handles arbitrary $\beta$.

Modules with central character $\psi$ are precisely modules over this full matrix algebra. Its column module is the unique simple module and has dimension $|R|$. Schur's argument makes intertwiners scalar; polar rescaling gives the $U(1)$ statement. Twisting the column module by an algebra automorphism shows that the automorphism is conjugation by an invertible operator. □

**PROVED** †

**Theorem 9.11** (Lagrangians and Schrödinger models over a local ring)**.** *Let $R$ be a finite commutative local ring, let $\psi \in \mathrm{Gen}(R)$, let $\beta \in \mathrm{Adm}(\omega)$, and suppose $2 \in R^\times$. Every Lagrangian $L \subseteq V(R)$ has $|L| = |R|$. The coordinate axes are Lagrangian, and for every ideal $I \subseteq R$ so is*

$$I \oplus \mathrm{Ann}(I).$$

*If $R$ is not a field, then*

$$L_* := \mathrm{soc}(R) \oplus \mathfrak{m}$$

*is a nonfree Lagrangian. Every algebra character $\chi : A_L \to \mathbb{C}$ produces a simple induced module $M_{L,\chi}$ of dimension $|R|$, and all these modules are isomorphic.*

*Scope.* This is a catalogue containing a general ideal family, not a classification of all Lagrangians over every local ring. In particular, it does not claim that every Lagrangian is free or belongs to the displayed ideal family.

**Provenance.** ids: D12–D16, FCR-POL; proved in: theory/fcr-local.md §7; tested by: theory/checks/fcr_local_check.py (G7), full submodule census at the two order-nine seeds; reviewed in: theory/verdicts/fcr-local-r1.md.

*Proof.* Character orthogonality in the two orders gives $|L|\,|L^{\perp_\psi}| = |R|^2$, so a self-perpendicular module has order $|R|$. Annihilator duality gives $\mathrm{Ann}(\mathrm{Ann}(I)) = I$, hence the ideal family. For a nonfield local ring, $\mathrm{Ann}(\mathfrak{m}) = \mathrm{soc}(R)$ and $\mathrm{Ann}(\mathrm{soc}(R)) = \mathfrak{m}$, so $L_*$ belongs to that family. Its quotient by $\mathfrak{m}L_*$ needs at least two generators, whereas a free module of order $|R|$ would have rank one.

The restriction of $\beta$ to a Lagrangian is symmetric, and $\chi_0(W(l)) = \psi(-\beta(l,l)/2)$ is an algebra character. The Weyl basis makes the algebra right-free over $A_L$ on representatives for $V(R)/L$, so the induced module has dimension $|R|$; uniqueness of the simple module then proves that all labels give isomorphic realizations. □

**PROVED** †

**Proposition 9.12** (The two presentation data and bare-algebra ambiguity)**.** *Let $R$ be a finite commutative local ring with $2 \in R^\times$ and $\mathrm{Gen}(R) \neq \varnothing$. Beyond $R$, the presentation uses exactly a generating character $\psi \in \mathrm{Gen}(R)$ and a cocycle $\beta \in \mathrm{Adm}(\omega)$. Distinct generating characters give inequivalent representations of the same group $H_\beta(R)$ because their central characters differ. All resulting bare algebras are isomorphic, but the torsor of bare-algebra isomorphisms has no point fixed by every target automorphism.*

*Scope.* The last assertion concerns all bare-algebra isomorphisms. It neither computes nor identifies a frame-preserving isomorphism torsor and makes no claim about a ring-side Weil or metaplectic symmetry.

**Provenance.** ids: D12–D16, FCR-CHOICE; proved in: theory/fcr-local.md §8; tested by: theory/checks/fcr_local_check.py (G8,G9); reviewed in: theory/verdicts/fcr-local-r1.md.

### 9.4 Clause-level return to an odd finite field

**PROVED** †

**Theorem 9.13** (Clause-level compatibility with the odd-field system)**.** *Let $R = \kappa$ be a finite field of odd characteristic and put $q := |\kappa|$. Specialization recovers on the nose:*

*(i) the nontrivial-character set and its $\kappa^\times$-torsor of order $q - 1$;*

*(ii) bilinearity, strong alternation, and nondegeneracy of $\omega$, together with the zero radical of $\psi \circ \omega$;*

*(iii) the Weyl commutator and conjugation identities;*

*(iv) simplicity, scalar centre, dimension $q^2$, and a noncanonical $M_q(\mathbb{C})$ realization;*

*(v) the unique irreducible unitary representation of dimension $q$, its $U(1)$-unique intertwiners, and innerness of algebra automorphisms;*

*(vi) exactly $q+1$ Lagrangian lines, $q$ algebra characters per line, $q(q+1)$ model labels, and one module-isomorphism class;*

*(vii) the torsor of admissible cocycles of order $q^3$, its unique antisymmetric member $\omega/2$, and the centre-fixing transports;*

*(viii) the two named presentation data, the distinct-central-character obstruction, and isomorphism of the resulting bare algebras.*

*The specialized definitions give the same $V(\kappa)$, form, admissibility equation, Weyl and group laws, and fixed operator $W_{\beta_0} = Z(-b)X(a)$ with the same sign.*

*Scope.* This is deliberately a clause-level comparison. It does not compare the field theory's frame-preserving isomorphism torsor of order $q^2|SL_2(\kappa)|$; the ring-side frame-preserving analysis is future work. It also does not compare the field theorem's cocycle or automorphism-group clauses, the $SL_2(\kappa)$-invariance of $\omega/2$, or the particular polarization-dependent matrix construction used in the field proof.

**Provenance.** ids: D12–D16, FCR-GEN, FCR-RAD, FCR-COMM, FCR-BETA-ODD, FCR-ALG, FCR-SVN, FCR-POL, FCR-CHOICE, FCR-REG; proved in: theory/fcr-local.md §9; tested by: theory/checks/fcr_local_check.py (G10), exact counts, form and radical tables, fixed matrices, rank, commutant, lines, half-census, and group profile; reviewed in: theory/verdicts/fcr-local-r1.md.

*Proof.* For a field, $\mathfrak{m} = 0$, $\mathrm{soc}(R) = R$, and the only ideals are 0 and $R$, so every nontrivial character is generating. Substitution in the preceding results proves every listed clause. A self-perpendicular subspace of the nondegenerate two-dimensional space is a line; nonzero vectors modulo $\kappa^\times$ give $(q^2 - 1)/(q - 1) = q + 1$ lines, and each line has $q$ additive characters. No excluded symmetry or frame clause follows from these calculations. $\square$

# 10 The five smallest local rings, worked in full

The three smallest orders of finite commutative local rings are 2, 3, and 4, and order 4 is a three-way tie. This section exercises every definition of Section 9 on the resulting five rings

$$\mathbb{F}_2, \qquad \mathbb{F}_3, \qquad \mathbb{F}_4, \qquad \mathbb{Z}/4, \qquad \mathbb{F}_2[\varepsilon]/(\varepsilon^2),$$

computing their Weyl–Heisenberg groups and complete irreducible representation catalogues. Every number in this section is verified by the exact-arithmetic census checker `theory/checks/small_rings_cat` written and run by an independent verifier lane that recomputed the orchestrator's catalogue blind; the checker runs green and carries eight mutation modes, each of which fails at its registered gate. Statements here are census facts about these five rings; no generality beyond them is claimed except where a claim identifier is cited.

## 10.1 The data, ring by ring

All five rings are Frobenius: the socle is one-dimensional over the residue field, so generating characters exist and form a single free orbit of the unit group (the existence criterion and torsor structure of Section 9).

| $R$ | $\mathfrak{m}$ | $\kappa$ | $\operatorname{soc}(R)$ | $\lvert R^\times\rvert = \lvert\operatorname{Gen}(R)\rvert$ | reference $\psi$ |
|---|---|---|---|---|---|
| $\mathbb{F}_2$ | $0$ | $\mathbb{F}_2$ | $R$ | 1 | $(-1)^x$ |
| $\mathbb{F}_3$ | $0$ | $\mathbb{F}_3$ | $R$ | 2 | $\zeta_3^x$ |
| $\mathbb{F}_4$ | $0$ | $\mathbb{F}_4$ | $R$ | 3 | $(-1)^{\operatorname{Tr}(x)}$ |
| $\mathbb{Z}/4$ | $(2)$ | $\mathbb{F}_2$ | $(2)$ | 2 | $i^x$ |
| $\mathbb{F}_2[\varepsilon]$ | $(\varepsilon)$ | $\mathbb{F}_2$ | $(\varepsilon)$ | 2 | $(-1)^{a+b}$ for $x = a + b\varepsilon$ |

**Provenance.** ids: FCR-GEN; proved in: theory/fcr-local.md §1; tested by: small_rings_catalogue_check.py (ring-data gates); reviewed in: theory/verdicts/smallest-rings-verification.md.

## 10.2 The Heisenberg groups are unitriangular matrix groups

**Proposition 10.1** (Matrix form of the reference-cocycle Heisenberg group)**.** *For every finite commutative unital ring $R$, the map*

$$(t,(a,b)) \longmapsto \begin{pmatrix} 1 & a & t \\ 0 & 1 & b \\ 0 & 0 & 1 \end{pmatrix}$$

*is an isomorphism from the Heisenberg group of the reference cocycle $\beta_0((a,b),(a',b')) = ab'$ onto the unitriangular group $\mathrm{UT}_3(R)$.*

*Proof.* Multiplying the two displayed matrices gives upper-right entry $t+t'+ab'$, which is exactly the group law of the Heisenberg group of $\beta_0$; the map is a bijection by construction. □

*Scope.* The identification is specific to the reference cocycle: at residue characteristic 2 other admissible cocycles can give non-isomorphic groups (the quaternion type over $\mathbb{F}_2$ below), and those are not unitriangular in this way.

**Provenance.** ids: D16; proved in: this section; tested by: small_rings_catalogue_check.py (cocycle gate); reviewed in: theory/verdicts/smallest-rings-verification.md.

The five groups, exactly:

| $R$ | $\lvert H\rvert$ | centre | exponent | order profile | identification |
|---|---|---|---|---|---|
| $\mathbb{F}_2$ | 8 | $\mathbb{Z}/2$ | 4 | 1:1, 2:5, 4:2 | dihedral $D_4$ |
| $\mathbb{F}_3$ | 27 | $\mathbb{Z}/3$ | 3 | 1:1, 3:26 | extraspecial $3^{1+2}_+$ |
| $\mathbb{F}_4$ | 64 | $(\mathbb{Z}/2)^2$ | 4 | 1:1, 2:27, 4:36 | characteristic-two Heisenberg |
| $\mathbb{Z}/4$ | 64 | $\mathbb{Z}/4$ | 8 | 1:1, 2:7, 4:40, 8:16 | $\mathrm{UT}_3(\mathbb{Z}/4)$ |
| $\mathbb{F}_2[\varepsilon]$ | 64 | $(\mathbb{Z}/2)^2$ | 4 | 1:1, 2:31, 4:32 | $\mathrm{UT}_3(\mathbb{F}_2[\varepsilon])$ |

The three order-64 groups are pairwise non-isomorphic: $\mathbb{Z}/4$ is separated by its cyclic centre and its sixteen elements of order 8, while $\mathbb{F}_4$ and $\mathbb{F}_2[\varepsilon]$ share centre and exponent but differ in involution count. The order-8 elements over $\mathbb{Z}/4$ come from the square law $(t, v)^2 = (2t + \beta_0(v, v),\, 2v)$: the fourth power is $(2\beta_0(v, v), 0)$, nonzero exactly when $ab$ is odd. Over $\mathbb{F}_2[\varepsilon]$ the identity $2v = 0$ kills this mechanism: the jet direction deepens the ring without deepening the clock. Mixed versus equal characteristic is thus visible in the bare group.

*Remark.* At $\mathbb{F}_3$ the group is independent of the cocycle choice (the odd-characteristic transport of Section 9). At $\mathbb{F}_2$ the eight admissible cocycles split six-to-two into a dihedral type and a quaternion type, separated by the Arf invariant of $Q_\beta(v) = \beta(v, v)$; the reference cocycle is of dihedral type. For $\mathbb{Z}/4$ and $\mathbb{F}_2[\varepsilon]$ the classification of the 64-element cocycle torsor is open and is the target of the running residue-characteristic-two increment.

**Provenance.** ids: WH-BETA-d, FCR-BETA-ODD; proved in: theory/wh-kappa-choice.md §6; theory/fcr-local.md §4; tested by: beta_census.py; small_rings_catalogue_check.py; reviewed in: theory/verdicts/wh-kappa-adjudication.md.

### 10.3 The complete representation catalogues

The commutator subgroup is the full centre, so the abelianization is the phase module $R \oplus R$, and irreducible representations are stratified by the character of the centre, $\psi_u = \psi(u\,\cdot)$ for $u \in R$.

**Proposition 10.2** (Catalogue census for the five smallest local rings)**.** *For each of the five rings above and each $u \in R$, the number of irreducible unitary representations of the Heisenberg group with central character $\psi_u$ is $\lvert \operatorname{Ann}(u)\rvert^2$, and each has dimension $\lvert R\rvert/\lvert \operatorname{Ann}(u)\rvert$. In particular the total count of irreducibles equals the number of conjugacy classes,*

$$\#\{\textit{classes}\} = \lvert R\rvert^2 + \sum_{u\neq 0} \lvert \operatorname{Ann}(u)\rvert^2 \qquad (5,\ 11,\ 19,\ 22,\ 22 \textit{ respectively}),$$

*and the sum of squared dimensions equals $\lvert R\rvert^3$.*

*Scope.* A census statement about these five rings only, established by exhaustive computation; the same stratification pattern for a general finite commutative local ring is expected but not claimed here — its general proof belongs to the planned non-Frobenius/collapse increment.

**Provenance.** ids: (census); proved in: small_rings_catalogue_check.py (class-count and catalogue gates); tested by: same checker, mutation modes red-class-count, red-catalogue; reviewed in: theory/verdicts/smallest-rings-verification.md.

The three strata carry distinct physics:

- *Bottom, $u = 0$:* the $\lvert R\rvert^2$ characters of the abelianization — the classical shadow.
- *Top, $u \in R^\times$:* one Schrödinger representation per generating character, of dimension $\lvert R\rvert$, in the fixed model $W(a, b) = Z(-b)X(a)$ on $\ell^2(R)$: the qubit Pauli pair for $\mathbb{F}_2$; the qutrit clock–shift for $\mathbb{F}_3$; two $\mathbb{F}_4$-labelled qubits (three inequivalent copies, one per generating character); the ququart clock–shift $Z = \mathrm{diag}(1, i, -1, -i)$, $ZX = iXZ$ for $\mathbb{Z}/4$ (two copies); and the jet-paired four-dimensional representation for $\mathbb{F}_2[\varepsilon]$ (two copies). Irreducibility is pinned by exact trace-orthogonality of the $\lvert R\rvert^2$ Weyl operators.

- *Middle,* $u \in \mathfrak{m} \setminus 0$ (only the two thickened rings): four two-dimensional representations each, with central character supported on the socle. These are qubit representations of the collapsed quotient $R/\operatorname{Ann}(u) \cong \mathbb{F}_2$ twisted by characters of the radical of the degenerate phase pairing. The verifier's decomposition of the four-dimensional model with this central character found exactly two inequivalent two-dimensional blocks, each of multiplicity one — the model contains two of the four classes.

A ring is thickened exactly when its catalogue has a middle stratum: the representation theory detects the nilpotents. Two honest cautions from the blind verification: $\mathbb{F}_4$ has *three* top-stratum representations, not two; and as bare dimension–multiplicity tables the $\mathbb{Z}/4$ and $\mathbb{F}_2[\varepsilon]$ catalogues coincide ($16 \times \dim 1$, $4 \times \dim 2$, $2 \times \dim 4$) — the catalogue separates the two rings only through its central-character labelling and the underlying groups, not through the numbers alone.

### 10.4 Lagrangian labels and models

| $R$ | Lagrangians | free | non-free | model labels |
|---|---|---|---|---|
| $\mathbb{F}_2$ | 3 | 3 | 0 | 6 |
| $\mathbb{F}_3$ | 4 | 4 | 0 | 12 |
| $\mathbb{F}_4$ | 5 | 5 | 0 | 20 |
| $\mathbb{Z}/4$ | 7 | 6 | 1 | 28 |
| $\mathbb{F}_2[\varepsilon]$ | 7 | 6 | 1 | 28 |

The fields show the $q + 1$ lines of the projective line; the thickened rings show the six free lines of $\mathbb{P}^1(R)$ plus the non-free witness $\operatorname{soc}(R) \oplus \mathfrak{m}$ of Section 9. All labels with a fixed generating central character present one representation, but the non-free label is a genuinely different presentation: position and momentum are each sampled at depth $\mathfrak{m}$, a mixed frame with no field analogue.

**Provenance.** ids: FCR-POL; proved in: theory/fcr-local.md §7; tested by: small_rings_catalogue_check.py (Lagrangian gate, mutation mode red-drop-nonfree); reviewed in: theory/verdicts/fcr-local-adjudication.md.

# 11 Sidequest: quantum mechanics over the field with one element

**Opened 6 September 2026.** This is a separate, literature-grounded exploration of arithmetic quantum mechanics over $\mathbb{F}_1$, also written $\mathbb{F}_{\mathrm{un}}$. Its north star is a precisely formulated quantum theory, with particular attention to the analogue of a qubit and to composition. The different established analogies are retained as separate candidates.

**North-star clarification, 7 September 2026.** The primary object is the family of subsystems and their assembly rules. An acceptable endpoint must admit a quantum operational realization by C*-algebras, normalized positive functionals or density operators, CP dynamics and Born probabilities. The proposed strategy extracts a compositional category that remembers the prime before formulating its field-with-one-element counterpart. Fibonacci is a test of the kind of composition sought, not an asserted endpoint. A strong monoidal fibre functor to ordinary Hilbert spaces is not a universal requirement: composite observable algebras may contain additional fusion observables. Compatible preparations and extensions of channels to composites are part of the target. Classical or combinatorial state sets alone do not satisfy this operational criterion.

The accompanying research map is `docs/sidequests/f1-qm.md`; primary source bodies, versions and retrieval hashes are indexed in `refs/LEDGER.md`. The finite-ring campaign is not superseded.

There is substantial relevant literature, but the constructions preserve different structures. A field may specify the arithmetic base, the state coefficients, a counting parameter, or a deformation parameter. Confusing these roles would make the comparison misleading. In the present campaign, the arithmetic base is a finite field or ring and the amplitudes remain complex. The direct "quantum $\mathbb{F}_{\mathrm{un}}$" papers instead change the coefficient object of the states themselves.

## 11.1 What could count as a qubit?

Two distinguishable states, a two-dimensional Hilbert space, a simple object of categorical dimension two, and two simple objects of a category are different notions. The following table makes the distinction explicit.

| Approach | Qubit analogue | Composition |
|---|---|---|
| Pointed $\mathbb{F}_1$ modules | A basepoint and two basis points. Complex linearization of normal partial maps gives $(\mathbb{C}^2, M_2(\mathbb{C}))$. | Smash product has four nonzero basis points and realizes as the Hilbert tensor product. Native basis states have no superpositions. |
| Cyclotomic modules | Two free $\mu_N$-orbits. For configuration $C_2$ and phase level two, clock and shift give the Heisenberg qubit. | Balanced smash of modules; central product of Heisenberg groups; ordinary tensor after complex realization. |
| Absolute Cartesian frames | Rank two has $N+2$ projective rays. At bare $\mathbb{F}_1$ these are $10, 01, 11$. | The rank-four frame has fifteen rays at $N=1$: nine product rays and six non-product rays. The operation on pairs is the Kronecker rule. |
| Thin symplectic geometry | Two coordinate polarizations, suggestive of position and momentum. These are not two Hilbert-state vectors. | Orthogonal sums of phase geometry; Hilbert tensor requires a realization. |

| Approach | Qubit analogue | Composition |
|---|---|---|
| Blueprint Heisenberg geometry | A chosen $\mathbb{F}_2$ fibre gives the reference order-eight group and its two-dimensional central-character module. | Ring realization and phase data precede the physical tensor product. |
| Bands and crowds | A relational Heisenberg object, with 27 signed or eight Krasner points. Neither number is a count of qubit states. | Geometric products exist; physical composition needs a specified phase and representation construction. |
| Hall/groupoid oscillator | No preferred qubit in one oscillator. Two colours give a two-dimensional one-particle sector after realization. | Full Fock spaces tensor on disjoint colour sets. Two identical bosons in two modes have dimension three; two distinguishable qubits have dimension four. |
| Representation categories | The two-dimensional simple object of $\mathrm{Rep}(D_8)$ or $\mathrm{Rep}(Q_8)$. | Internal fusion splits its square into four charges. Independent copies instead use an external tensor and a matching-central-character quotient. |
| Quantum torus | $VU = -UV$, $U^2 = V^2 = 1$ gives $M_2(\mathbb{C})$. | Two selected central fibres give $M_4(\mathbb{C})$. |
| Arithmetic descent and Bost–Connes | A specified finite fibre or a chosen two-level encoding; no canonical qubit is supplied by the global arithmetic data. | Frobenius and transfer correspondences; tensor products of statistical systems after realization. |
| Tropical and motivic routes | Support or incidence information, without a specified positive two-state quantum system. | The relevant semiring, geometric or categorical tensor must be named. |

## 11.2 The developed geometric and physical analogies

**Counting and thin geometry.** The projective count $1 + q + \cdots + q^{r-1}$ specializes to $r$ at $q = 1$; Gaussian coefficients specialize to subset counts. Tits's thin geometry therefore retains coordinate points, subsets and permutation symmetries. For symplectic rank $r$ the reflection group is $W(C_r) = (C_2)^r \rtimes S_r$, acting on symplectic pairs. A coordinate Lagrangian selects one direction from each pair, giving $2^r$ choices. This is the incidence content of the familiar specialization of $\prod_{i=1}^{r}(q^i + 1)$. The geometric analogies are developed in Deitmar's *Schemes over* $\mathbb{F}_1$, section 5, and Lorscheid's Tits–Weyl models. Here a *Weyl group* is a reflection group, not the group of Weyl displacement operators. Moreover, $q^3 \mapsto 1$ for the order of $\mathrm{UT}_3(\mathbb{F}_q)$ is a polynomial specialization, not a limit of groups or a proof that its $\mathbb{F}_1$ geometry is empty.

**Direct quantum $\mathbb{F}_{\mathrm{un}}$.** Chang, Lewis, Minic and Takeuchi construct a $q = 1$ coordinate-state specialization of finite-field amplitude models: superposition disappears. This is different from this campaign's complex Hilbert space $\ell^2(\mathbb{F}_q)$. Thas's *Absolute Quantum Theory* uses larger Cartesian frames and develops monomial dynamics and quantum-information analogies. Its Hermitian expression is partial: a sum is allowed only when at most one term is nonzero. Orthogonality is disjointness of supports. The involution determines the unitary subgroup; inversion of phases makes all phase-permutation matrices unitary in this sense. Its cloning discussion distinguishes simple projective rays, which can be copied, from all frame states. These statements do not supply an ordinary positive-definite Hilbert inner product on the frame.

**An existing geometric Heisenberg model.** Lorscheid's Proposition 5.5 constructs Tits–Weyl models for standard parabolic unipotent radicals of $\mathrm{GL}_n$. For the upper-triangular Borel of $\mathrm{GL}_3$, this is the reference Heisenberg group

$$h(a,b,t) = \begin{pmatrix} 1 & a & t \\ 0 & 1 & b \\ 0 & 0 & 1 \end{pmatrix}, \qquad h(a,b,t)h(c,d,u) = h(a+c, b+d, t+u+ad).$$

Its underlying blue scheme is affine three-space. Its Weyl extension has one point, while ring base extension recovers the full unipotent group. Thus a positive $\mathbb{F}_1$ model already exists; selecting a central character and constructing its representation are further steps. The whole polarizing-cocycle torsor, especially its characteristic-two types, is not supplied by that reference model. Likewise, a Tits lift of a reflection is not automatically a metaplectic or Weil lift.

**Cyclotomic conventions.** For the raw phase monoid $\{0\} \cup \mu_N$, integral realization gives $\mathbb{Z}[\mu_N] \simeq \mathbb{Z}[t]/(t^N - 1)$. A cyclotomic blueprint may instead impose additive relations giving $\mathbb{Z}[\zeta_N]$. A faithful complex phase character is another named realization datum. These conventions are compared in Lorscheid's *A blueprinted view on $\mathbb{F}_1$-geometry*, section 1.1.4. The group $\mu_{q-1}$ reflects multiplicative field data; finite-field Weyl phases come from additive characters and lie in $\mu_p$. These two phase counts must not be identified.

## 11.3 Pointed modules, ordinary qubits and absolute frames

**Definition 11.1** (Normal pointed maps and their two products)**.** For $r \geq 0$ let $V_r = \{*, 1, \dots, r\}$. A normal pointed map preserves $*$ and has at most one preimage of each non-basepoint; it is a partial injection of the nonzero sets. Wedge identifies basepoints, while smash collapses all pairs containing a basepoint in the Cartesian product. Put $L(S) = \mathbb{C}[S \setminus \{*\}]$, with its basis orthonormal, and let $B(S)$ be the complex span of realized normal endomorphisms in $\mathrm{End}_{\mathbb{C}} L(S)$.

*Scope.* These objects have no native addition of vectors. The image algebra $B(S)$ is explicitly a complex realization.

**Provenance.** ids: D1010; proved in: definitions.md; tested by: f1_check.py (M13); reviewed in: supporting comparison.

**Proposition 11.2** (The pointed-set qubit after operator realization)**.** SKETCH *For every positive-rank finite pointed set $S$, $B(S) = \mathrm{End}_{\mathbb{C}} L(S)$. For finite pointed sets $S, T$,*

$$L(S \wedge T) \simeq L(S) \otimes L(T).$$

*For positive ranks this identifies the image algebras with their tensor product. In particular, $V_2$ realizes to $(\mathbb{C}^2, M_2(\mathbb{C}))$, and $V_2 \wedge V_2 = V_4$ realizes to two qubits.*

*Scope.* The realization retains a distinguished basis. The concrete image algebra is not the unreduced partial-injection monoid algebra. Algebras are transported along bijections; arbitrary partial maps do not give unital maps between full endomorphism algebras.

**Provenance.** ids: F1-NORMAL; proved in: f1-comparisons.md, section 3; tested by: f1_check.py (M13); reviewed in: supporting SKETCH.

*Proof.* The partial map taking $j$ to $i$ and everything else to the basepoint realizes as $E_{ij}$. These matrix units span the full algebra, and their partial inverses realize adjoints. Smash products of these maps give $E_{ij} \otimes E_{k\ell}$. The basis identification proves the tensor claim. □

At rank two the partial-injection monoid has seven elements, including zero. Its contracted monoid algebra has dimension six, whereas its image has dimension four: linear realization imposes relations such as $I = E_{00} + E_{11}$. Keeping only the two bijections gives the commutative span of $I$ and the swap. Thus the class of allowed morphisms matters. The pointed-module, normal-map and base-change conventions come from Szczesny, pages 5–8.

**Definition 11.3** (Cyclotomic projective Cartesian frames)**.** Fix an abstract cyclic group $\mu_N$ of order $N \geq 1$ and put $F_N = \{0\} \sqcup \mu_N$, with absorbing zero. For $r \geq 1$, define

$$P_N(r) = \big(F_N^r \setminus \{(0, \ldots, 0)\}\big)/\mu_N$$

by simultaneous scalar multiplication. The product-state map $P_N(r) \times P_N(s) \to P_N(rs)$ sends $([x], [y])$ to $[(x_i y_j)_{i,j}]$. Rays outside its image are called nonproduct rays in this factorization sense.

*Scope.* A Cartesian frame is different from a free pointed module. No probability rule or total Hermitian form is stipulated.

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**Provenance.** ids: D1009; proved in: definitions.md; tested by: f1_check.py (M11); reviewed in: supporting comparison.

**Proposition 11.4** (Frame qubits and their nonproduct rays)**.** SKETCH *For every $N \geq 1$,*

$$|P_N(r)| = \frac{(N+1)^r - 1}{N}.$$

*The rank-two product-state map is injective, so the rank-four frame has $(N+2)^2$ product rays and $N(N+1)(N+2)$ nonproduct rays.*

*Scope.* Nonproduct means failure of tensor factorization. It does not by itself establish a Born rule, interference or Bell correlations.

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**Provenance.** ids: F1-FRAME; proved in: f1-comparisons.md, section 2; tested by: f1_check.py (M11); reviewed in: supporting SKETCH.

*Proof.* Scalar multiplication acts freely on any nonzero tuple. For a nonzero product array, ratios within a nonzero row and column recover both factors up to scalars. The ray counts then follow by subtraction. $\square$

For $N = 1$ the one-system rays are $10, 01, 11$; the composite has fifteen rays, nine product and six nonproduct. The diagonal array $\left(\begin{smallmatrix} 1 & 0 \\ 0 & 1 \end{smallmatrix}\right)$ is a nonproduct because product supports are rectangular. This larger frame is the reason a nonadditive theory can retain a factorization notion of entanglement while the coordinate-only model does not. In the strict pointed-map category, only simple vectors are morphisms from the unit; the additional frame states require an explicitly enlarged state notion.

## 11.4 A finite cyclotomic Weyl–Heisenberg formulation

**Definition 11.5** (Cyclotomic modules and named complex realization)**.** Fix $N \geq 1$, an abstract cyclic group $\mu_N$ of order $N$, and a named faithful character $\iota : \mu_N \to \mathbb{C}^\times$. Let $\mathrm{Free}_*^{\mu_N}$ be finite pointed sets with a $\mu_N$ action fixing the basepoint and free on its complement; maps are pointed equivariant maps. Define

$$\mathbb{C}_\iota(S) = \mathbb{C}[S]/(e_0,\ e_{us} - \iota(u)e_s)$$

and use the induced linear maps on morphisms.

*Scope.* The phase embedding is part of the realization datum. The objects are free phase sets, not Cartesian frames.

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**Provenance.** ids: D1001; proved in: definitions.md; tested by: f1_check.py; reviewed in: f1-r1.md.

**Definition 11.6** (Hyperbolic phase data and the pointed Schrödinger action)**.** For a finite abelian group $A$ with exponent dividing $N$, put

$$A^\vee = \mathrm{Hom}(A, \mu_N), \quad V_A = A \times A^\vee, \quad c_A((a,\chi),(b,\eta)) = \eta(a)^{-1},$$
$$\kappa_A((a,\chi),(b,\eta)) = \chi(b)\eta(a)^{-1}.$$

Define $H_N(A) = \mu_N \times V_A$ by $(u,v)(u',w) = (uu'c_A(v,w), v+w)$ and adjoin an absorbing zero to get $H_N(A)^0$. Put $S_A = \{0\} \sqcup (\mu_N \times A)$ and stipulate

$$(u,a,\chi)[z,x] = [uz\chi(x+a), x+a], \qquad 0_H s = 0_S.$$

On its realization let $X(a)e_x = e_{x+a}$, $Z(\chi)e_x = \iota(\chi(x))e_x$, and $W(a,\chi) = Z(\chi)X(a)$.

*Scope.* The abelian configuration group is extra data. No field structure, division by two or identification $A \simeq A^\vee$ is assumed.

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**Provenance.** ids: D1002,D1003; proved in: definitions.md; tested by: f1_check.py (M1–M3); reviewed in: f1-r1.md.

**Theorem 11.7** (The finite cyclotomic Schrödinger system)**.** **PROVED** *Fix $N \geq 1$, cyclic $\mu_N$ of order $N$, a faithful $\iota : \mu_N \to \mathbb{C}^\times$, and a finite abelian $A$ of exponent dividing $N$. For the associated phase system, $|A^\vee| = |A|$, characters extend from subgroups and separate points, and $\kappa_A$ is perfect. The displayed multiplication makes $H_N(A)$ a group with centre exactly $\mu_N$, acting faithfully on $S_A$. After the named complex realization,*

$$W(a,\chi)W(b,\eta) = \iota(\eta(a)^{-1})W(a+b,\chi\eta), \qquad \mathrm{Tr}(W(v)^*W(w)) = |A|\delta_{v,w}.$$

*These operators span* $\mathrm{End}_{\mathbb{C}}(\mathbb{C}[A])$*. The irreducible unitary representation of $H_N(A)$ with central character $\iota$ is unique up to equivalence, has dimension $|A|$, and has intertwiners unique up to $U(1)$.*

*Scope.* This is finite Heisenberg theory for abelian configurations, with an explicit pointed module. A universal representation theorem for all proposed $\mathbb{F}_1$ geometries is not asserted.

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**Provenance.** ids: F1-DUAL,F1-WEYL,F1-REAL; proved in: f1-cyclotomic.md, sections 0–2; tested by: f1_check.py (M1–M3); reviewed in: f1-adjudication.md.

*Proof.* Character extension follows by adjoining a cyclic generator and choosing a root in $\mu_N$; the exponent hypothesis ensures the required root exists. Restriction and induction on the group order give $|A^\vee| = |A|$. A nontrivial character sums to zero by translating its sum. Character evaluation verifies the cocycle identity and action. Testing the two coordinate subgroups proves perfection. The trace vanishes unless both translation and character are trivial; the resulting $|A|^2$ orthogonal operators form a basis of the full matrix algebra. Matrix units then prove uniqueness of its simple module and scalar uniqueness of intertwiners. □

**Definition 11.8** (Fixed-level isomorphisms and balanced composition)**.** Let $\mathrm{FinAb}_N^{\simeq}$ have the above configurations and their group isomorphisms. For $f : A \to B$, put $f_*\chi = \chi \circ f^{-1}$, $S(f)[u,x] = [u, f(x)]$ and $H(f)(u,a,\chi) = (u, f(a), f_*\chi)$. Define

$$S \wedge_{\mu_N} T = (S \wedge T)/([us,t] \sim [s,ut]), \qquad H \boxdot K = (H \times K)/\{(u,u^{-1}) : u \in \mu_N\}.$$

In the last expression the two groups have specified central copies of $\mu_N$.

*Scope.* Functoriality is asserted on isomorphisms at a common phase level. Arbitrary configuration homomorphisms are not silently included.

---

**Provenance.** ids: D1004; proved in: definitions.md; tested by: f1_check.py (M6); reviewed in: f1-r1.md.

**Theorem 11.9** (Composition of independent phase systems). **PROVED** *Fix $N \geq 1$ and a faithful phase realization $\iota$. The module, group and Hilbert assignments on* $\mathrm{FinAb}_N^{\simeq}$ *are strong symmetric monoidal, with comparisons*

$$S_A \wedge_{\mu_N} S_B \simeq S_{A\times B}, \qquad H_N(A) \boxdot H_N(B) \simeq H_N(A \times B),$$
$$\mathbb{C}[A] \otimes \mathbb{C}[B] \simeq \mathbb{C}[A \times B].$$

*At $N = 1$ the exponent condition forces $A = 0$ and a one-dimensional phase-system realization, while free pointed modules of arbitrary rank still exist.*

*Scope.* The group tensor is a central quotient, not a Cartesian product retaining two independent centres. The level-one statement is restricted to perfect phase systems, not all $\mathbb{F}_1$ quantum routes.

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**Provenance.** ids: F1-FUNCT,F1-ONE; proved in: f1-cyclotomic.md, sections 3,5; tested by: f1_check.py (M1,M6); reviewed in: f1-adjudication.md.

*Proof.* The phase-set comparison sends $[[u, a], [v, b]]$ to $[uv, (a, b)]$. The corresponding group map multiplies central phases and takes the external character. Its kernel is the antidiagonal centre. Every associativity diagram multiplies the same phases and associates the same tuple, proving coherence. The Hilbert comparison is the basis map. □

**Definition 11.10** (Central pushout from a finite local ring). Let $R$ be finite commutative unital local with $1 \neq 0$, and let $\psi : (R, +) \to \mu_N$ be named, with exponent of $(R, +)$ dividing $N$. Assume $\iota\psi$ is generating, meaning its kernel contains no nonzero ideal. Let $H_{\beta_0}(R)$ have triples $(t, a, b)$ and product $(t + u + ad, a + c, b + d)$. Put

$$\chi_b(x) = \psi(-bx), \qquad P_\psi = \big(\mu_N \times H_{\beta_0}(R)\big)/\{(\psi(t)^{-1}, (t, 0, 0)) : t \in R\}.$$

*Scope.* A generating character need not be faithful on the additive group. The raw Heisenberg group and its phase image are distinguished.

---

**Provenance.** ids: D1005,D12,D13,D16; proved in: definitions.md; tested by: f1_check.py (M4,M5); reviewed in: f1-r1.md.

**Theorem 11.11** (Exact recovery of the arithmetic reference convention). **PROVED** *Let $R$ be a finite commutative unital local ring with $1 \neq 0$, exponent of $(R, +)$ dividing $N \geq 1$, and a named $\psi : (R, +) \to \mu_N$ such that the faithful complex realization $\iota\psi$ is generating. Then $b \mapsto \chi_b$ identifies $(R, +)$ with its phase dual and $P_\psi \simeq H_N((R, +))$ by $[u; (t, a, b)] \mapsto (u\psi(t), a, \chi_b)$. The raw group map has central kernel* $\ker\psi$. *Its realized operators are*

$$W(a, \chi_b)e_y = (\iota\psi)(-b(y + a))e_{y+a} = Z(-b)X(a)e_y,$$

*with product cocycle $(\iota\psi)(ab')$, uniformly including characteristic two.*

*Scope.* Only the reference cocycle is recovered here. This does not identify all raw polarizing-cocycle groups with one phase group.

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**Provenance.** ids: F1-RING; proved in: f1-cyclotomic.md, section 4; tested by: f1_check.py (M4,M5); reviewed in: f1-adjudication.md.

For example, over $\mathbb{F}_4$ a nontrivial trace character has a two-element kernel. Its raw Heisenberg group has order 64 and its phase image order 32. Over $\mathbb{F}_2$, the order-eight reference group acts on the usual qubit. This supplies a checked bridge to the main campaign.

## 11.5 Fourier transform and a precisely scoped additive extension

**Definition 11.12** (Quadratic phases, strict normalizer and Fourier kernel)**.** A quadratic phase is $\theta : A \to \mu_N$, $\theta(0) = 1$, for which $B_\theta(a, x) = \theta(a + x)\theta(a)^{-1}\theta(x)^{-1}$ is a bicharacter. For $t \in A$, $f \in \mathrm{Aut}(A)$ put $T_{t,f,\theta}[u, x] = [u\theta(x), f(x) + t]$. The strict normalizer is the normalizer of the $\mu_N$-level Weyl group in $\mathrm{Aut}_{\mathrm{Free}_*^{\mu_N}}(S_A)$. Define

$$F_A e_x = \sum_{\rho \in A^\vee} \iota(\rho(x))e_\rho, \quad \mathrm{ev}_a(\rho) = \rho(a), \quad J_A(a, \chi) = (\chi^{-1}, \mathrm{ev}_a), \quad r_A(a, \chi) = \chi(a).$$

*Scope.* A self-Fourier endomorphism additionally needs a pairing identifying $A$ with its dual. No such pairing is suppressed.

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**Provenance.** ids: D1006; proved in: definitions.md; tested by: f1_check.py (M7); reviewed in: f1-r1.md.

**Theorem 11.13** (What remains monomial and what Fourier requires)**.** PROVED *Fix $N \geq 1$, a faithful $\iota$, and finite abelian $A$ of exponent dividing $N$. Its strict normalizer consists exactly of the $T_{t,f,\theta}$ and central phases. Its phase-space maps are*

$$(a, \chi) \longmapsto \big(f(a), (\chi B_\theta(a, -)) \circ f^{-1}\big).$$

*For $|A| > 1$, $F_A$ has full column support and cannot be the realization of a strict pointed map. Nevertheless,*

$$F_A^* F_A = |A| I, \qquad F_A W(v) = \iota(r_A(v))W(J_A v)F_A.$$

*Scope.* Only bicharacters admitting a $\mu_N$-valued quadratic refinement occur in the strict subgroup. Enlarging phases alone does not make Fourier monomial. No canonical Weil splitting is asserted.

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**Provenance.** ids: F1-MON; proved in: f1-cyclotomic.md, section 6; tested by: f1_check.py (M7, Fourier only); reviewed in: f1-adjudication.md.

**Definition 11.14** (Framed cyclotomic kernels and lifted phase isomorphisms)**.** Put $K_N = \mathbb{Q}[\mu_N]/\ker\iota$. The category $\mathrm{Mat}(K_N)$ has finite sets $I$ as objects and $J \times I$ matrices over $K_N$ as morphisms; tensor is Cartesian product of sets and Kronecker product of matrices. The associated modules are explicitly framed by $[1, i]$ in $S_I = \{0\} \sqcup (\mu_N \times I)$. Let $\mathrm{PMat}(K_N)$ be invertible matrices modulo $K_N^\times$, realized by entrywise evaluation.

The groupoid $\mathrm{LiftHyp}_N$ has configurations $A$ as objects. An arrow is $(g, r)$ with $g : V_A \to V_B$ a group isomorphism preserving $\kappa$ and $r : V_A \to \mu_N$ satisfying

$$r(v)r(w)c_B(gv, gw) = c_A(v, w)r(v + w).$$

It denotes the centre-fixing map $(u, v) \mapsto (ur(v), g(v))$; composition composes these maps. Tensor is $A \times B$ on objects and $(g \times h, (v, w) \mapsto r(v)s(w))$ on arrows, with the canonical reordering of phase factors. The unit is $A = 0$ and the symmetry is the swap.

*Scope.* This target is an additive cyclotomic envelope. It is not claimed to be a category of $\mathbb{F}_1$ schemes; arbitrary free torsors are not given coordinate matrices without a frame.

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**Provenance.** ids: D1007; proved in: definitions.md; tested by: no general checker; reviewed in: f1-adjudication.md, O1–O3 repair.

**Proposition 11.15** (Projective realization of lifted phase symmetries)**.** SKETCH *For $N \geq 1$ and a named faithful $\iota$, there is a projective strong symmetric monoidal functor*

$$Q_N : \mathrm{LiftHyp}_N \longrightarrow \mathrm{PMat}(K_N)$$

*defined by the Egorov intertwining equations. Its complex realization is projectively unitary; affine-quadratic maps and Fourier have the values above.*

*Scope.* The lift is input data. Geometric descent of the additive envelope and a lift of every symplectic transformation remain separate questions.

**Provenance.** ids: F1-CORR; proved in: f1-cyclotomic.md, section 7; tested by: not claimed as computationally proved; reviewed in: supporting SKETCH after scope repair.

*Proof.* The intertwining equations have coefficients in $K_N$ and a one-dimensional solution space after embedding in $\mathbb{C}$. Gaussian elimination has the same rank over either field, so there is a nonzero solution over $K_N$. Irreducibility makes it invertible; uniqueness up to scalar gives projective composition and tensor coherence. Unitary normalization is performed only after complex realization. □

### 11.6 Relational Heisenberg geometry over bands

**Definition 11.16** (Bands and the upper-unitriangular crowd)**.** A band is a commutative pointed multiplicative monoid $B$ with a null ideal $N_B$ of formal sums, in which every $a$ has a unique $-a$ with $a + (-a) \in N_B$; morphisms preserve null sums. The regular partial field $\mathbb{F}_1^{\pm} = \{0, 1, -1\}$ uses sums vanishing in $\mathbb{Z}$. The Krasner hyperfield $\mathbb{K} = \{0, 1\}$ declares a sum null exactly when it has zero or at least two nonzero terms. Define $H_{\text{crowd}}(B) = B^3$, represented by $h(a, b, t)$ as above, with identity $(0, 0, 0)$. Its colaw consists of triples for which

$$\sum_i a_i, \qquad \sum_i b_i, \qquad t_1 + t_2 + t_3 + a_1 b_2 + a_1 b_3 + a_2 b_3$$

and both cyclic variants of the last sum lie in $N_B$. The inverse set of $h$ consists of $k$ with $(h, k, 1)$ in the colaw; the product set $h * k$ consists of $c$ for which some $d$ has both $(h, k, d)$ and $(c, d, 1)$ in the colaw.

*Scope.* Nullity is part of the band, not an invented binary addition. The signed band and the raw pointed phase monoid have different structure.

**Provenance.** ids: D1008; proved in: definitions.md; tested by: f1_check.py (M10); reviewed in: supporting comparison.

**Proposition 11.17** (A band-level Heisenberg crowd and its ring fibres)**.** **SKETCH** *The functor $B \mapsto H_{\text{crowd}}(B)$ on bands, with the displayed colaw, is an affine algebraic crowd. For every commutative ring $R$, its fibre is exactly $\mathrm{UT}_3(R)$. The signed fibre has 27 points, 319 colaw triples and 23 points admitting an inverse inside that fibre. The Krasner fibre has eight points and 152 colaw triples, with multivalued inverses.*

*Scope.* These finite fibres are not ordinary groups. Even a crowd product with the identity may be empty when its required inverse mediator is absent. No Hilbert representation is inferred from the point counts.

**Provenance.** ids: F1-CROWD; proved in: f1-comparisons.md, section 1; tested by: f1_check.py (M10); reviewed in: supporting SKETCH.

This specializes the algebraic $\mathrm{SL}_3$ crowd of Lorscheid–Thas (2023), section 5.4. Restricting to an identity-containing subset preserves the crowd axioms; the displayed coordinate relations are affine. Over rings they say $h_1 h_2 h_3 = I$. The signed example retains $h(1, 0, 0) * h(0, 1, 0) = \{h(1, 1, 1)\}$, and hence the central cross-term, although $h(1, 0, 0)^2$ is absent. This provides a constructive second Heisenberg formulation, with a representation theory still to specify.

### 11.7 Tensor categories and the qubit fusion object

**Definition 11.18** (The categorical phase system)**.** Let $\mathcal{C}_N(A) = \mathrm{Rep}_{\mathbb{C}}(H_N(A))$, with all linear intertwiners and internal tensor using the diagonal group action. Let $\mathcal{C}_N(A)_\lambda$ be the full subcategory on which central $u$ acts as $\lambda(u)I$. The unitary version chooses invariant Hermitian inner products and has the same intertwining spaces. Write $\omega_A$ for the forgetful tensor functor. For groups $H, K$ with specified central $\mu_N$, define $\mathrm{Rep}(H) \boxtimes_{\mathrm{match}} \mathrm{Rep}(K)$ as the full subcategory of $\mathrm{Rep}(H \times K)$ on which all $(u, u^{-1})$ act trivially.

*Scope.* The matching condition is explicit. It is not an unspecified relative tensor product, and it is different from internal fusion.

**Provenance.** ids: D1011; proved in: definitions.md; tested by: f1_check.py (examples); reviewed in: supporting comparison.

**Proposition 11.19** (Central sectors explain why the category matters)**.** SKETCH *For $N \geq 1$ and finite abelian $A$ of exponent dividing $N$, $\mathcal{C}_N(A)$ is a symmetric fusion category with a faithful tensor functor $\omega_A$, and*

$$\mathcal{C}_N(A) = \bigoplus_{\lambda \in \widehat{\mu_N}} \mathcal{C}_N(A)_\lambda, \qquad \mathcal{C}_N(A)_\lambda \otimes \mathcal{C}_N(A)_\nu \subseteq \mathcal{C}_N(A)_{\lambda\nu}.$$

*Moreover, for finite groups $H, K$ with specified central embeddings of $\mu_N$,*

$$\mathrm{Rep}(H \boxdot K) \simeq \mathrm{Rep}(H) \boxtimes_{\mathrm{match}} \mathrm{Rep}(K).$$

*On simple external tensors, the matching condition is equality of the two central characters. In particular a fixed nontrivial sector is not a tensor category by itself.*

*Scope.* The unit of matching composition on this family is $\mathrm{Rep}(\mu_N)$, with one simple per grade. It is not the unit for the unrestricted Deligne product.

**Provenance.** ids: F1-CAT; proved in: f1-comparisons.md, section 5; tested by: f1_check.py (M6,M12; examples); reviewed in: supporting SKETCH.

*Proof.* Average an inner product over the finite group to obtain semisimplicity. Central character eigenspaces give the grading, and diagonal tensor multiplies eigencharacters. A representation factors through the central quotient precisely when its antidiagonal centre acts trivially; inflation and descent are inverse tensor functors. On an external simple the condition is $\lambda(u)\nu(u)^{-1} = 1$. □

**Definition 11.20** (Dihedral and quaternion qubit data)**.** For $e \in \{0, 1\}$ let $G_e = \mathbb{F}_2^3$ with

$$(t, a, b)(u, c, d) = (t + u + ad + e(ac + bd), a + c, b + d).$$

Put

$$\sigma_e(t, a, b) = (-1)^t i^{e(a+b)} Z^b X^a, \quad X = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad Z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix},$$

using representatives $a, b \in \{0, 1\}$ in the exponent of $i$. Define $\nu_2(\sigma_e) = |G_e|^{-1} \sum_g \mathrm{Tr}(\sigma_e(g^2))$.

*Scope.* The group law distinguishes the two cocycle choices. The quaternion matrices require fourth-root phases in this monomial realization.

**Provenance.** ids: D1012; proved in: definitions.md; tested by: f1_check.py (M12); reviewed in: supporting comparison.

**Proposition 11.21** (Two qubit objects with the same fusion rules). SKETCH *$G_0 = D_8$ of order eight and $G_1 = Q_8$. Each has four one-dimensional characters and the single two-dimensional irreducible $\sigma_e$. Writing the four characters as $1, \alpha, \beta, \alpha\beta$,*

$$\sigma_e \otimes \sigma_e \simeq 1 \oplus \alpha \oplus \beta \oplus \alpha\beta, \qquad \nu_2(\sigma_0) = 1, \quad \nu_2(\sigma_1) = -1.$$

*Their sourced tensor-category identifications are*

$$\mathrm{Rep}(D_8) \simeq \mathrm{TY}(\mathbb{F}_2^2, \chi_{\mathrm{alt}}, 1/2), \qquad \mathrm{Rep}(Q_8) \simeq \mathrm{TY}(\mathbb{F}_2^2, \chi_{\mathrm{alt}}, -1/2).$$

*Scope.* The displayed tensor square uses one diagonal symmetry group. It does not replace the external tensor describing independent qubits.

**Provenance.** ids: F1-FUSION; proved in: f1-comparisons.md, section 6; Shimizu, p. 20; tested by: f1_check.py (M12); reviewed in: supporting SKETCH.

The two-dimensional character is $2, -2$ on the two central elements and zero elsewhere. Squaring it gives the sum of the four quotient characters. Six elements square to 1 in $D_8$ and only two do in $Q_8$, giving the indicator signs. The categorical identifications and their distinction are in Shimizu's *Frobenius–Schur indicators in Tambara–Yamagami categories*, Definition 3.1, Table 1 and page 20.

For two independent reference qubits the relevant group instead is $H_2(C_2) \boxdot H_2(C_2) = H_2(C_2 \times C_2)$, of order 32. Its nontrivial-central-character irreducible has dimension four. Both this representation and internal fusion use the underlying Hilbert space $\mathbb{C}^2 \otimes \mathbb{C}^2$, but their symmetry actions differ. This is a concrete reason to retain the tensor category, its sectors and its fibre functor as part of the quantum object.

The fibre functor is essential to this particular formulation:

$$\mathrm{End}_{\mathrm{Rep}(H)}(\sigma) = \mathbb{C}, \qquad \mathrm{Hom}_{\mathrm{Rep}(H)}(1, \sigma) = 0$$

for the nontrivial central character. The equivariant category alone does not give every state vector or every qubit observable. Those are recovered as vectors of $\omega(\sigma) = \mathbb{C}^2$ and elements of $\mathrm{End}_{\mathbb{C}}(\omega(\sigma)) = M_2(\mathbb{C})$. An alternative categorical state notion would have to be specified.

## 11.8 The independent Hall and categorical-oscillator route

**Definition 11.22** (Pointed-set Hall algebra and Fock realization). On the normal pointed-set category above, the rational Hall vector space has basis $u_r$ indexed by rank. Its product counts pointed subsets $T$ with the prescribed isomorphism classes of $T$ and $S/T$. Put $\mathcal{F}_{\mathrm{alg}} = \mathbb{C}[x]$ with $\langle x^m, x^n \rangle = \delta_{mn} n!$, define $a^\dagger f = xf$, $af = df/dx$, and let $\mathcal{F}_{\mathrm{bos}}$ be the Hilbert completion.

*Scope.* Hall coefficients and the Fock space are additive realizations of the pointed-set data. Creation is unbounded.

**Provenance.** ids: D1010; proved in: definitions.md; tested by: f1_check.py (M8,M9); reviewed in: supporting comparison.

**Proposition 11.23** (A bosonic Heisenberg algebra from finite sets). SKETCH *The rational Hall algebra of finite normal pointed sets is $\mathbb{Q}[x]$ under*

$$u_m u_n = \binom{m+n}{m} u_{m+n}, \qquad u_n \longmapsto x^n/n!.$$

*On $\mathcal{F}_{\mathrm{alg}}$, creation and annihilation are adjoints and $[a, a^\dagger] = I$. In the normalized basis $x^n/\sqrt{n!}$ their coefficients are $\sqrt{n+1}$ and $\sqrt{n}$.*

*Scope.* The polynomial Hall algebra alone is not the Weyl algebra; annihilation is additional operator structure. No finite-dimensional characteristic-zero CCR representation is asserted.

**Provenance.** ids: F1-HALL; proved in: f1-comparisons.md, section 4; tested by: f1_check.py (M8,M9); reviewed in: supporting SKETCH.

The subset count proves the Hall formula. The operator calculation is $d(xf)/dx - x\,df/dx = f$. The source construction is Szczesny's Hall algebra, section 3. Baez–Dolan give the factorial groupoid weights and the positive categorical relation

$$AA^\dagger \simeq A^\dagger A \oplus \mathrm{id}.$$

Morton–Vicary realize the categorified Heisenberg algebra using spans of finite-set groupoids: add/remove histories split according to whether the removed point was just added. Their two-linearization retains the tower of $\mathrm{Rep}(S_n)$, with the stated finiteness/completion caveats. The connection to $\mathbb{F}_1$ is the equivalence between finite sets and the core of finite pointed modules. It is not a claim that these authors construct one universal $\mathbb{F}_1$ scheme theory.

This supplies a categorical north star different from the finite Weyl system. Degree two of the representation-category tower has two simples in $\mathrm{Rep}(S_2)$; that is not automatically a qubit. With two colours, the one-particle complex space is a qubit and the two-boson sector is $\mathrm{Sym}^2(\mathbb{C}^2)$, of dimension three. A one-mode cutoff to $|0\rangle, |1\rangle$ instead has projected commutator $\mathrm{diag}(1, -1)$, and is not closed under creation. The Hall operation uses wedge and particle-number addition; the finite configuration tensor uses smash and rank multiplication.

## 11.9 Toric, analytic and arithmetic-statistical formulations

**Definition 11.24** (A quantum torus and its selected central fibre)**.** For a named $\xi \in \mathbb{C}^\times$ put

$$\mathcal{T}_\xi = \mathbb{C}\langle U^{\pm 1}, V^{\pm 1}\rangle/(VU - \xi UV).$$

If $\xi$ is a primitive $N$th root, put $\mathcal{T}_\xi(1,1) = \mathcal{T}_\xi/(U^N - 1, V^N - 1)$.

*Scope.* The parameter $\xi$, its order $N$ and a finite-field cardinality are different data. Central values are explicitly selected.

**Provenance.** ids: D1013; proved in: definitions.md; tested by: cyclic Weyl checks; reviewed in: supporting comparison.

**Proposition 11.25** (The qubit as a quantum-torus fibre)**.** SKETCH *For $\xi$ primitive of order $N$, $\mathcal{T}_\xi(1,1) \simeq M_N(\mathbb{C})$ via $Ue_j = e_{j+1}$ and $Ve_j = \xi^j e_j$ on $\mathbb{Z}/N$. For $N = 2$ this is a qubit algebra; two independent fibres tensor to $M_4(\mathbb{C})$.*

*Scope.* The universal quantum torus has not been identified with a finite matrix algebra. At $\xi = 1$ it is commutative.

**Provenance.** ids: F1-TORUS; proved in: f1-comparisons.md, section 7; tested by: f1_check.py (cyclic cases); reviewed in: supporting SKETCH.

The $N^2$ clock/shift words are trace-orthogonal and span, which proves the selected-fibre statement. This is the twisted-lattice picture of Neeb's rational quantum tori. Monoidal tori retain character lattices; a chosen skew pairing is additional symplectic data. Manin's cyclotomic analytic geometry supplies Habiro completions and evaluation or Taylor maps at roots of unity. It suggests an arithmetic interpolation problem for quantum fibres, but does not itself give a common Hilbert space or compatible Weil intertwiners. A deformation limit $\xi \to 1$ is different from all the preceding $q \to 1$ operations.

**Frobenius descent.** Borger's lambda geometry treats lambda structure as descent data below $\operatorname{Spec}\mathbb{Z}$. For flat schemes it can be expressed as commuting Frobenius lifts. Monoid algebras have the lift $m \mapsto m^p$. Its naive coordinatewise application to the integral Heisenberg group fails already on the additive subgroup: $(1+1)^p \neq 1^p + 1^p$ in $\mathbb{Z}$. This rejects that particular group-compatible lift, not all arithmetic descent. A quantum extension should specify whether arithmetic maps act by homomorphisms, noninvertible operators or correspondences. No qubit or Born rule is selected by lambda structure alone.

**A published quantum-statistical realization.** Connes–Consani–Marcolli's *Fun with* $\mathbb{F}_1$, Proposition 6.1 and Theorem 6.2, constructs an $\mathbb{F}_1$ model of the Bost–Connes endomotive, with the rational system recovered by scalar extension. It uses a cyclotomic tower and Frobenius/transfer correspondences. Its complex realization has the time evolution

$$\sigma_t(\mu_n) = n^{it}\mu_n, \qquad \sigma_t(e(r)) = e(r).$$

In the standard representation on $\ell^2(\mathbb{N}_{>0})$, $He_n = (\log n)e_n$ and $\operatorname{Tr}(e^{-\beta H}) = \zeta(\beta)$ for $\beta > 1$; see Bost–Connes (1995). This is genuine quantum statistical dynamics, not a finite Stone–von Neumann representation. Choosing two energy levels provides an encoding but not a sector invariant under its full observable algebra.

Lieber–Manin–Marcolli, published in 2022, develop lifts to Grothendieck rings, assembler categories, spectra and Nori motives. This is an independent literature-based reason to explore categorical targets, without asserting that these categories are the fusion categories of the finite Heisenberg construction.

**Tropical and motivic boundaries.** The band/Krasner framework retains combinatorial flags beyond a single coordinate apartment. Boolean semirings, Krasner hyperfields, signed partial fields and pointed monoids have different additive information. Connes–Consani's Arithmetic Site uses a tropical semiring on a multiplicative-semigroup topos to realize arithmetic correspondences. It does not prescribe a local Weyl algebra or positive quantum states. Relative and homotopical geometries broaden the candidate bases and targets, but no general Heisenberg representation theorem for those bases is established in this sidequest. Finally, Bejleri–Marcolli's *Quantum field theory over* $\mathbb{F}_1$ concerns torifications and varieties attached to Feynman integrals; its introduction explicitly distinguishes that problem from physical Lagrangians and Feynman rules over $\mathbb{F}_1$.

**An additional tensor analogy: fermionic occupancy.** Lin–Wang–Wu–Yang (2013) categorify the one-mode fermion algebra with $c^2 = (c^\dagger)^2 = 0$ and $cc^\dagger + c^\dagger c = 1$. Occupancy subsets suggest a combinatorial exterior-power route; this is our proposed $\mathbb{F}_1$ bridge, not their claim. Hadzihasanovic–de Felice–Ng (2018), section 2, give a precise monoidal category of local fermionic modes and maps of definite parity. Its basic object has underlying space $\mathbb{C}^{1|1}$; tensor adds parity and the physical fermionic swap negates $|11\rangle$. The two occupancy labels therefore do not supply unrestricted qubit operations. Parity-preserving one-mode observables are diagonal, while a fixed-parity two-mode sector can encode a qubit. Signed phases are a candidate carrier for exchange signs over $\mathbb{F}_1$; the cited complex categorical construction does not prove that descent. This is a further composition law to compare with the bosonic and distinguishable-system routes.

### 11.10 Precisely formulated outputs and open comparisons

The finite arithmetic candidate is the configuration-to-phase-system functor, with a named phase realization and a checked tensor law. The categorical candidate retains

$$(\mathcal{C}_N(A),\ \mathcal{C}_N(A)_\iota,\ \omega_A)$$

together with internal fusion and matching external composition. The pointed normal-map candidate already realizes an ordinary qubit over the bare combinatorial base. The Heisenberg crowd is a geometric object with correct ring fibres, while the Hall/groupoid route gives an independent categorical oscillator. Bost–Connes supplies the arithmetic quantum-statistical test case. These are several precise formulations, not evidence that all the analogies must coincide.

The next comparison problems are concrete: relate a chosen Tits reflection lift to its projective Fourier operator; extend the geometric Heisenberg model to nonreference cocycles; identify a genuine geometric home for the additive phase kernels; compare finite cyclotomic fibres with arithmetic transfer maps; and distinguish tensor products of distinguishable systems from bosonic particle-number sectors. A theory on arbitrary $\mathbb{F}_1$ schemes still needs its input category, morphisms and realization functor specified. It is a research target, not a theorem or a well-formed universal conjecture supplied by the present literature.

Seven finite-kernel claims are **PROVED**; the extensions remain **SKETCH**. Thirteen exact example gates and their mutations check the finite computations. The cited literature and our deductions are distinguished throughout.